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Diffusional Coupling between Micro and Macroporosity for NMR Relaxation in Sandstones and Grainstones¹

Vivek Anand² and George J. Hirasaki³

ABSTRACT

The interpretation of NMR measurements with fluid saturated rocks assumes that the T_1 or T_2 distribution is directly related to the pore size distribution. In many cases, this assumption is valid. However, the assumption breaks down if the fluid in different sized pores is coupled through diffusion. In such cases, the estimation of formation properties such as permeability and irreducible water saturation using the traditional $T_{2,cutoff}$ method gives erroneous results. Several techniques like “spectral” BVI and tapered $T_{2,cutoff}$ were introduced to take into account the effects of diffusional coupling for better estimation of properties.

This paper aims to provide a theoretical and experimental understanding of NMR relaxation in systems with diffusional coupled micro and macropores. Relaxation is modeled such that the fluid molecules relax at the surface of the micropore and simultaneously diffuse between the two pore types. The T_2 distribution of the pore is a function of several parameters including micropore surface relaxivity, fluid diffusivity and pore geometry. The governing parameters are combined in a single coupling parameter (α) that is defined as the ratio of the characteristic relaxation rate of the coupled pore to the rate of diffusional mixing of magnetization between micro and macropore. Depending on the value of α , the two pore types can communicate through total coupling, intermediate coupling or decoupled regimes.

The model is applied to treat diffusional coupling in sandstones with a distribution of macropores lined with clay flakes. Simulations are verified by comparing with experimental results for chlorite-coated, North Burbank sandstone. It is observed that the relaxation time distribution shows a bimodal distribution at 100% water saturation but a unimodal distribution when saturated with hexane. This occurs because the extent of coupling is higher for hexane than for water due to lower relaxivity and higher diffusivity of hexane. The α values indicate intermediate coupling for water and strong coupling for hexane.

The model is also applied to explain pore coupling in grainstone carbonates with intra and intergranular porosity. In this case, α is shown to have a quadratic dependence on grain radius and inverse dependence on micropore radius. The theory is experimentally validated on several systems with microporous particles of varying grain diameters and known microporosities. Here too, the T_2 distribution at 100% water saturation varies from bimodal for coarse-grained particles to unimodal for fine-grained particles. The transition from bimodal to unimodal distribution is also predicted theoretically from the values of α .

Keywords: NMR, diffusional coupling, microporosity, relaxation time cutoffs

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INTRODUCTION

NMR T_2 measurements are often used to estimate the pore structure and formation properties such as porosity, permeability and irreducible water saturation. NMR estimation of pore size distribution assumes that in the fast diffusion limit, T_2 of a fluid in a single pore is given as

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \rho_2 \left(\frac{S}{V} \right)_{pore} \quad (1)$$

where T_{2B} is the bulk relaxation time, ρ_2 is the transverse surface relaxivity and $(S/V)_{pore}$ is the pore surface-to-volume ratio. For a rock sample having a pore size distribution, each pore size is assumed to be associated with a T_2 component and the net magnetization relaxes as a multi-exponential decay.

$$M(t) = \sum_j f_j \exp\left(-\frac{t}{T_{2,j}}\right) \quad (2)$$

where f_j is the amplitude of each $T_{2,j}$. Such interpretation of NMR measurements assumes that pores of different sizes relax independent of each other. In many cases, this assumption is valid. However, it is observed that the assumption often fails for shaly sandstones and carbonates especially in grainstones and packstones. Ramakrishnan et al. (1999) explained that the failure could be understood by considering the diffusion of magnetization between intra (micro) and intergranular (macro) pores. The resulting T_2 distribution is influenced by surface-to-volume ratio of both micro- and macropores and thus, the correspondence between T_2 and pore size distribution is lost. In such cases, traditional method of employing a sharp $T_{2,cutoff}$ for estimating bound fluid fractions and permeability would not be applicable.

The effect of diffusional coupling on accurate estimation of irreducible saturation is illustrated for the case of clay

lined pores in sandstones. Figure 1 shows the schematic diagram of a clay lined pore (Straley et al., 1995) (a) at 100% water saturation and (b) at irreducible saturation after capillary drainage of the macropore. First consider the case of the pore at irreducible saturation in Figure 1(b). Since the macropore is drained, there is no diffusional exchange between fluid in micro- and macropore. Thus, the fluid in micropores relaxes with a rate proportional to the surface-to-volume ratio of the micropores. Now consider the case of 100% water-saturated pore in Figure 1(a). If the fluid in micropores is in diffusional exchange with that in the macropore, its apparent volume is larger even though the surface area for relaxation remains same. Thus, the fluid in micropores relaxes with a longer relaxation time than at irreducible saturation. If a sharp $T_{2,cutoff}$ based on the relaxation time of micropores at irreducible saturation is employed, it may underpredict the irreducible saturation because a fraction of fluid in micropores is relaxing slower than $T_{2,cutoff}$. Techniques like “spectral” BVI and tapered $T_{2,cutoff}$ (Coates et al., 1998; Kleinberg et al., 1997) usually provide better estimates in such cases by assuming an irreducible fraction which is diffusively coupled with bulk pore fluid for each pore size. However, a theoretical basis for the application of these techniques needs to be established.

DIFFUSIONAL COUPLING BETWEEN MICRO AND MACROPORES

Mathematical modeling

We begin by numerically simulating the decay of magnetization in the symmetry element between two clay flakes in the pore model of Figure 1. Figure 2 shows the simplified

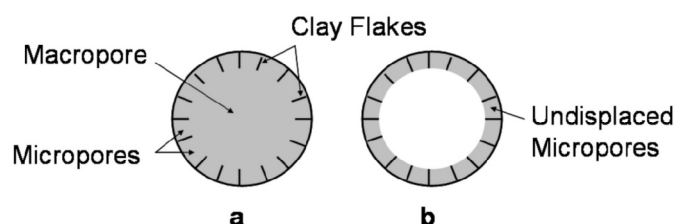


FIG. 1 Conceptual model of pore coupling in clay lined pore in sandstones (Straley et al., 1995). (a) 100% water saturated pore. Fluid in micropores relaxes with $T_2 > T_{2,cutoff}$ due to diffusional exchange with the macropore. (b) Pore at irreducible water saturation. Fluid relaxes with a rate proportional to surface to volume ratio of the micropore.

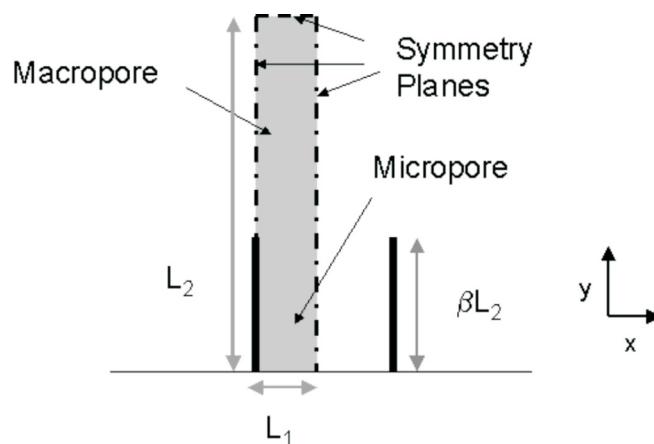


FIG. 2 Physical model of coupled pore geometry. Fluid molecules relax at the micropore surface while diffusing between the micro and macropore.

pore model consisting of a clay-bounded micropore adjacent to a macropore. The curvature of the outer grain surface and the thickness of the clay flake are neglected. The fluid molecules relax at the surface of the micropore and simultaneously diffuse between the two pore types. As a result, the T_2 distribution of the pore is determined by several parameters such as micropore surface relaxivity, diffusivity of the fluid and geometry of the pore system. The simple model helps to keep the analysis tractable and also captures the essential features of more complicated pore coupling models (Toumelin et al., 2003).

The coupled pore is defined by three geometrical parameters: half length to the middle of macropore (L_2), half-distance between clay flakes (L_1) and microporosity fraction (β). The decay of magnetization per unit volume (M) in the pore is given by the Bloch-Torrey equation

$$\frac{\partial M}{\partial t} = D \nabla^2 M - \frac{M}{T_{2,B}} \quad (3)$$

where D is the diffusivity of the fluid. The boundary conditions are

$$D \vec{n} \cdot \nabla M + \rho M = 0 \text{ at micropore surface} \quad (4)$$

$$\vec{n} \cdot \nabla M = 0 \text{ at symmetry planes} \quad (5)$$

where $\vec{n}$ is the unit normal pointing outwards from the pore surface and ρ is the surface relaxivity. A uniform magnetization is assumed in the entire pore initially. In addition, the bulk relaxation rate is assumed to be very small in comparison to surface relaxation and is neglected.

We introduce characteristic parameters to make the governing equations (3-5) dimensionless. Expressing the equations in dimensionless form helps to characterize the systems with fewer parameters. The spatial variables are, thus, normalized with respect to the half-length of the pore (L_2), magnetization with respect to initial magnetization and time with respect to a characteristic relaxation time, $T_{2,c}$ defined as

$$T_{2,c} = \frac{V_{total}}{\rho S_{active}} = \frac{L_1 L_2}{\rho \beta L_2} = \frac{L_1}{\rho \beta} \quad (6)$$

In the above equation, S_{active} refers to the surface of the micropore at which relaxation is taking place and V_{total} refers to the total volume of the pore. An analogous relaxation time of the micropore, $T_{2,\mu}$, is defined as

$$T_{2,\mu} = \frac{1}{\rho} \left(\frac{V}{S} \right)_\mu = \frac{\beta L_2 L_1}{\rho \beta L_2} = \frac{L_1}{\rho} \quad (7)$$

where $(V/S)_\mu$ refers to volume-to-surface ratio of the

micropore. The characteristic relaxation time $T_{2,c}$ is related to $T_{2,\mu}$ by comparing equations (6) and (7)

$$\Rightarrow T_{2,c} = \frac{T_{2,\mu}}{\beta} \quad (8)$$

We also introduce two dimensionless groups, namely the aspect ratio of the pore η and Brownstein number μ (Brownstein and Tarr, 1979), defined as

$$\eta = \frac{L_2}{L_1} \quad (9)$$

$$\mu = \frac{\rho L_2}{D} \quad (10)$$

The dimensionless groups can be combined into a single parameter called coupling parameter (α) defined as

$$\alpha = \beta \eta \mu = \frac{\rho \beta L_2^2}{D L_1} \quad (11)$$

η is the aspect ratio of the characteristic dimension of the macropore to that of the micropore. μ and α are ratios of the relaxation rate to the diffusion rate but μ treats the system as a single macropore while α includes the contribution of the micropore to the total surface-to-volume ratio. The physical significance of the parameters is detailed in the Results section. The governing equations can thus be expressed in terms of above mentioned dimensionless parameters as given in Appendix A. A finite difference Alternating Direction Implicit technique (Peaceman and Rachford, 1955) is employed for the numerical solution of the dimensionless equations. The details of the numerical technique are described in Appendix B.

Results

The decay of magnetization in the coupled pore is characterized by three parameters; aspect ratio of the pore (η), microporosity fraction (β) and Brownstein number (μ). Depending on the value of μ , defined as,

$$\mu = \frac{\text{Relaxation rate}}{\text{Diffusion rate}} = \frac{\rho / L_2}{D / L_2^2} = \frac{\rho L_2}{D} \quad (12)$$

the decay can be classified into fast, intermediate and slow diffusion regimes. In the fast diffusion regime ($\mu \ll 1$), the lowest eigen value of equation (3) completely dominates and the decay curve is mono-exponential. In the slow diffusion regime ($\mu \gg 10$), the higher modes also contribute to the relaxation and the decay curve is multi-exponential (Brownstein and Tarr, 1979). These diffusion regimes are

visualized with the help of snapshots of magnetization in the pore at intermediate decay times. Figure 3 shows the contour plots of magnetization for $\beta = 0.5$ and $\eta = 10$ at dimensionless time $t = 1$ for various values of μ . For μ small compared to 1 ($\mu = 0.1$), fast diffusion leads to nearly homogenous magnetization in the entire pore. With the increase in the value of μ , gradients in magnetization along the longitudinal direction become substantial. These gradients imply that the micropore is relaxing much faster than the macropore.

Note that μ is based on an analysis for a one-dimensional system with a single characteristic length (L_2). For a one dimensional pore, μ can also be expressed as the square of the ratio of the pore length to the diffusion length as shown

$$\mu = \frac{\rho L_2}{D} = \frac{L_2^2}{DL_2/\rho} = \left(\frac{L_2}{\sqrt{DT_{2,1-D}}} \right)^2. \quad (13)$$

Here $1/T_{2,1-D}$ is the relaxation rate of the one-dimensional pore given as

$$\frac{1}{T_{2,1-D}} = \rho \left(\frac{S}{V} \right)_{pore} = \frac{\rho}{L_2}. \quad (14)$$

The assumption of a single surface-to-volume ratio inherent in the definition of μ fails for a system of coupled micro and macropore with different surface-to-volume ratios. Figure 4 shows the contour plots of magnetization at dimensionless time $t = 1$ for three systems with $\beta = 0.5$ and $\mu = 0.1$ but with increasing aspect ratios. Even though μ remains same, the

systems with larger η show larger gradients in magnetization. This increase in gradients is because as η increases, the dimension of micropore decreases ($\eta = L_2/L_1$) and thus, the relaxation rate of micropore increases. Since μ is independent of the micropore dimension, it can not characterize relaxation regimes in coupled pore systems.

Coupling parameter

Two processes characterize the decay of magnetization in the coupled geometry: relaxation of spins at the micropore surface and diffusion of spins between the micro- and macropores. If the relaxation of spins in the micropore is much faster than the inter-pore diffusion, coupling between the two pore types is small. On the other hand, if the diffusion rate is much greater than the relaxation rate, the two pores are significantly coupled with each other. Thus, the extent of coupling is characterized with the help of a coupling parameter (α) that compares the characteristic relaxation rate of the pore (equation (6)) to the rate of diffusional mixing of spins between the micro and macropore, i.e.

$$\alpha = \frac{1/T_{2,c}}{D/L_2^2} = \frac{\rho\beta/L_1}{D/L_2^2} = \frac{\rho\beta L_2^2}{DL_1}. \quad (15)$$

The physical significance of α is illustrated with the help of simulated T_2 distributions for the previously mentioned case of $\beta = 0.5$ and $\eta = 10$ as shown in Figure 5. For small α ($= 0.5$), relaxation at the micropore surface is small compared to diffusional mixing between the two pore types.

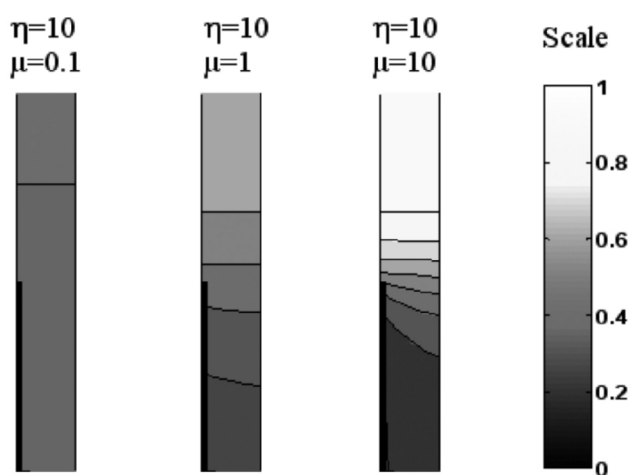


FIG. 3 Contour plots of magnetization at $t = 1$ in coupled pores with $\beta = 0.5$ and $\eta = 10$ and different values of μ . The gradients along the longitudinal direction for larger μ imply that micropore is relaxing faster compared to macropore.

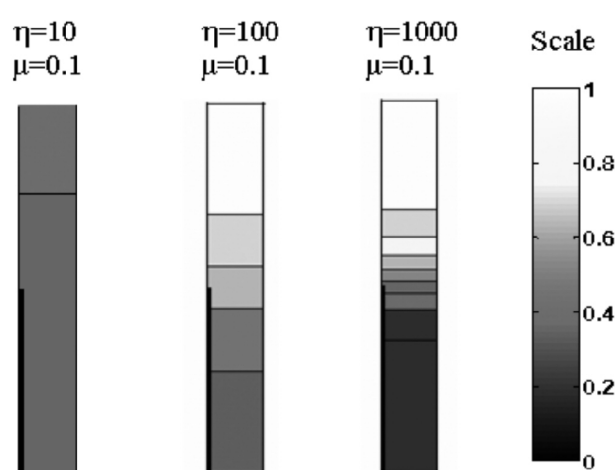


FIG. 4 Contour plots of magnetization at $t = 1$ in coupled pores with $\beta = 0.5$ and $\mu = 0.1$ and different values of η . For comparison, the aspect ratios are not drawn to scale. Note the difference in contour plots for systems with same μ .

Thus, the micro and macropore relax at the same rate and the T_2 distribution shows a single peak. As the value of α increases ($\alpha = 5$), some spins in the micropore are able to relax faster than they can diffuse into the macropore. This results in the appearance of a peak at short relaxation times (micropore peak). In addition, the spins in the macropore diffuse to the micropore slowly and thus, the macropore peak shifts towards longer relaxation times. As a still weaker coupling regime is approached ($\alpha = 50$), inter-pore diffusion becomes negligible and the entire micropore relaxes independent of the macropore.

α can also be expressed as the square of the ratio of the length of the pore to the diffusion length in characteristic relaxation time $T_{2,c}$ (equation (6)), as shown below

$$\alpha = \frac{\rho\beta L_2^2}{DL_1} = \frac{L_2^2}{D(L_1/\rho\beta)} = \left(\frac{L_2}{\sqrt{DT_{2,c}}} \right)^2. \quad (16)$$

If the macropore length is much larger than the diffusion distance in characteristic time, i.e. $L_2 \gg \sqrt{DT_{2,c}} \Rightarrow \alpha \gg 1$, the

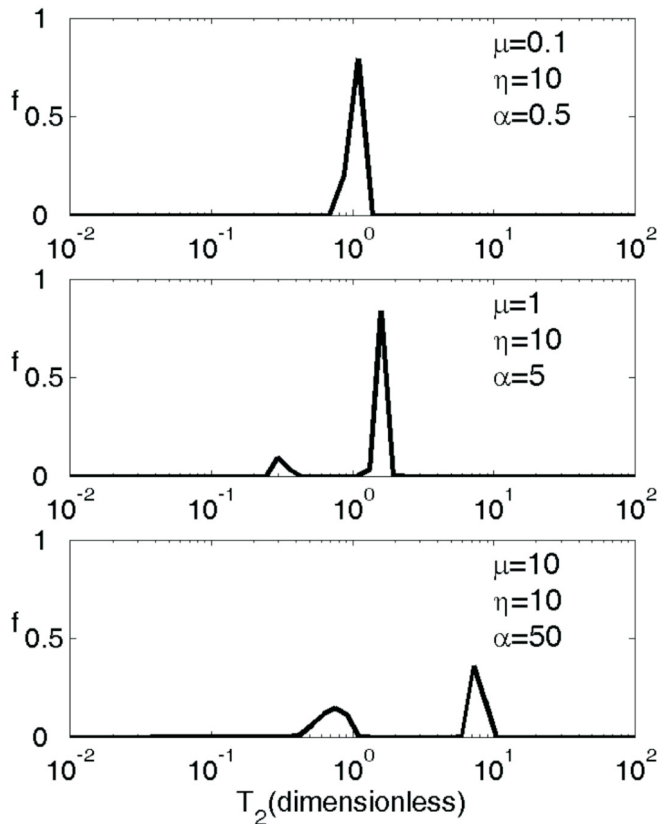


FIG. 5 Simulated T_2 distributions of coupled pores with $\beta = 0.5$ and $\eta = 10$ and different values of μ . The systems transition from unimodal to bimodal distribution with increase in α .

pores are decoupled and vice versa. Since the characteristic relaxation time takes into account the effective surface-to-volume ratio of the coupled pore, α provides a better metric than μ to quantify coupling between micro and macropore. Figure 6 shows the simulated T_2 distributions of systems with $\beta = 0.5$ and $\mu = 0.1$ but with increasing η . The systems progressively transition from unimodal to bimodal distribution as η increases even though μ remains same. The decrease in coupling is however, quantified by increasing values of α .

The relaxation characteristics of micro- and macropores, as demonstrated in T_2 distributions of Figures 5 and 6, are separately analyzed in terms of α as described below.

Micropore relaxation

The fraction of total area under the micropore peak, ψ , in the T_2 distribution gives the magnitude of the fraction of the microporosity which is decoupled from the rest of the pore. Thus, ψ serves as the criterion to quantify the extent of coupling between micro- and macropores. For totally coupled

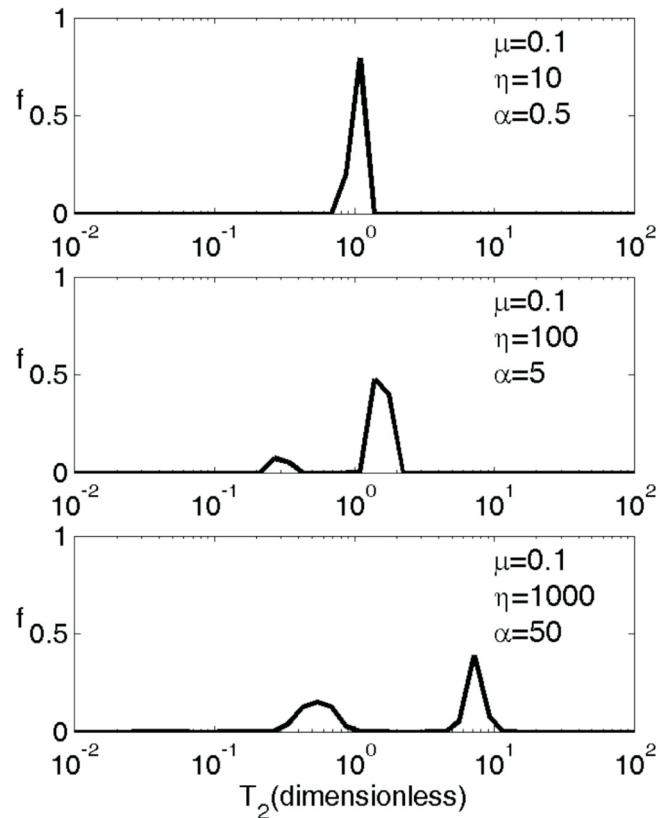


FIG. 6 Simulated T_2 distributions of coupled pores with $\beta = 0.5$ and $\mu = 0.1$ and different values of η . The systems transition from unimodal to bimodal distribution with increase in α even though μ remains same.

micro and macropore $\psi = 0$ while for decoupled pores $\psi = \beta$. A value of ψ between 0 and β indicates intermediate state of diffusional coupling. Figure 7 shows the plot of ψ normalized by β (henceforth referred to as independent microporosity fraction) with α . The curves correspond to different β and span a range of η from 10 to 1000. The results show that depending on the value of α , the micro and macropore can be in one of the three states of:

1. Total coupling ($\alpha < 1$): For values of α less than 1, diffusion is much faster than the relaxation of the magnetization in the micropore. Thus, the micropore is totally coupled with the macropore and the entire pore relaxes with a single relaxation rate.
2. Intermediate coupling ($1 < \alpha < 250$): In this case, diffusion is just fast enough to couple part of the micropore with the macropore. The T_2 distribution consists of distinct peaks for the two pore types but the amplitudes of the peaks are not proportional to the porosity fractions.
3. Decoupled ($\alpha > 250$): The two pore types relax independent of each other and the T_2 spectrum consists of separate peaks with amplitudes representative of the porosity fractions (β and $1-\beta$ for micro and macroporosity, respectively). Further, the dimensionless relaxation time of the micropore peak reaches a value β (Appendix C) indicating complete independence of the two pores.

The independent microporosity fraction correlates more strongly with α than with μ (inset in Figure 7). This is because α has dependence on the length scale of both micro and macropore and thus, provides a better measure to quantify the extent of coupling.

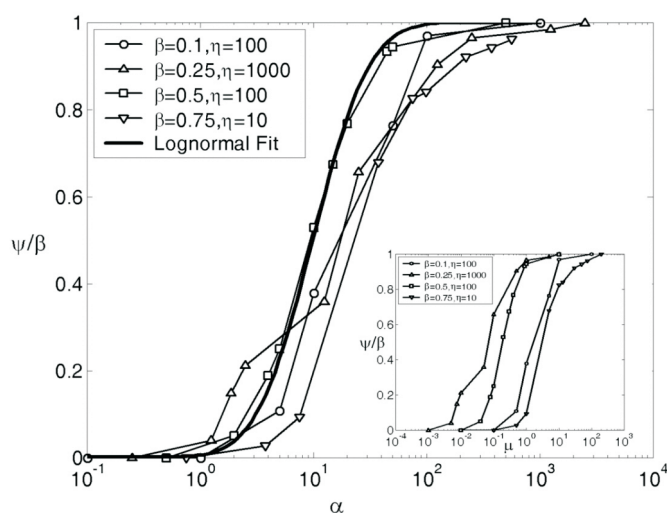


FIG. 7 Independent microporosity fraction (ψ/β) shows lognormal relationship with α . The inset shows plot of (ψ/β) with μ .

The sigmoidal character of the curves in Figure 7 suggests that we can establish a lognormal relationship between the independent microporosity fraction and α . Mathematically, the relationship can be expressed as

$$\frac{\psi}{\beta} = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{\log \alpha - 2.29}{0.89\sqrt{2}} \right) \right]. \quad (17)$$

The choice of mean and standard deviation of the lognormal relationship is governed by experimental results, as shown later.

Macropore relaxation

Since the relaxation of both micro and macropore is governed by the same Bloch equation (3), we expect the relaxation time of the macropore to also correlate with α . It is however, found that the macropore relaxation time correlates with the product of α and square of macroporosity fraction ($1 - \beta$). This is because the product $(1 - \beta)^2 \alpha$ represents the normalized diffusion time (t_d) within the macropore as described below

$$(1 - \beta)^2 \alpha = \frac{((1 - \beta)L_2)^2 / D}{L_1 / \rho\beta} = \frac{t_d}{T_{2,c}} \quad (18)$$

Figure 8 shows the plot of dimensionless relaxation time of the macropore with $\nu = (1 - \beta)\sqrt{\alpha}$ for different parameter values. Here, the relaxation time is correlated with ν instead of its square because ν is proportional to the length scale, L_2 , of macropore (equation (18)). We can establish a cubic

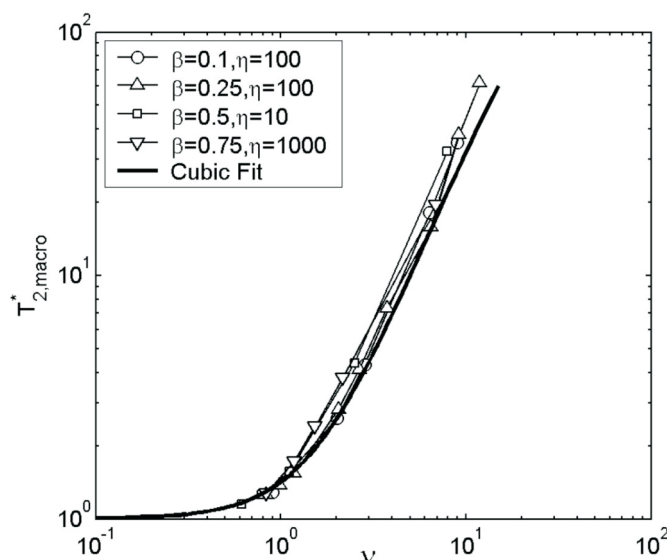


FIG. 8 Dimensionless relaxation time of macropore shows cubic relationship with $\nu = (1 - \beta)\sqrt{\alpha}$.

relationship between the dimensionless macropore relaxation time and ν as shown in equation (19). The cubic relationship provides a better statistical correlation with the experimental results than a quadratic one. The functional relationship, although fitted to experimental results for sandstones and grainstones as shown later, closely match the simulation results.

$$T_{2,macro}^* = 1 + 0.025\nu + 0.4\nu^2 - 0.009\nu^3 \quad (19)$$

where $10^{-1} < \nu < 10^1$.

SANDSTONES

In this section, we extend the ideas developed in the previous section to describe diffusional coupling in clay-lined pores in sandstones. Straley et al. (1995) modeled the clay flakes as forming micro-channels perpendicular to the pore walls such that each micropore opens to a macropore (Figure 9). The two dimensional structure of the clay-lined pore can be modeled as a periodic array of rectangular flakes arranged along the wall of the macropore (Zhang et al., 2001; Zhang et al., 2003). Since the model is periodic, the relaxation process can be adequately modeled by considering only the symmetry element between two clay flakes. The model is further simplified to the one in Figure 2 by approximating the flakes to be needle shaped with negligible thickness.

Pore size distribution

To experimentally validate the theoretical model, the NMR response of North-Burbank (NB) sandstone with pores lined with chlorite flakes is simulated (Trantham and Clampitt, 1977). Analysis of the sandstone cores yielded an average porosity of 0.22 and air/brine permeability of 220 mD. Figure 10 shows the pore throat distribution obtained by mercury porosimetry for one of the cores. The bimodal structure of the pore size distribution arises due to the pres-

ence of pore-lining chlorite flakes. Mercury first invades the macropores giving rise to the peak at larger pore radii. The clay flakes, being closely spaced, are invaded by mercury at high capillary pressures which gives rise to the peak at smaller pore radii.

A lognormal pore throat distribution with mean of $8 \mu\text{m}$ and standard deviation of $0.135 \mu\text{m}$ is simulated to approxi-

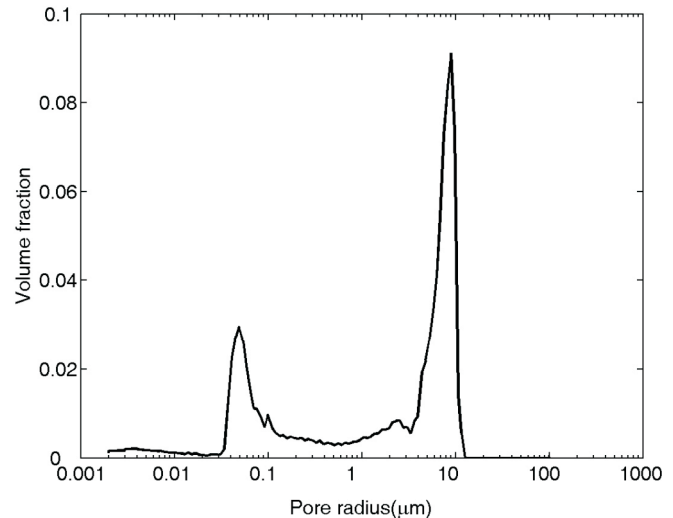


FIG. 10 Pore throat distribution of North-Burbank sandstone obtained using mercury porosimetry. The bimodal structure arises due to pore-lining chlorite clay flakes.

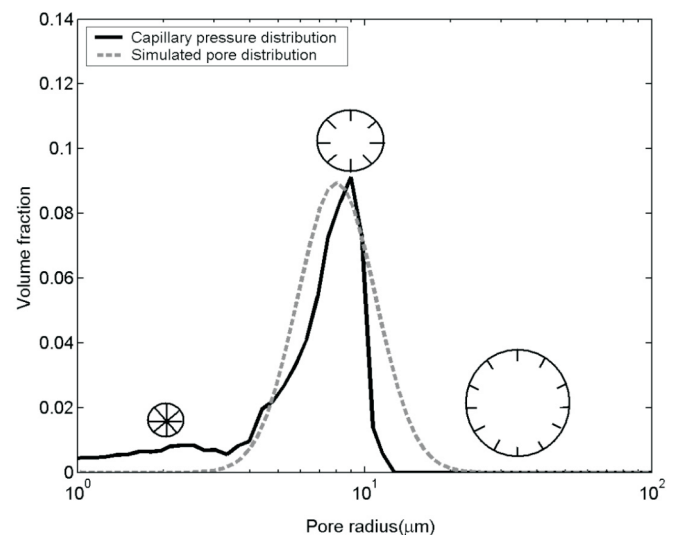


FIG. 11 Simulated lognormal pore size distribution to approximate the distribution of clay-lined macropores. Also shown are the pores with changing proportion of pore volume occupied by the clay flakes.

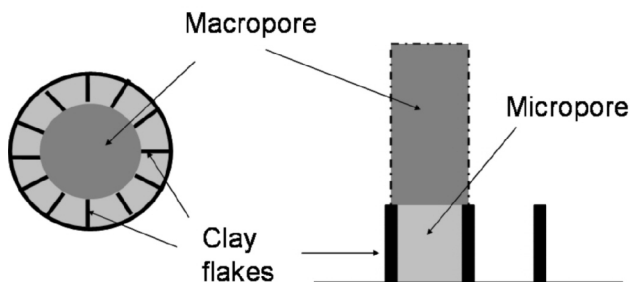


FIG. 9 a) Model of a clay lined pore showing micropores opening to a macropore b) Simplified model with rectangular clays arranged along macropore wall.

mate the distribution of macropores (Figure 11). Since mercury porosimetry measures the distribution of pore throats, the distribution of pore bodies is obtained by assuming a fixed pore body to pore throat ratio of 3 (Lindquist et al., 2000). Thus, the most abundant pore has the pore body radius (L_2) of $24\text{ }\mu\text{m}$. Each pore is then modeled to be lined with clay flakes which are assumed to be of constant length and equally spaced in all pores. As a result, the flakes completely occupy the small pores and form a thin rim on the surface of larger pores. The distance between the flakes (L_1) is given by the peak at smaller pore radius in the pore size distribution ($\approx 0.03\text{ }\mu\text{m}$).

Numerical solution

In order to solve equation (3) for the decay of magnetization in the i^{th} pore, we need three parameters:

1. Microporosity fraction β_i
2. Aspect ratio $\eta_i = L_{2,i}/L_{1,i}$
3. Brownstein number $\mu_i = \rho L_{2,i}/D$

The parameters in different pores are, however, not totally independent of each other since they are constrained by the assumptions of constant length and equal spacing between clay flakes in all pores. Mathematically, the constraints imply

$$L_{1,i} = L_{1,c} = \text{const.} \quad (20)$$

$$\beta_i L_{2,i} = \beta_c L_{2,c} = \text{const.} \quad (21)$$

where the subscript “c” refers to the characteristic or most abundant pore. If we specify the parameters for the characteristic pore, the parameters for the rest of the pores are calculated by making use of the constraints. Similar to the analysis of a single pore in the previous section, the governing equations (3-5) for each pore are normalized with respect to common characteristic parameters. The spatial variables and time are respectively normalized by the radius and characteristic relaxation time (equation (6)) of the most abun-

dant pore. The normalized equations are then solved for the decay of magnetization in each pore individually. The magnetization in the entire pore structure is computed by interpolating the individual magnetization values at some common values of time and then summing them over the entire volume. The total magnetization (M_{tot}) in the pore structure is given as

$$M_{\text{tot}}(t) = \sum_i^{N_p} V_{p,i} M_i(t) \quad (22)$$

where N_p is the number of pores, $V_{p,i}$ is the volume fraction of the i^{th} pore and M_i is the magnetization in the i^{th} pore at dimensionless time t . The T_2 distribution for the pore structure is obtained by fitting a multi-exponential distribution to the total magnetization.

Results

Since each pore in the pore size distribution has a different value of α , a volume averaged $\bar{\alpha}$ for the distribution is defined as

$$\bar{\alpha} = \sum_i \alpha_i V_{p,i} = \sum_i \beta_i \eta_i \mu_i V_{p,i} \quad (23)$$

The simulated T_2 distributions for the pore size distribution with typical values of β_c and η_c ($\beta_c = 0.3$ and $\eta_c = 100$) are shown in Figure 12 as a function of $\bar{\alpha}$. The T_2 distribution changes from unimodal to bimodal with the increase in the values of $\bar{\alpha}$. This is because when $\bar{\alpha} < 1$, the pores are in total coupling regime and each pore relaxes single exponentially with the dimensionless relaxation time, $T_{2,i}$, given as

$$T_{2,i} = \frac{(V/\rho S)_i}{T_{2,c}} = \frac{\beta_c}{\beta_i} = \frac{L_{2,i}}{L_{2,c}} \quad (24)$$

Thus, for $\bar{\alpha} < 1$, the T_2 distribution exactly replicates the unimodal lognormal distribution of the pore radii. As the pores enter the intermediate coupling regime ($\bar{\alpha} > 1$), a fraction of microporosity starts relaxing independently of the macroporosity, thereby giving the T_2 distributions a bimodal shape.

Experimental verification

The simulations with characteristic parameters representative of the core properties are compared with the experimental results for North Burbank. The value of β_c is calculated such that the microporosity fraction of the simulated pore size distribution corresponds to the irreducible water saturation, i.e

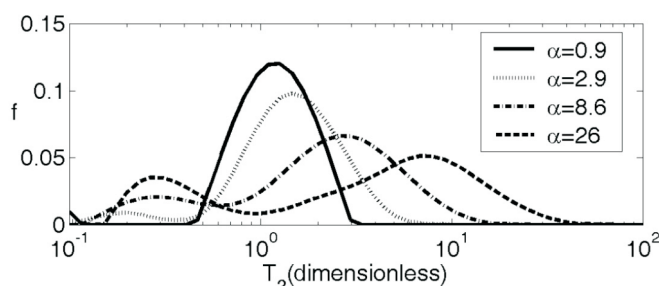


FIG. 12 Simulated T_2 distributions for a pore size distribution with $\beta_c = 0.3$ and $\eta_c = 100$ showing transition from unimodal to bimodal distribution with increase in $\bar{\alpha}$.

TABLE 1 Characteristic parameters for the simulations for three NB cores.

Core	β_c	η_c	μ_c	$\bar{\alpha}$
NB1	0.3	800	0.048	12.2
NB2	0.28	800	0.065	15
NB3	0.21	800	0.094	16.6

$$\sum_i^{N_p} \beta_i V_{p,i} = S_{w,irr} \quad (25)$$

$$\Rightarrow \beta_c = \frac{S_{w,irr}}{L_{2,c} \sum_i^{N_p} \frac{V_{p,i}}{L_{2,i}}} \quad (26)$$

The aspect ratio η_c is calculated from the ratio of macropore and micropore radii obtained from mercury porosimetry. The third parameter μ_c is specified such that the simulations best match with the experimental results.

Figure 13(a) shows the comparison of the T_1 distribution of a water-saturated NB core with the corresponding simu-

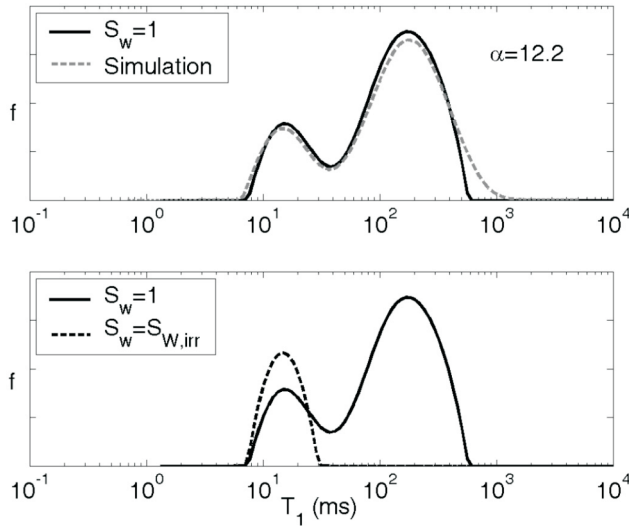


FIG. 13 a) Comparison of simulated and experimental T_1 distributions for 100% water saturated NB core. The value of $\bar{\alpha}$ indicate intermediate coupling regime for water saturated core. b) Comparison of T_1 distributions of the core at 100% water saturation and at irreducible saturation. Increase in the amplitude of the micropore peak at irreducible saturation also illustrates intermediate coupling regime.

lated distribution. The comparison is made with the T_1 (instead of the T_2) distributions since the T_2 relaxation is additionally influenced by diffusion in internal gradients induced by chlorite flakes (Zhang et al., 2001; Zhang et al., 2003). Diffusional coupling, however, affects both T_1 and T_2 relaxation since it arises due to diffusion between pores of different surface-to-volume ratios. The characteristic parameters for the simulations are shown in Table 1. The dimensionless simulated distributions are dimensionalized by choosing $T_{2,c} = 50$ ms, which gives the best overlay of the simulated and experimental distributions. The value of $\bar{\alpha}$ ($= 12.2$) indicates that the micro and macropores are in intermediate coupling regime. The intermediate coupling regime is also demonstrated in Figure 13(b), which shows the comparison of T_1 distributions of the core at 100% water saturation and at irreducible saturation. **An increase in the amplitude of the micropore peak is observed at irreducible saturation. This increase in the amplitude is observed because at irreducible saturation, there is no diffusional exchange of the fluid in micro and macropores and the fluid relaxes with a rate proportional to the surface-to-volume ratio of the micropores. However, at 100% water saturation, a fraction of the fluid in micropores is exchanging with the macropores.** The apparent volume of the fluid in micropores is larger thereby resulting in a smaller surface to volume ratio, which increases its relaxation time.

Figure 14(a) shows the plot of the amplitude of the micropore peak at 100% water saturation normalized with

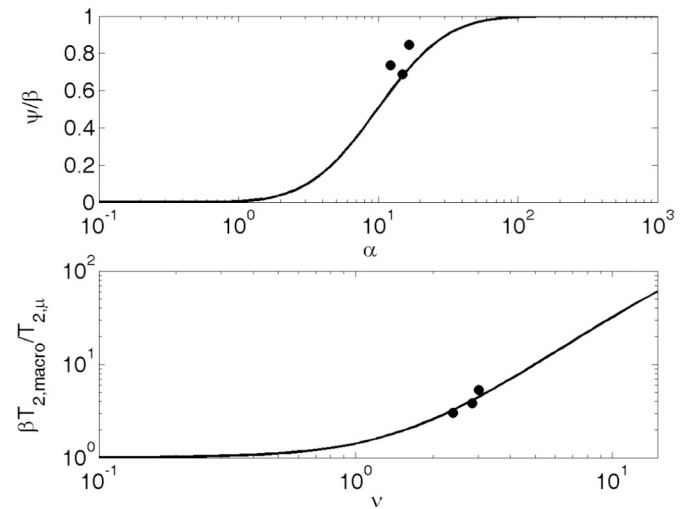


FIG. 14 a) Plot of independent microporosity fraction with α for three water saturated NB cores. The measurements fall in intermediate coupling regime of the lognormal relationship of equation (17). b) Plot of normalized relaxation time with $\beta = (1 - \beta)\sqrt{\alpha}$ for the three cores.

the total microporosity fraction with $\bar{\alpha}$ for three NB cores considered in this study. The simulation parameters required for the calculation of $\bar{\alpha}$ for the cores are also mentioned in Table 1. The measurements fall on the intermediate coupling regime of the lognormal relationship (equation (17)) for all cores. Figure 14 (b) shows that the cubic relationship (equation (19)) for the normalized macropore relaxation time also holds for the cores. In the figure, the relaxation time of the macropore is normalized by a characteristic relaxation time defined by equation (8).

To explore another coupling regime, measurements were made with dry cores saturated with hexane. Higher extent of coupling is expected with hexane than with water due to higher diffusivity and lower surface relaxivity for hexane. The lower surface relaxivity for hexane is due to intrinsic smaller hydrocarbon relaxivity of the sandstone surfaces (Chen et al., 2005). Figure 15 shows the T_1 distribution of the hexane saturated core and the corresponding simulated distribution. The dimensionless simulated distribution is dimensionalized by choosing $T_{2,c} = 450$ ms. In this case, the T_1 distribution is unimodal implying the merger of the micro and macropore peak. The smaller value of $\bar{\alpha}$ suggest stronger coupling for hexane than for water as can be seen by comparing the values in Figures 13 and 15 respectively.

Estimation of surface relaxivity

The values of the surface relaxivity for the cores can be calculated from the corresponding values of $\bar{\alpha}$. For the values of parameters $L_{2,c} = 24 \mu\text{m}$, $L_1 = 0.03 \mu\text{m}$, diffusivity for water and hexane $D_W = 2.5 \times 10^{-5} \text{ cm}^2/\text{s}$ and $D_H = 4.2 \times 10^{-5} \text{ cm}^2/\text{s}$ (Reid et al., 1987), the average value of relaxivity is found to be $7.1 \mu\text{m}/\text{sec}$ for water and $1.6 \mu\text{m}/\text{sec}$ for hexane. The lower surface relaxivity of hexane is not due to water-wetness of the sandstone since hexane was in direct contact with the mineral surfaces and no water was present. Another estimate of relaxivity is obtained by comparing the cumulative pore size distributions obtained

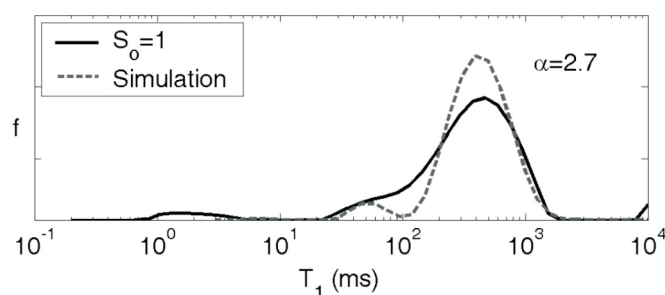


FIG. 15 Comparison of simulated and experimental T_1 distributions for hexane saturated NB core. The value of $\bar{\alpha}$ indicates strong coupling for hexane saturated core.

by T_1 relaxation and mercury porosimetry. However, the estimates from the latter method are about three times ($20 \mu\text{m}/\text{sec}$ for water) as high as those calculated from simulations. This is because mercury porosimetry does not take into account the large surface area provided by the clay flakes in the estimation of relaxivity.

GRAINSTONES

The analysis of the first section on diffusional coupling is also applied to describe pore coupling in grainstone carbonates. Ramakrishnan et al. (1999) modeled the grainstones as microporous spherical grains surrounded by intergranular pores. This three dimensional model can be mapped into a two-dimensional model of a periodic array of slab-like grains separated by intergranular macropores as shown in Figure 16. We transform this model to the one in Figure 2 by neglecting the thickness of grain between the micropores and assuming the pores to be linear in shape. Note that in this model relaxation at the outer surface of the grains is neglected. We are justified in making these assumptions if the surface-to-volume ratio of the micropore is much larger compared to the external surface-to-volume ratio of the spherical grains.

Coupling parameter for grainstones

The transformation of the spherical grain model to the 2-D model in Figure 2 enables us to define a coupling parameter for grainstones through a mapping of characteristic parameters. In Figure 2, L_1 was defined to be the half-width of the micropore. Hence for microporous grains, L_1 corresponds to the radius of the micropore (R_μ) i.e.

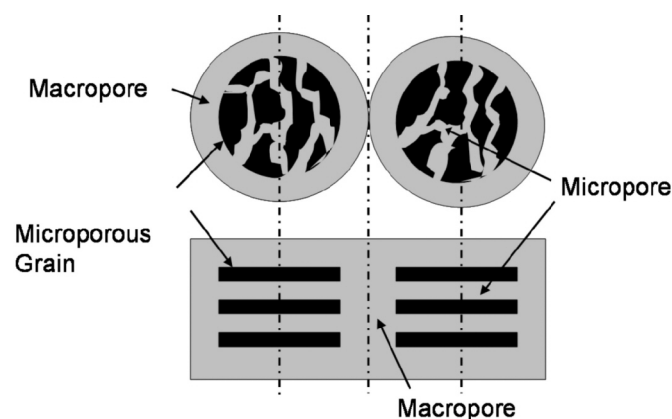


FIG. 16 Pore coupling model in grainstone systems. The three-dimensional model can be mapped to a two-dimensional model of periodic array of microporous grains separated by macropores.

$$L_1 = R_\mu . \quad (27)$$

Also, as a first approximation, L_2 can be taken to be equal to the grain radius (R_g)

$$L_2 = R_g . \quad (28)$$

Substituting equations (27) and (28) in equation (11), we get the definition of α for grainstones as

$$\alpha_{\text{grain}} = \frac{\rho \beta R_g^2}{DR_\mu} \quad (29)$$

α_{grain} , thus, shows a quadratic dependence on the grain radius and inverse dependence on the micropore radius.

This definition suggests that grainstones with large grain radius and/or small micropore radius are expected to show less effect of diffusional coupling.

The above definition of coupling parameter also helps us to understand the analysis of grainstone model developed by Ramakrishnan et al. (1999). They suggested that in the case when the decay of magnetization in macropore occurs on a time scale much larger than that for the decay of magnetization in micropore, relaxation in the coupled geometry can be expressed as a bi-exponential decay with amplitudes representative of the micro and macroporosity fractions as shown

$$M(t) = \phi_m \exp\left(-\frac{\rho_a t}{V_{sm}}\right) + (\phi - \phi_m) \exp\left(-\frac{t}{T_{2,\mu}}\right). \quad (30)$$

In the above equation, V_{sm} is the macropore volume-to-surface ratio, ϕ and ϕ_m are the total porosity and macroporosity respectively and ρ_a is the apparent relaxivity for the macropore. This bi-exponential model is valid when the diffusion length of magnetization within the microporous grain is much smaller than the grain radius, i.e.

$$\sqrt{\frac{DT_{2,\mu}}{\phi_\mu F_\mu}} \ll R_g \quad (31)$$

where F_μ is the formation factor. We can understand the above condition by substituting the expressions for the parameters from the two dimensional model of Figure 16. The relaxation rate of the micropore is related to the micropore radius, assuming cylindrical pores, as

$$\frac{1}{T_{2,\mu}} = \rho \left(\frac{S}{V} \right)_\mu = \frac{2\rho}{R_\mu}. \quad (32)$$

The microporosity fraction β is equal to the product of grain

fraction $(1 - \phi_m)$ and porosity of grains ϕ_m normalized by total porosity ϕ ,

$$\beta = \frac{\phi_\mu (1 - \phi_m)}{\phi}. \quad (33)$$

Substituting the expressions for $1/T_{2,\mu}$ and ϕ_μ from equations (32) and (33) in (31), we get

$$\sqrt{\frac{DR_\mu (1 - \phi_m)}{2\rho\beta\phi F_\mu}} \ll R_g \quad (34)$$

$$\Rightarrow \alpha_{\text{grain}} \equiv \frac{\rho\beta R_g^2}{DR_\mu} \gg \frac{(1 - \phi_m)}{2\phi F_\mu}. \quad (35)$$

The above condition implies that the micropore relaxes independently of the macropore for large values of α , which is the same condition for the decoupled regime obtained for our model. However, for typical values of grainstone parameters, the value of apparent relaxivity is much larger than the micropore relaxivity (Ramakrishnan et al., 1999). Thus, even though the surface-to-volume ratio of macropore is significantly smaller than that of the micropore, diffusional coupling can result in the decay of macro and micropore at comparable time scales. For such cases, the pores are in intermediate coupling regime and the amplitudes of the bi-exponential fit are not representative of the actual micro and macroporosity fractions as was observed in the numerical simulations of Ramakrishnan et al. (1999).

Experimental validation

In order to experimentally validate the grainstone model, NMR response of three systems — microporous chalk, silica gels and alumino-silicate molecular sieves — was studied as a function of grain radius. These systems with varying physical properties help us to systematically analyze the effect of different governing parameters on pore coupling. The physical properties of the systems are listed in Table 2.

Chalk

Crushed microporous chalk (Crayola) was sieved into five fractions with average grain radii of 335 μm , 200 μm , 112 μm , 56 μm and 11 μm . Known quantities of sorted fractions were water saturated in 1" by 1" Teflon sleeves whose bases were sealed with a covering of Teflon tape. After measuring the NMR response at 100% water saturation, the systems were centrifuged in a Beckman rock core centrifuge at an air/water capillary pressure of 100psi for 3 hours to drain the macropores. The Teflon base is permeable to water but prevents any grain loss during centrifugation.

Note that even though the sieve fractions are unconsolidated, they have irreducible intragranular microporosity which is not displaced on capillary drainage.

The T_2 distributions of the five fractions at 100% water saturation and the corresponding distributions at irreducible saturation are shown in Figure 17. The values of α_{grain} calculated using equation (29) and $\mu_{grain}(=\rho R_g/D)$ are also mentioned for each sieve fraction. We can see that for the

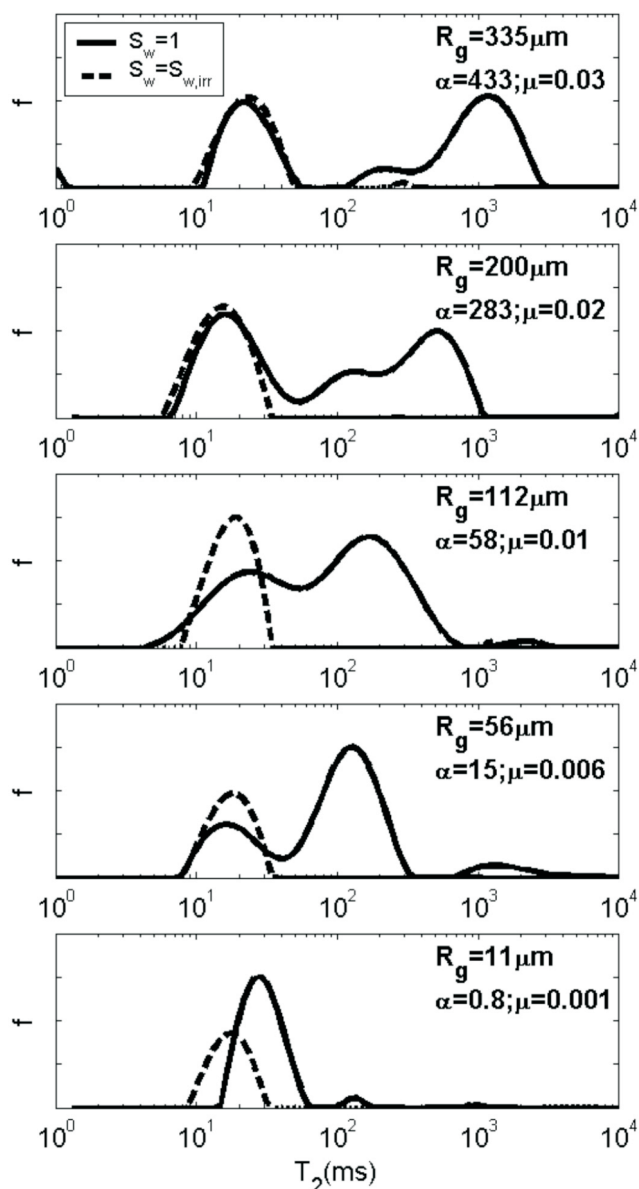


FIG. 17 T_2 distributions of microporous chalk as a function of grain radius. The transition from decoupled ($R_g = 335 \mu m$) to total coupling regime ($R_g = 11 \mu m$) is predicted by the values of α .

TABLE 2 Physical properties of the grainstone systems.

	Chalk	Silica Gels	Molecular Sieves
BET Surface Area (m^2/g)	4.1	300	20
Micropore Diameter ($\text{\AA}$)	185	150	4
Surface Relaxivity ($\mu m/sec$)	0.27	0.06	0.04

two coarsest fractions ($R_g = 335 \mu m$ and $200 \mu m$), the T_2 distributions show distinct peaks for micro and macropores and the area under the micropore peak is the same as that at irreducible conditions. This implies that the systems are in the decoupled regime, which is verified by large values of α_{grain} . The effect of coupling becomes more pronounced for systems with $R_g = 112 \mu m$ and $56 \mu m$, which show a build up of micropore peak amplitude at irreducible saturation. The amplitude of the micropore peak increases because at irreducible condition, there is no diffusional exchange of the fluid in micro and macropores. Thus, the relaxation rate of the fluid is proportional to the surface-to-volume ratio of the micropores. However, at 100% saturation, part of the fluid in micropores is in diffusional exchange with the macropores. Thus, the apparent volume of the fluid in micropores is larger which decreases its relaxation time. This is the same explanation given by Coates et al. (1998) for the observed increase in amplitude of short T_2 components at irreducible saturation in sandstone cores. The values of α_{grain} for $R_g = 112 \mu m$ and $56 \mu m$ correspond to the intermediate coupling regime and thus, quantitatively supports the explanation. The unimodal T_2 distribution of the finest fraction ($R_g = 11 \mu m$) at 100% water saturation shows that the system is in total coupling regime ($\alpha_{grain} = 0.8 < 1$).

Silica gels

A homologous series of silica gels (provided by Sigma-Aldrich) with grain radii of $168 \mu m$, $55 \mu m$ and $28 \mu m$ constituted the second system. Figure 18 shows the T_2 distributions at 100% water saturation and at irreducible saturation for the three fractions. Similar to the response of chalk, the distributions change from being bimodal to unimodal with the decrease in particle diameter indicating increased coupling. The values of α_{grain} suggest intermediate coupling regime for the two coarsest fractions ($R_g = 168 \mu m$ and $55 \mu m$) and total coupling regime for the finest fraction ($R_g = 28 \mu m$).

Molecular sieves

Alumino-Silicate molecular sieves with nominal pore diameter of $4\text{\AA}$, supplied by Fisher Chemicals, was crushed and sieved into four fractions with average grain radii of $200\text{ }\mu\text{m}$, $112\text{ }\mu\text{m}$, $56\text{ }\mu\text{m}$ and $16\text{ }\mu\text{m}$. The T_2 distributions of the four fractions are shown in Figure 19 at 100% water saturation and at irreducible saturation. The response shows similar trend of narrowing T_2 distributions with decrease in the grain diameter. An increase in amplitude of the micropore peak at irreducible conditions is observed for fractions with $R_g = 56\text{ }\mu\text{m}$ and $16\text{ }\mu\text{m}$, which shows that these fractions are in intermediate coupling regime. The values of α_{grain} for the respective fractions predict the transition of the coupling regimes.

Figure 20 shows that the lognormal and cubic relationships of equations (17) and (19) also hold for the three systems, thereby establishing the validity of the grainstone model.

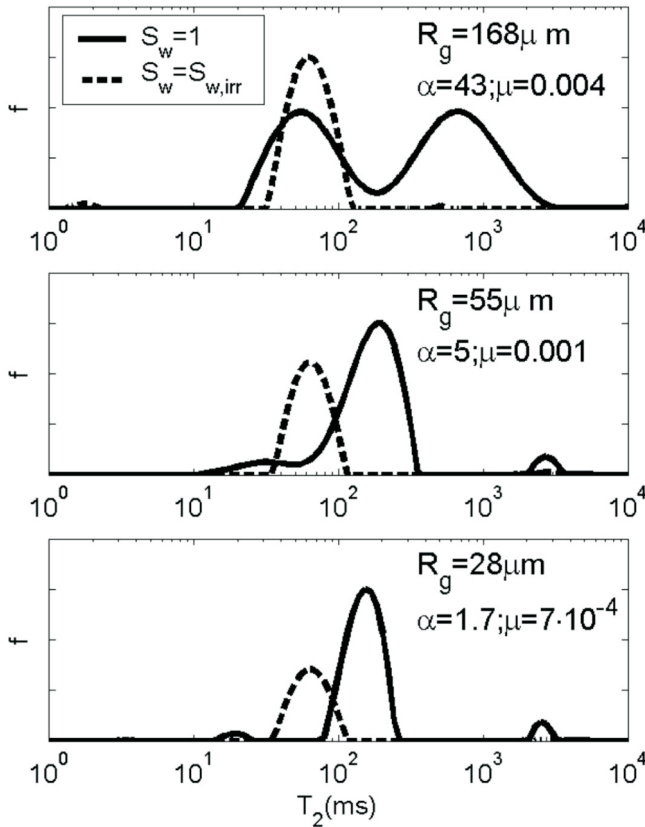


FIG. 18 T_2 distributions of silica gels as a function of grain radius. The transition from almost decoupled ($R_g = 168\text{ }\mu\text{m}$) to total coupling regime ($R_g = 55\text{ }\mu\text{m}$) is predicted by the values of α .

ESTIMATION OF IRREDUCIBLE WATER SATURATION

The sharp cutoff method of estimating $S_{w,irr}$ employs a lithology-specific sharp $T_{2,cutoff}$ to partition the T_2 spectrum into free fluid and bound fluid saturations. For formations with diffusively coupled micro and macropores, the use of a sharp cutoff may give incorrect estimates since in such cases the direct relationship between pore size and T_2 distribution no longer holds.

In the case of pore coupling, the estimation of $S_{w,irr}$ amounts to the calculation of microporosity fraction β for a given T_1 or T_2 distribution at 100% water saturation. The solution of this inverse problem is obtainable by making use of the correlations for independent microporosity fraction and normalized macropore relaxation time (equations (17) and (19)). Three parameters are required for the solu-

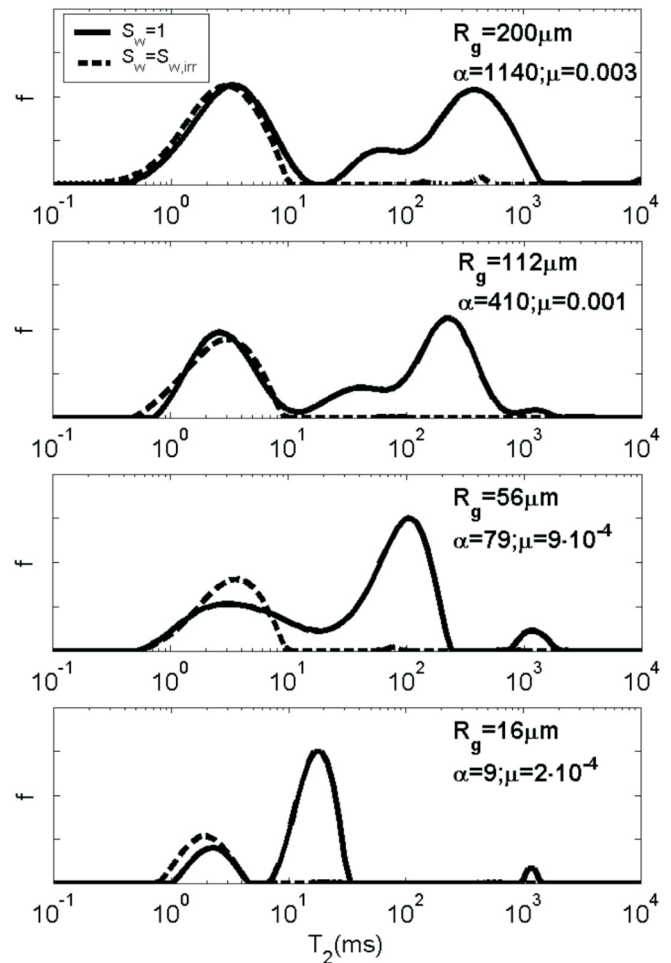


FIG. 19 T_2 distributions of alumino-silicate molecular sieves as a function of grain radius. The values of α_{grain} for the sieve fractions predict the transition of the coupling regimes.

tion: micropore peak amplitude (ψ), relaxation time of the micropore ($T_{2,\mu}$) and relaxation time of the macropore ($T_{2,macro}$). It is assumed that $T_{2,\mu}$ is known from laboratory core analysis and is same for the formation. This assumption is justified if the formation has similar relaxivity and micropore structure as the cores. From the T_2 spectrum at 100% water saturation, the values of ψ and $T_{2,macro}$ can be calculated as the area under the micropore peak and the relaxation time of the mode of the macropore peak, respectively. Hence, for the estimated parameter values, the correlations can be simultaneously solved for the values of α and β . Graphically, the solution involves determining the intersection point of contours of ψ and $T_{2,macro}/T_{2,\mu}$ on the α and β parameter space as shown in Figure 21. The values of con-

tour lines for $T_{2,macro}/T_{2,\mu}$ differ by a factor of 2 and those for ψ differ by 0.1. The coordinates of the intersection point of the contours for experimentally determined values of ψ and $T_{2,macro}/T_{2,\mu}$ estimates the value of α and β for the formation. For a unimodal distribution with a zero value of ψ (total coupling regime), the microporosity fraction can be calculated from the ratio of the relaxation times of micro and macropore, i.e.

$$\beta = \left(\frac{T_{2,\mu}}{T_{2,macro}} \right), \psi = 0. \quad (36)$$

In this case, the value of α is indeterminate and can be anything less than 1. This is because as ψ approaches 0, the contours for $T_{2,macro}/T_{2,\mu}$ asymptote to the reciprocal β value independent of α . (Note $\psi = 0$ implies totally coupled micro and macropore and not necessarily the absence of microporosity).

Figure 22 shows the comparison of the calculated values of β and α with the values determined experimentally for the grainstone and sandstone systems. An average value of $T_{2,\mu}$ obtained from the individual values for different sieve fractions or cores is used for calculations. The estimates lie within an average absolute deviation of 4% and 11% for β and α , respectively. This indicates that the technique is applicable to all the systems studied irrespective of the properties and coupling regimes.

Unification of spectral and sharp cutoff theory

The estimation of $S_{w,irr}$ using spectral or tapered $T_{2,cutoff}$ is

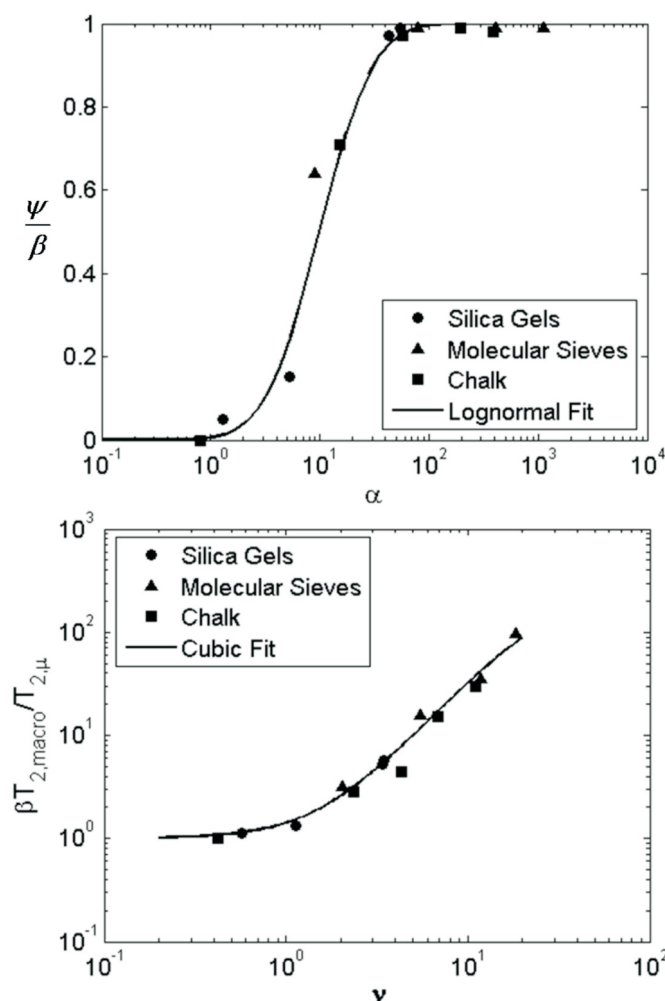


FIG. 20 Plot of independent microporosity fraction with α for model grainstone systems (Upper panel). Plot of normalized macropore relaxation time with $\nu = (1 - \beta)\sqrt{\alpha}$ for the systems (Lower Panel). The experimental points follow the lognormal and cubic relationships of equations (17) and (19).

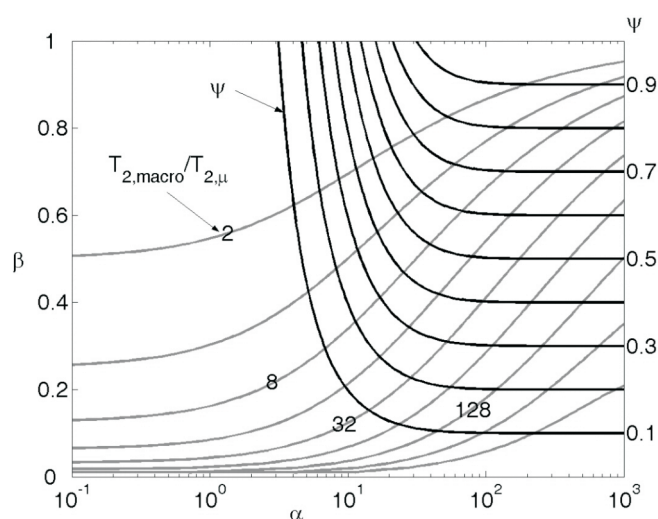


FIG. 21 Contour plots of correlations for ψ and $T_{2,macro}/T_{2,\mu}$ in α and β parameter space. Intersection point of contours estimates the value of α and microporosity fraction for the formation.

based on the premise that each pore size has its own inherent irreducible water saturation. The fraction of bound water associated with each pore size is defined by a weighting function $W(T_{2,i})$ where $0 \leq W(T_{2,i}) \leq 1$. The $S_{w,irr}$ is then given as

$$S_{w,irr} = \sum_j^n W_j f_j \quad (37)$$

where n is the number of bins and f_j is the amplitude of each bin. The weighting factors are determined using empirical permeability models or cylindrical pore models (Coates et al., 1998; Kleinberg and Boyd, 1997).

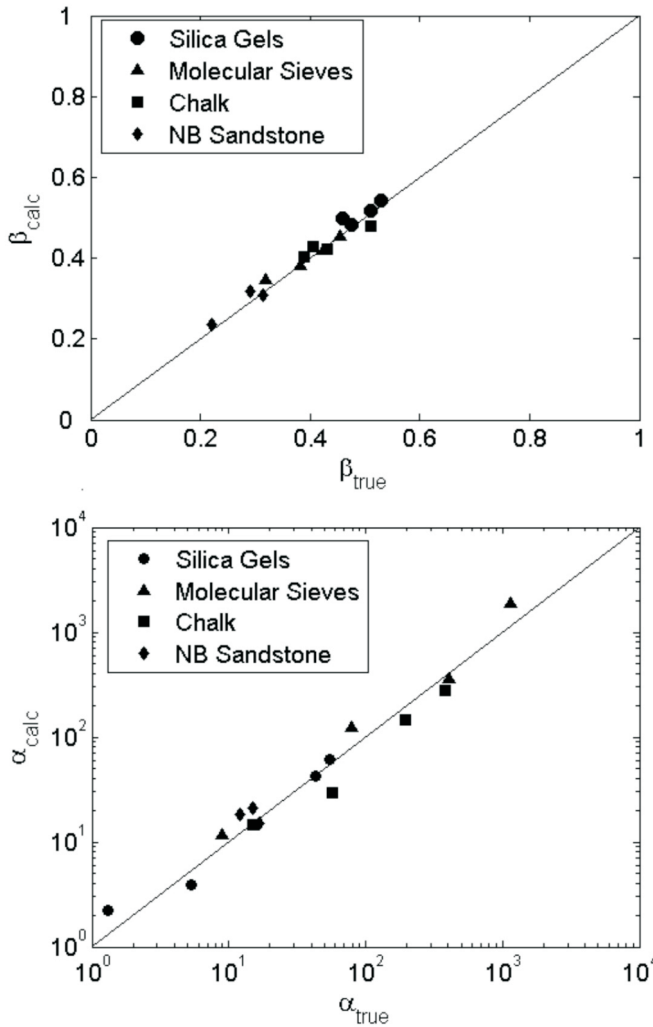


FIG. 22 Comparison of calculated and experimentally measured values of α and β for the grainstone and sandstone systems. The values are estimated within 11% and 4% error respectively.

An implicit assumption of the above-mentioned technique is that the producible and irreducible fractions of each pore have same relaxation time at 100% water saturation. However, the analysis of a single pore (see section on diffusional coupling) shows that micro and macropore can communicate through decoupled and intermediate coupling regimes as well. Thus, in a general coupling scenario, the response of the pore shows distinct peaks for micro and macropore with amplitudes ψ and $(1 - \psi)$ respectively. The amplitude ψ can vary from 0 to β depending on the coupling regime. Therefore, the portion of the microporosity coupled with the macropore divided by the macropore amplitude is given as

$$\Phi\left(\frac{T_{2,macro}}{T_{2,\mu}}, \alpha\right) = \frac{\beta - \psi}{1 - \psi}. \quad (38)$$

Φ is a function of the ratio of macro to micropore relaxation times ($T_{2,macro}/T_{2,\mu}$) and α which determines the microporosity portion coupled with the macropore response. As α increases, the extent of pore coupling decreases and thus, the microporosity fraction coupled with the macropore response also decreases. Figure 23 show the plot of Φ with $(T_{2,macro}/T_{2,\mu})$ for different values of α . The procedure for estimating Φ as a function of $(T_{2,macro}/T_{2,\mu})$ is as follows. For a known value of α , Φ is calculated from equation (38) for several values of β using equation (17). Similarly, $T_{2,macro}/T_{2,\mu}$ is calculated for the known α and same values of β by substituting the expression of $T_{2,c}$ from equation (8) in equation (19)

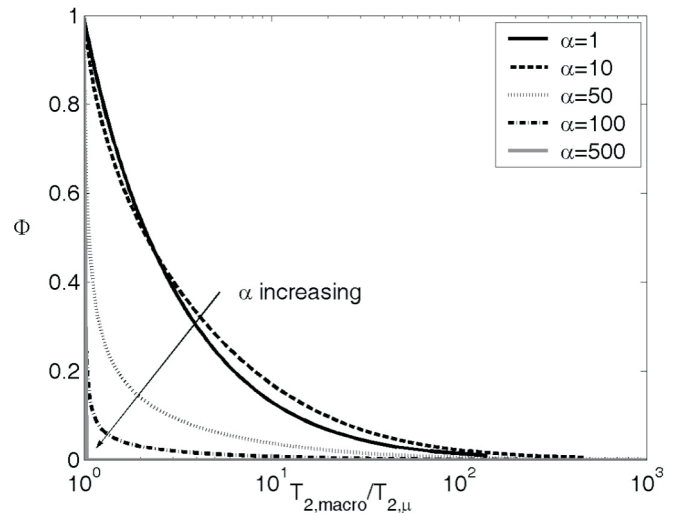


FIG. 23 Plot of Φ versus $T_{2,macro}/T_{2,\mu}$ for different α . A spectral or tapered cutoff is required for the estimation of irreducible saturation in intermediate coupling regime. A sharp cutoff is applicable for decoupled regime.

$$\frac{T_{2,macro}}{T_{2,\mu}} = \frac{1 + 0.025\nu + 0.4\nu^2 - 0.009\nu^3}{\beta}. \quad (39)$$

Thus, the values of Φ and $(T_{2,macro}/T_{2,\mu})$ for the same α and β can be cross-plotted as shown in Figure 23. The curves show that a spectral or tapered cutoff is required for the estimation of irreducible saturation in total or intermediate coupling regime. As α increases, Φ decreases for same $T_{2,macro}/T_{2,\mu}$ indicating lesser correction for diffusional coupling is required for larger α . Once the pores are decoupled, a sharp cutoff is suitable for estimating irreducible fraction as illustrated by sharp fall of Φ curve to zero for all $\alpha > 200$. This could also probably explain the suitability of a single lithology-specific $T_{2,cutoff}$ for estimating irreducible saturations when the formation is in decoupled regime irrespective of the properties. More experiments are, however, needed to prove this postulate.

CONCLUSIONS

The direct correspondence between pore size and T_2 distribution fails if pores of different sizes are coupled by diffusion. A theoretical model is developed that provides a quantitative understanding of the effect of physical and geometrical parameters on diffusional coupling. It is shown that the Brownstein number fails to characterize relaxation regimes in coupled pores due to its dependence on macropore length scale only. Instead, a coupling parameter (α) is introduced that is defined as the ratio of the characteristic relaxation rate of the coupled pore to the rate of diffusional mixing of magnetization between micro and macropore. α includes the contribution of both micro and macropore to the total surface-to-volume ratio and provides a better measure to quantify diffusional coupling. Depending on the value of α , micro- and macropores can communicate through total coupling, intermediate coupling or decoupled regimes.

An inversion technique for the estimation of microporosity fraction and α for formations with unknown geometrical parameters is also introduced. The required parameters for the estimation are easily obtainable from laboratory core analysis and the T_2 (or T_1) spectrum at 100% water and irreducible saturation. It is assumed that the relaxation time of micropore is the same in the cores and in the field. α also provides a quantitative basis for the application of spectral or sharp cutoffs. A sharp $T_{2,cutoff}$ is applicable for the estimation of irreducible microporosity fraction in the decoupled regime. However, a spectral or tapered cutoff is required in total coupling or intermediate coupling regimes. This is because in these regimes and at 100% water saturation, the amplitudes of micro and macropore peaks of the T_2 distribution are not representative of true

porosity fractions. Experiments with sandstones and microporous grainstones show that the application of inversion technique provides accurate estimate of microporosity fraction in all coupling regimes.

NOMENCLATURE

D	Self diffusion coefficient
f	Amplitude of the T_2 bins
L_1, L_2	Length scales of micro and macropore, respectively
M	Magnetization
R_g	Grain radius
R_μ	Micropore radius
$S_{w,irr}$	Irreducible water saturation
S/V	Surface to Volume ratio of pore
t	Time
T_2	Transverse relaxation time
$T_{2,\mu}, T_{2,macro}$	Transverse relaxation time of micro and macropore, respectively
T_{2B}	Bulk transverse relaxation time
V_p, N_p	Volume fraction and number of pores
W	Weight fraction of each T_2 bin in the SBVI model
ρ, ρ_a	True and Apparent surface relaxivity
x, y	Spatial variables
α	Coupling parameter
β	Microporosity fraction
η	Aspect ratio of the pore
μ	Brownstein number
λ	Regularization parameter
ψ	Micropore peak amplitude
ϕ, ϕ_m, ϕ_μ	Total, macro and microporosity, respectively

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APPENDIX A

The governing equations (3-5) in terms of dimensionless variables (with superscript *) are

$$\frac{\partial^2 M^*}{\partial x^{*2}} + \frac{\partial^2 M^*}{\partial y^{*2}} = \alpha \frac{\partial M^*}{\partial t^*} \quad (\text{A.1})$$

$$\frac{\partial M^*}{\partial x^*} - \mu S(y^*) M^* = 0 \text{ at } x^* = 0, 0 \leq y^* \leq 1 \quad (\text{A.2})$$

$$\frac{\partial M^*}{\partial x^*} = 0 \text{ at } x^* = \eta^{-1}, 0 \leq y^* \leq 1 \quad (\text{A.3})$$

$$\frac{\partial M^*}{\partial y^*} = 0 \text{ at } y^* = 0 \text{ and } 1, 0 \leq x^* \leq \eta^{-1}. \quad (\text{A.4})$$

In the above system, the discontinuous boundary condition along the y -axis is combined into a single equation by using the step function $S(y^*)$ defined as

$$\begin{aligned} S(y^*) &= 1 \text{ for } 0 \leq y^* \leq \beta \\ &= 0 \text{ for } \beta < y^* \leq 1 \end{aligned} \quad (\text{A.5})$$

APPENDIX B

Numerical solution

The non-dimensionalized equations (A.1-A.5) are numerically integrated for the time evolution of magnetization in the pore. However, the direct numerical integration leads to non-physical oscillations in the solution due to round off errors. Instead, the governing equations are expressed in residual form by expressing the unknown as the change in magnetization from the previous iteration as shown

$$\delta M^{k+1} = M^{k+1} - M^k \quad (\text{B.1})$$

where k refers to the iteration index. The equations are then solved for the change in magnetization at each iteration step and the solution added to magnetization at previous step to yield the magnetization at the current step. The iterations are continued till the residual falls below the error tolerance. An Alternating-Direction-Implicit finite difference technique (Peaceman and Rachford, 1955) is employed for the numerical integration. A sequence of four iteration parameters 0.75, 0.075, 0.0075, 0.00025 is found to be optimal in reducing the number of iterations per time step. The time truncation errors are controlled by using an automatic time step (Δt) selector algorithm (Todd et al., 1972). Δt at the next time step is scaled by the ratio of maximum change in M desired to maximum change in M over the entire domain at the previous time step. Thus, the time truncation errors are limited due to small Δt in the beginning of the simulation (when the rate of change of M is large) and large Δt towards the end.

The decay curve is obtained by summing the magnetization values over the entire domain at each time step. Simulated decay data are sampled at the times corresponding to

0.5% change in the average magnetization and fitted to a multi-exponential distribution to obtain the T_2 distribution

$$\overline{M(t_i)} \approx \sum_j f_j \exp(-t_i / T_{2,j}) \quad (\text{B.2})$$

where $\overline{M(t_i)}$ is the average magnetization in the entire domain at discrete times (t_i). The coefficients f_j are obtained by minimizing the following objective function (Dunn et al., 1994)

$$\sum_i \left[\overline{M(t_i)} - \sum_j f_j \exp(-t_i / T_{2,j}) \right]^2 + \lambda \sum_j f_j^2. \quad (\text{B.3})$$

In the above expression, λ is the regularization parameter. The numerical scheme is validated by comparing the numerical solution with the analytical solution obtained by Brownstein and Tarr (1979) for the case of $\beta = 1$ for different values of μ . In all cases, the two solutions match within an accuracy of 0.1% (maximum relative error) indicating the correctness of the numerical solution.

APPENDIX C

We resolve the issue of faster relaxation of micropore in the coupled case than in the decoupled case, observed in our simulations. For the case of no diffusional coupling, the dimensionless relaxation time of the micropore is inversely proportional to its surface-to-volume ratio i.e.

$$\lim_{\alpha \rightarrow \infty} T_{2,\mu} = \lim_{\alpha \rightarrow \infty} \frac{T_{2,\mu}}{T_{2,c}} = \frac{L_1 / \rho}{L_1 / \rho \beta} = \beta. \quad (\text{C.1})$$

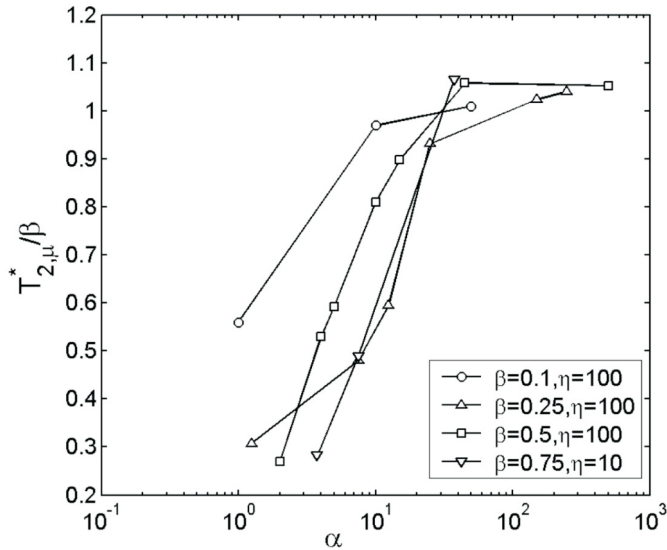


FIG. C.1 Plot of micropore relaxation time with α for different simulation parameters. Micropore appears to relax faster when coupled with macropore.

Hence, when coupling between micro and macropore is allowed, the micropore is expected to relax slower than β . Figure C.1 shows the normalized relaxation time of micropore peak as a function of α for different simulation parameters. We see that for the decoupled regime, the normalized micropore relaxation time tends to 1 as expected. But for the intermediate coupling regime, it appears that the micropore is relaxing faster than the decoupled rate i.e. $T_{2,\mu}^* < \beta$. This artifact of faster relaxation of micropore in the coupled case was also observed in Ramakrishnan's analysis (1999). No explanation was, however, offered in their paper.

The apparent contradiction is resolved by studying the early relaxation data of the coupled pore. Analysis of the initial slope of the decay curve reveals that the system relaxes with a unit relaxation time. This is because at very short times, the fluid in micro and macropore does not have sufficient time to exchange by diffusion with each other. Thus, the micropore decays with relaxation time of β while macropore relaxes with infinite relaxation time.

$$\begin{aligned} M^* &= \beta \exp(-t / \beta) + (1 - \beta) \exp(-t / \infty) \\ &\approx \beta(1 - t / \beta) + (1 - \beta) \quad (t < 1) \quad (\text{C.2}) \\ &= 1 - t \approx \exp(-t). \end{aligned}$$

However, at longer times, macropore relaxes with a finite relaxation time due to diffusional coupling with the micropore. This decrease in relaxation time of the macropore has an apparent effect of reducing the relaxation time of the micropore. Figure C.2 shows the simulated

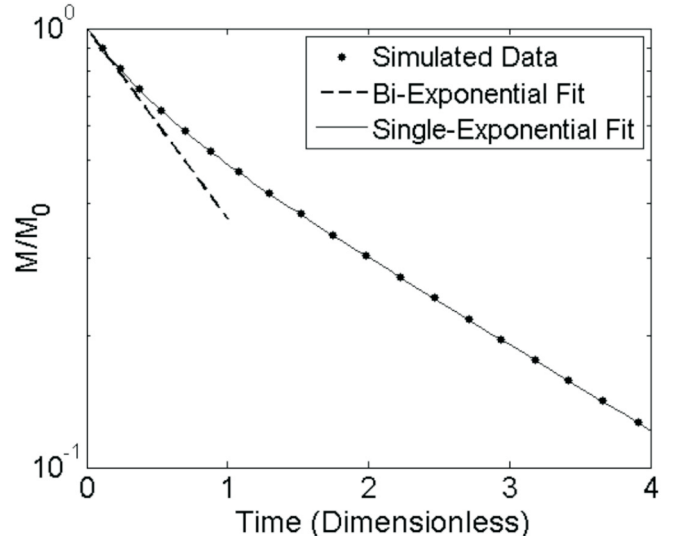


FIG. C.2 Simulated relaxation data for $\alpha = 10$ ($\beta = 0.5$, $\eta = 100$ and $\mu = 0.2$). The black dashed line is an exponential decay with relaxation time of 1. The solid line is a bi-exponential fit ($0.26 \exp(-t/0.4) + 0.74 \exp(-t/2.2)$) of the data.

relaxation data for the case of $\beta = 0.5$, $\eta = 100$ and $\mu = 0.2$ ($\alpha = 10$, intermediate coupling regime). The system initially decays with unit relaxation time as shown by the dashed line. However, a bi-exponential fit (solid line) to the data estimates the relaxation time of the micropore to be 0.4 which is less than the expected value of 0.5. The inconsistency arises due to decrease in relaxation time of the macropore from infinity at short times to 2.2 at longer times.

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A laboratory study of NMR relaxation times and pore coupling in heterogeneous media

Elliot Grunewald¹ and Rosemary Knight¹

ABSTRACT

Nuclear magnetic resonance (NMR) relaxation times of geologic materials are closely related to pore geometry. In heterogeneous media, however, the details of this relationship are poorly understood because of a phenomenon known as pore coupling, which arises when diffusing protons sample multiple pores before relaxing. Laboratory experiments allow us to explore whether surface geochemistry can influence pore coupling and how this process affects the observed relaxation-time distribution. Measurements of the NMR response for microporous silica gel packs, treated with varying amounts of surface-coating iron, demonstrate that samples with less iron exhibit stronger pore coupling than those with abundant iron. When pore coupling is strong, the relaxation-time distribution grossly misrepresents the underlying bimodal pore-size distribution of micropores and macropores. Specifically, the bimodal relaxation-time distribution becomes merged and the relative amplitude of the peaks fails to reflect the true macropore and micropore volume. A reduction in pore coupling, observed with increasing iron content, is attributed to a decrease in the distance protons are able to diffuse before relaxing. Basic parameters describing the shape of the relaxation-time distributions for this range of samples are well-predicted by a 1D analytical model. Experimental results conclusively demonstrate that surface geochemistry is an important factor determining the degree to which pore coupling occurs and illustrate how this phenomenon can affect the interpretation of NMR relaxation measurements in heterogeneous porous media.

INTRODUCTION

Nuclear magnetic resonance (NMR) relaxation measurements provide an unparalleled view into the pore space of geologic materi-

als. The proton NMR method probes the time required for hydrogen nuclei in the pore water to relax to equilibrium through diffusion and interaction with the pore-scale physicochemical environment. NMR relaxation-time measurements have been used for decades in geophysical well logging to estimate hydrocarbon-reservoir properties such as porosity, pore-size distribution, and permeability (Seevers, 1966; Kenyon et al., 1988). Recent research has extended NMR geophysical techniques to near-surface environmental applications and has led to the development of a surface-based NMR method — magnetic resonance sounding (MRS) — to measure the NMR response of groundwater aquifers (Legchenko and Valla, 2002; Hertrich, 2008; Walsh, 2008). Our current study seeks to improve the interpretation of near-surface NMR measurements by exploring the link between NMR relaxation times and pore-scale properties in heterogeneous porous media.

The analysis of NMR relaxation measurements in geologic materials requires an accurate model of the physical processes governing relaxation. Conventional interpretation of these data assumes a model of isolated pores in which each diffusing proton samples only one pore. Thus, it is assumed that each pore contributes independently to the NMR signal, with a characteristic relaxation time. In many near-surface materials, however, pores are sufficiently connected that diffusing protons can readily sample multiple pores during NMR measurement. For materials in which pores are strongly coupled by diffusion, the resulting relaxation-time distribution reflects a complex averaging of the pore network that can be difficult to interpret.

Understanding the conditions in which pore coupling occurs and how this process can influence NMR relaxation measurement is fundamental to the successful use of NMR data in geophysical applications. Several studies have explored this issue theoretically, using models of idealized pore networks to determine how factors such as pore geometry (Cohen and Mendelson, 1982; McCall et al., 1991; Zielinski et al., 2002) and surface geochemistry (Ramakrishnan et al., 1999; Ramakrishnan et al., 2001) might control the extent of pore coupling. Despite the clear need to account for pore-coupling effects, relatively few laboratory studies have explored this phenomenon. Toumelin et al. (2002) show that interpore diffusion can strong-

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ly influence NMR measurements in carbonates and note that the degree of coupling generally increases with temperature as pore-fluid diffusivity increases. Anand and Hirasaki (2007) investigate geometric controls on pore coupling in microporous grain packs; they find that coupling between micropores and larger intergranular pores becomes more significant as grain diameter decreases. However, no laboratory study has explored the geochemical controls on pore coupling suggested by Ramakrishnan (1999).

The aim of our study is twofold: (1) to improve understanding of geochemical controls on pore coupling and (2) to determine how the coupling process affects the interpretation of NMR relaxation measurements. We first establish that surface geochemistry can, in fact, control the strength of pore coupling. In our laboratory approach, we study a well-characterized porous media, silica gel, and measure the NMR response of samples with varied geochemistry. Utilizing a demonstrated link between surface iron concentration and pore coupling strength, we are then able to identify specific effects of pore coupling on the relaxation-time distribution. Finally, we compare our experimental results with theoretical predictions from Ramakrishnan et al. (1999). Our study of NMR relaxation measurement in a controlled system contributes to our ultimate goal of improving the interpretation of NMR measurements in heterogeneous near-surface sediments.

BACKGROUND

In the standard NMR relaxation experiment, a static magnetic field is applied to a water-saturated sample, causing an equilibrium alignment of the magnetic moments associated with protons in the pore water. The system is then perturbed by a secondary radio-frequency pulse that excites the protons to a higher energy state, and we monitor the magnetization M over time as the system relaxes to equilibrium.

Two time constants describe NMR relaxation: T_1 and T_2 . The longitudinal relaxation rate T_1^{-1} is the rate at which the protons return to their equilibrium state. In the commonly assumed case of fast diffusion (Senturia and Robinson, 1970), the net observed relaxation rate T_1^{-1} for a single pore is given by

$$T_1^{-1} = T_{1B}^{-1} + \rho_1 \left(\frac{S}{V} \right), \quad (1)$$

where T_{1B}^{-1} is the relaxation rate of the bulk water, S/V is the surface-area-to-volume ratio of the pore (inversely proportional to the pore radius r_{pore}), and ρ_1 is the longitudinal surface relaxivity (Brownstein and Tarr, 1979). Surface relaxivity is a geochemical property describing the capacity of the grain surface to enhance relaxation; it generally increases with the concentration of paramagnetic impurities on a surface (e.g., manganese(II) $[\text{Mn}^{2+}]$ or iron(III) $[\text{Fe}^{3+}]$) (Foley et al., 1996; Bryar et al., 2000). Together, the product $\rho(S/V)$ is called the surface relaxation rate; it reflects the combined effect of geometry and geochemistry.

For practical reasons, T_1 measurements are difficult to make with borehole or surface NMR instruments, so field applications of NMR typically involve measurements of T_2 . The transverse relaxation rate T_2^{-1} is the rate at which protons dephase as they return to equilibrium. The expression for T_2^{-1} thus includes an additional term to account for internal magnetic field gradients G experienced by diffusing protons:

$$T_2^{-1} = T_{2B}^{-1} + \rho_2 \left(\frac{S}{V} \right) + \frac{D}{12} (\gamma G t_E)^2. \quad (2)$$

Here, ρ_2 is the transverse surface relaxivity, T_{2B}^{-1} is the relaxation rate of the bulk water, D is the self-diffusion coefficient of the water, γ is the gyromagnetic ratio of the hydrogen nucleus, and t_E is the echo time, a timing parameter of the measurement. Relaxation in geologic materials with negligible magnetic susceptibility is dominated by surface relaxation, in which case the expression for T_2^{-1} is reduced to

$$T_2^{-1} = \rho_2 \left(\frac{S}{V} \right). \quad (3)$$

If we consider the NMR relaxation response of a single water-filled pore, the transverse magnetization signal is described by an exponential decay:

$$M(t) = A e^{-t/T_2}, \quad (4)$$

where the initial signal amplitude A is proportional to the number of hydrogen nuclei (directly related to the pore volume). In a rock or sediment with a distribution of pore types, it conventionally is assumed that each pore contributes to the NMR signal in isolation and the net observed signal is modeled as multiexponential decay:

$$M(t) = \sum_i A_i \exp\left(\frac{-t}{T_{2i}}\right), \quad (5)$$

where the amplitude A_i is proportional to the volume of water-filled porosity associated with a corresponding relaxation time T_{2i} . Using this model of isolated pores, the $M(t)$ data are inverted to obtain a T_2 distribution, which is scaled according to equation 3 to estimate a pore-size distribution assuming uniform surface relaxivity and a simple pore shape (e.g., $S/V = 3/r_{\text{pore}}$ for spherical pores).

The simplest conceptualization of the isolated pore assumption is that each proton samples only one pore during the NMR measurement. D'Orazio et al. (1989) explain more completely that the isolated pore model is valid as long as the distance over which a proton will diffuse is smaller than the scale over which dissimilar pores (with different T_2) may be encountered. Although this condition generally is satisfied for NMR measurements in consolidated sandstones with tight pore throats and relatively uniform pore bodies (Davies, 1990; Borgia et al., 1996), the model often breaks down for materials with well-connected heterogeneous pores. Hinedi et al. (1993) and Hinedi et al. (1997) study the NMR response of microporous near-surface materials, including soil aggregates and analogous synthetic silica gels, and note that the size of larger intergranular pores is grossly underestimated. Although not recognized by researchers, this error is most likely a consequence of pore coupling.

During a given time period T , a proton will sample the pore network over an average diffusion length ℓ , approximated as

$$\ell = \sqrt{6DT}, \quad (6)$$

where D is the self-diffusion coefficient of water (at 30°C, $D = 2.46 \times 10^{-9} \text{ m}^2/\text{s}$; Simpson and Carr, 1958). Considering the duration of NMR decay as represented by T_2 (typically 10–1000 ms), a proton may diffuse more than 100 μm during NMR measurement. As the diffusion length approaches the scale at which dissimilar pores are encountered, pore coupling will begin to complicate the NMR response. In the limiting case of infinite coupling when the dif-

fusion length is unbounded, the resulting T_2 distribution will exhibit a single peak representing $\rho S/V$ of the entire pore network (McCall et al., 1991); however, the effect of pore coupling at intermediate strengths is poorly understood.

Ramakrishnan et al. (1999) model diffusional coupling in microporous grains and argue that the degree to which pores are coupled may be highly dependent on surface relaxivity. When ρ_2 is high, the diffusion length is reduced because enhanced relaxation limits the time and distance over which protons can travel before relaxing. Bryar et al. (2000) and Daughney et al. (2000) note consistent results for laboratory measurements of T_1 in iron-coated silica gels. Surface coatings of paramagnetic iron(III) (Fe^{3+}) increase ρ_1 of the gels by more than two orders of magnitude and significantly alter the shape of the relaxation-time distributions. Our experiments here use a similar system of iron-coated silica gels to develop a range of samples between which the degree of pore coupling is expected to vary with iron concentration. The NMR T_2 relaxation measurements on these samples should thus allow us to directly investigate the effects of pore coupling and explore the link between T_2 and pore geometry in heterogeneous materials.

METHODS AND MATERIALS

Sample characterization and preparation

The four samples used in this study are derived from a high-purity silica gel (Davisil® 150 Å 60–200 mesh, manufactured by Grace Davison) that has a bimodal pore-size distribution comprised of intragranular micropores (within the grains), reported to be approximately 15 nm in diameter, and larger intergranular macropores (between the grains). We first sieved the pure gel to achieve a narrower grain-size distribution of 53–105 μm . The diameter of intragranular pores in a random packing is estimated to be about 20 μm , assuming a ratio of average pore size to grain size of 0.225 (Barker and Mehta, 1992). The micropore volume, expressed as milliliters/gram gel, was provided from manufacturer specifications as 1.2 ml/g. We measured the surface area of the gels ($315.1 \pm 2.5 \text{ m}^2/\text{g}$) by the Brunauer-Emmett-Teller (BET) nitrogen adsorption method using a Micromeritics ASAP 2020 accelerated surface area and porosimetry system.

Our experimental approach was based on the assumption that increasing the amount of paramagnetic Fe^{3+} on the gel surface would reduce pore coupling by increasing the surface relaxation rate; therefore, we closely controlled surface geochemistry. We rinsed the gel repeatedly with 10% hydrogen chloride (HCl) and distilled, deionized (DDI) water (18 M Ω cm) to remove paramagnetic impurities, and we retained a portion of the cleaned gel to be used as the highest purity sample for NMR analysis. We then synthesized three additional gels with varied amounts of paramagnetic Fe^{3+} oxyhydroxide surface coating from the cleaned gel using methods of Grantham et al. (1997). Samples were placed under vacuum and saturated with aqueous solutions of ferrous sulfate (FeSO_4). The mixtures were then oxidized by adding excess 30% hydrogen peroxide (H_2O_2) and were placed on an orbital shaker to equilibrate overnight. Following the iron coating treatment, the gels were rinsed with DDI water and oven dried overnight at 45°C.

We determined the amount of paramagnetic Fe^{3+} sorbed to each gel from subsamples by atomic adsorption spectroscopy; concentrations of Fe^{3+} by mass in the four samples ranged from 0.00% in the pure sample to 0.10% in the most heavily coated sample. Qualitative analysis by scanning electron microscopy (SEM) and energy disper-

sion spectrometry (EDS) confirmed that iron was effectively sorbed to the interior and exterior surfaces of the gels. Neither the porosity nor the surface area of the gels was changed, within experimental error, by the iron coating process.

NMR measurements

For NMR measurements, we packed dry samples of each gel in consistent weights and volumes into cylindrical Teflon™ sample holders with an interior diameter of 2.1 cm and height of 6 cm. In the first series of measurements, samples were completely saturated with DDI water. To ensure maximum saturation, we first evacuated each packed sample in a dry vacuum (75 mm mercury) for 30 min and then submerged the sample in a water bath before atmospheric pressure was restored. Samples were allowed to saturate for 2 hr while submerged. The volume of water held in micropores for each sample was calculated from the mass of dry sample (using manufacturer-specified 1.2 ml/g gel); the volume of water in macropores was determined gravimetrically as the difference between total water volume and micropore volume. We determined that of the total water volume, 58% was held in micropores and 42% was held in macropores.

We acquired a first series of NMR measurements directly after saturation. Following measurement at saturated conditions, we centrifuged the packed samples to drain water from the large intergranular pores. Centrifuging removed approximately 98% of the intergranular pore water (estimated gravimetrically); the intragranular porosity was left undrained by this process as a result of the extremely high capillary potential of the micropores. We then acquired a second series of NMR measurements, immediately after centrifuging, at which point the NMR signal was presumed to be primarily sensitive to water in the undrained intragranular micropores.

All NMR relaxation data were collected using a 2.2-MHz Maran Ultra NMR core analyzer (Resonance Instruments) with a CPMG pulse sequence. This instrument has a dead time of approximately 60 μs . Data were acquired consistently at 30°C and were stacked 100 times to improve the signal-to-noise ratio (S/N). Measurements at varied echo-time t_E spacing indicated that magnetic gradient effects were negligible; therefore, the shortest available spacing, $t_E = 150 \mu\text{s}$, was used to maximize data sampling at short times. Independent measurements on each sample were repeated three times to determine experimental variance.

We inverted each measured magnetization decay signal for a T_2 relaxation-time distribution using a regularized nonnegative least-squares algorithm developed after Whittall et al. (1991). In this approach, the data are fit to 200 logarithmically spaced T_2 values, from 1 ms to 10 s, and the inverted model is regularized for smoothness. Regularization parameters were selected such that the data were fit to a tolerance equal to the measured noise level. This statistically robust approach limits inversion artifacts and is required in order to ascribe physical significance to observed differences in the smoothness of the T_2 distribution.

RESULTS AND DISCUSSION

Laboratory NMR results

The four gels provide a group of samples with identical geometry and varied geochemistry, with ρ_2 expected to increase as iron content increases. An increase in ρ_2 is predicted to lead to a decrease in pore coupling as faster surface relaxation reduces the distance protons

may diffuse before relaxing. Thus, analyzing these samples should allow us to assess (1) the extent to which pore coupling may be controlled by surface geochemistry and (2) how the coupling process affects NMR data interpretation.

The resulting T_2 distributions for each of the gel samples are shown in Figure 1, with iron concentrations increasing from top (0.000% Fe^{3+}) to bottom (0.010% Fe^{3+}). Solid curves show distributions for NMR measurements at fully saturated conditions; dashed curves are for measurements collected after centrifuging. Distributions for each sample are normalized by the total signal amplitude at full saturation, which is consistent between all samples to within experimental error.

We first consider the T_2 distributions for all four samples obtained after centrifuging, when the NMR measurement is sensitive exclusively to the saturated micropores. The integrated amplitude of each distribution reflects 58% of the normalized signal; thus, the total volume of microporosity is captured accurately. Despite the reported uniform intragranular pore size, we observe slightly broad peaks. Regularization of the inverted distributions may result in some peak broadening; however, we have consistently found that inversions of data with high S/N (e.g., ~ 130 here) can produce narrow peaks when warranted by the data. We therefore interpret the breadth of the peaks as indicating that the size and surface-iron concentration of the

micropores are not entirely uniform. For samples with increasing iron concentrations, we observe a shift of the centrifuged T_2 peak to shorter times, reflecting an expected increase in ρ_2 . We calculate ρ_2 for each sample (Table 1) from equation 3, using the logarithmic mean relaxation time for each centrifuged distribution and independent measurements of S/V .

Analyzing the T_2 distributions for measurements at full saturation allows us to investigate the extent to which coupling occurs between the macropores and micropores. We begin with the results for the low- ρ_2 pure gel, which we expect to exhibit the strongest degree of pore coupling. Given that the macropores and micropores differ in size by approximately three orders of magnitude, the isolated pore model should predict two distinct peaks in the relaxation-time distribution. Instead, we observe one broad peak with two marginally discernible modes separated by less than two orders of magnitude. This apparent averaging of the underlying pore-size distribution indicates that the two pore types are strongly coupled by diffusion. In the other three samples, as the concentration of iron coating the gel surface is increased, we observe a separation of the distribution into two clearly discernible modes or peaks, reflecting the anticipated decrease in pore coupling. However, even for the most heavily iron-treated gel, the distance separating these peaks is less than would be predicted by a purely isolated pore model.

Let us now explore the meaning of the relaxation-time distributions for these samples, spanning a range of coupling strengths. We refer to the two modes in each distribution as the fast relaxing signal (at lower T_2 values) and the slow relaxing signal (at higher T_2 values). For all samples, the fast relaxing signal corresponds directly to the signal from the microporosity, represented in the centrifuged measurements. Thus, the relaxation time of the micropores is reflected accurately despite the presence of pore coupling and could be used to estimate a pore size, assuming a given ρ_2 value. On the other hand, the location of the slow relaxing signal, which should reflect the relative size of the macropores in the absence of pore coupling, overlaps the fast-relaxing micropore signal when pore coupling is strong and successively separates as the degree of pore coupling is reduced. Thus, when pore coupling is dominant, the relative size of the macropores would likely be dramatically underestimated.

In addition to this apparent shift in the position of the T_2 distributions, we observe that pore coupling affects the relative amplitudes of the fast and slow relaxing signals. We estimate the relative volume of water relaxing at fast T_2 ($\theta_{w\text{-fast}}$) versus slow T_2 ($\theta_{w\text{-slow}}$) for each saturated sample by first splitting the distribution at the point of minimum slope and then integrating the distribution to the left and right, respectively. Values for the ratio $\theta_{w\text{-fast}}/\theta_{w\text{-slow}}$, estimated in this manner for each sample, are listed in Table 1. In the absence of pore coupling, we would expect $\theta_{w\text{-fast}}/\theta_{w\text{-slow}}$ to reflect the true ratio of mi-

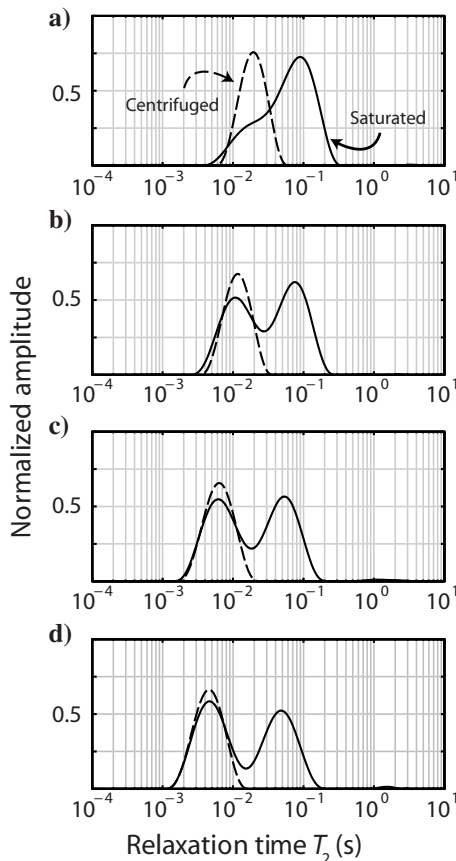


Figure 1. NMR relaxation-time distributions for the four silica gel samples: (a) pure gel, (b) 0.002% Fe^{3+} , (c) 0.004% Fe^{3+} , and (d) 0.010% Fe^{3+} . Iron concentrations increase from top to bottom. Solid curves are for measurements at full saturation; dashed curves are for measurements after centrifuging (signal exclusively from micropores). Curves for each sample are normalized by the initial signal amplitude for the fully saturated measurements.

Table 1. Tabulated estimates of geochemical properties for each sample and the estimated $\theta_{w\text{-fast}}/\theta_{w\text{-slow}}$ for laboratory measurements and model data.

% Fe^{3+} by mass	ρ_2 ($\mu\text{m/s}$)	$\theta_{w\text{-fast}}/\theta_{w\text{-slow}}$ laboratory	$\theta_{w\text{-fast}}/\theta_{w\text{-slow}}$ modeled
0.000	0.201 ± 0.007	0.61 ± 0.04	0.64
0.002	0.345 ± 0.004	0.93 ± 0.05	1.00
0.004	0.592 ± 0.010	1.12 ± 0.06	1.21
0.010	0.953 ± 0.004	1.26 ± 0.06	1.34

cropore volume to macropore volume (1.37 ± 0.04). For the pure gel, which exhibits the greatest degree of pore coupling, $\theta_{w-fast}/\theta_{w-slow}$ underpredicts the true ratio by 55%, grossly misrepresenting the true pore geometry. As pore coupling weakens with increasing iron content, we see that the accuracy of this estimated ratio improves, eventually reaching 94% of the true value.

These laboratory observations are compatible with a conceptual model of pore coupling proposed by Ramakrishnan et al. (1999), in which protons move freely between intragranular micropores and intergranular macropores before relaxing. The leftward merging of the slow relaxing peak for samples with low iron concentrations occurs because most protons that originate in macropores can diffuse freely into the microporous grains before relaxing. These protons thus relax at a much faster rate than would be predicted for isolated macropores and exhibit an averaging of rates for the two pore types. When surface relaxivity is high, fewer protons that originate in the macropores can efficiently sample the microporous region before relaxing, lessening the degree to which the slow relaxing peak is skewed leftward.

The relative amplitude of the two peaks is likewise affected by the relative number of protons that sample both pore types. When relaxation occurs slowly, many protons are able to diffuse out of the smaller intragranular pores before relaxing and so decay at an averaged rate. For samples with substantial iron surface coatings, relaxation occurs rapidly, so more protons originating in micropores relax before diffusing out of the grain. Thus, the amplitude of the fast relaxing T_2 peak more accurately represents the true micropore volume in these samples. We anticipate that the presence of other paramagnetic ions such as Mn^{2+} on the grain surface would produce similar effects.

Comparison to analytical models

We can further compare our results quantitatively with the 1D model of pore coupling, presented and described by Ramakrishnan et al. (1999). This model applies the NMR diffusion-reaction equation in two separate domains, one representing a microporous grain and one representing a macropore. The microporous grain, with radius R_g and porosity ϕ_μ , is treated as a continuum in which surface relaxation occurs at a specified rate $T_{2S\mu}$. To account for restricted diffusion occurring within the micropores, the self-diffusion coefficient of water D is scaled by the formation factor of the microporous grain F_μ . The total porosity of the system ϕ_t determines the dimensions of the two domains. The magnetization in the macropore domain M_m is governed by the standard NMR diffusion equation:

$$D \frac{d^2 M_m}{dx^2} - M_m T_{2B}^{-1} = \frac{dM_m}{dt}, \quad (7)$$

where T_{2B} is the bulk water relaxation rate.

Equation 8 describes M_μ , the magnetization in the domain of the microporous grain:

$$\frac{D}{F_\mu} \frac{d^2 M_\mu}{dx^2} - \phi_\mu M_\mu (T_{2S\mu}^{-1} + T_{2B}^{-1}) = \phi_\mu \frac{dM_\mu}{dt}. \quad (8)$$

A boundary condition between domains,

$$D \frac{d^2 M_m}{dx^2} = \frac{D}{F_\mu} \frac{d^2 M_\mu}{dx^2} + \rho_2 M_m (1 - \phi_\mu), \quad (9)$$

matches flux into and out of the microporous grain and allows for relaxation at the grain wall with a surface relaxivity of ρ_2 .

We fully parameterized four models to be representative of our measured samples. A value of $T_{2B} = 3.05$ s, consistently measured for all centrifuged fluids, was used across all models. Values for $(T_{2S\mu}^{-1} + T_{2B}^{-1})$ in equation 8 are derived from the mean log relaxation time for the micropores in each sample, measured after centrifuging. Representative values of ρ_2 , obtained as described previously, are listed in Table 1. Geometric parameters $R_g = 30$ μm , $\phi_t = 0.83$, and $\phi_\mu = 0.725$ reflect manufacturer specifications confirmed by independent measurements. It was not possible to measure F_μ directly, so we selected a value of 1.4 ($1/\phi_\mu$), consistent with an open, weblike network of pores typical of silica gels. We used a standard value of $D = 2.46 \times 10^{-9}$ m^2/s for water at 30 °C (Simpson and Carr, 1958).

Ramakrishnan (1999) explains that exact analytic solutions to the parameterized equations can be obtained using Laplace transforms; the solutions yield two principal exponential-decay modes (smoothing effects are not modeled). However, obtaining analytic solutions is very challenging, so we used Comsol Multiphysics® to obtain accurate numerical solutions. In Figure 2, we plot the position and magnitude of the resulting principal decay modes, superimposed

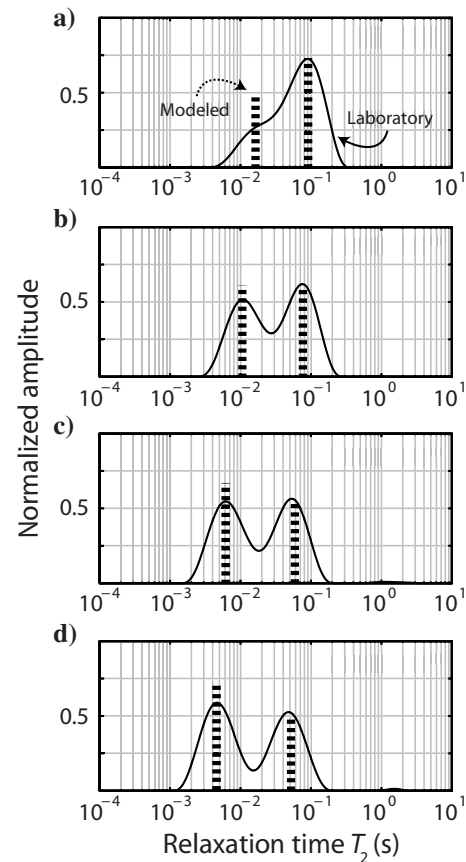


Figure 2. Comparison between laboratory measurements and the simulated relaxation response based on the Ramakrishnan et al. (1999) 1D model: (a) pure gel, (b) 0.002% Fe^{3+} , (c) 0.004% Fe^{3+} , and (d) 0.010% Fe^{3+} . Solid curves are for laboratory measurements on gels at saturated conditions. Dashed bars show the position and amplitude of the primary exponential relaxation modes derived from the model. Mode amplitudes are all normalized such that, for the pure gel, the long-time peak is exactly matched by the long-time mode.

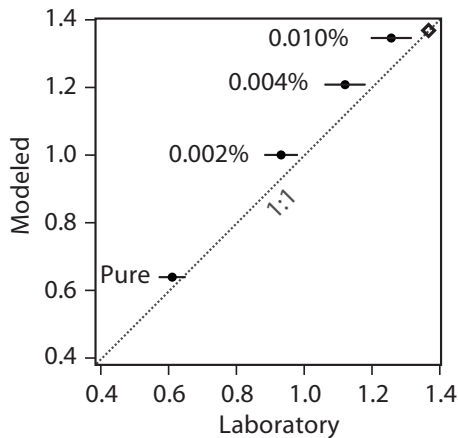


Figure 3. Comparison of $\theta_{w-fast}/\theta_{w-slow}$ derived from laboratory- and model-generated results. Estimates from laboratory measurements represent the ratio of integrated amplitude for the short-time peak to that of the long-time peak. Error bars show the standard deviation for this estimate from repeated measurements. Estimates from modeled data represent the ratio of the short-time mode amplitude to that of the long-time mode. The hollow black diamond (top right) represents the actual value of the micropore-to-macropore volume, which would be reflected by $\theta_{w-fast}/\theta_{w-slow}$ in the absence of pore-coupling effects.

over the experimental T_2 distributions for each sample. In general, modeled relaxation times closely predict the location of the T_2 peaks for laboratory measurements. However, the relaxation time for the slow relaxing peak is slightly overestimated in models representing the 0.004% and 0.010% Fe^{3+} gels.

Plotted in Figure 3 and listed in Table 1 is a comparison of $\theta_{w-fast}/\theta_{w-slow}$, determined for each of the laboratory and modeled distributions (direct comparison is not possible in Figure 2 because the width of the experimental T_2 peaks is not constant). We plot modeled and experimental values along a 1:1 line in Figure 3 as filled circles with experimental error bars; the hollow diamond (top right) represents the true ratio value (1.37), which would be observed for zero pore coupling. The modeled and experimental data agree quite well; however, as iron concentrations increase, the modeled $\theta_{w-fast}/\theta_{w-slow}$ increasingly exceeds the experimental result. These minor disparities between modeled and laboratory results at higher iron concentrations suggest that the model may slightly overpredict the extent to which increased surface relaxivity can reduce pore-coupling effects. Nonetheless, general agreement with our experimental results indicates that this relatively simple 1D model provides an accurate and extremely useful tool for predicting NMR relaxation behavior in this class of heterogeneous media.

CONCLUSIONS

The NMR relaxation times of porous geologic materials are closely linked to the volume and geometry of the water-saturated pore space. Thus, there is tremendous potential for using NMR relaxation measurements, acquired with borehole or surface-based instruments, in near-surface applications to estimate subsurface properties. It is clear, however, that substantial research is needed to fully understand the processes governing NMR relaxation in near-surface materials. This study has focused on the impact of pore coupling, a process that can significantly affect NMR relaxation measurements in systems with well-connected pores.

Our experimental results demonstrate conclusively that surface geochemistry can be a significant factor determining the strength of pore coupling and thus the relationship between pore geometry and NMR relaxation times. By controlling for surface geochemistry in well-characterized samples, we directly assessed the impact of pore coupling on the observed T_2 distribution in samples representing a range of coupling strengths. For materials with intragranular microporosity and intergranular macroporosity, pore coupling reduces the slowest relaxation time associated with macropores and decreases the amplitude of the fast relaxing signal associated with the micropores. If unaccounted for in the interpretation of the NMR relaxation data, the presence of pore coupling would lead to an underestimation of the total micropore volume and the size of the largest pores. Good agreement between our laboratory results and an analytical model suggests that a theoretical approach may help predict the extent to which pore coupling will affect NMR measurements in other porous media.

Our study has only considered the behavior of silica gels, which have smaller pores and a lower ρ_2 value than is typical for most geologic materials. The small pores tend to limit the extent of the diffusion, whereas the low ρ_2 value tends to enhance the extent of diffusion. Although these properties of silica gel are different from those of a typical geologic material, the well-characterized geometry of the gel allows us to investigate the physics linking surface chemistry and pore coupling. Our findings lead us to anticipate that for a wide class of unconsolidated materials with well-connected pores, the NMR relaxation-time distribution will not accurately convey the underlying pore structure because of pore coupling effects. Additional research is required to explore the impact of pore coupling in a broader range of materials to improve the interpretation of NMR data and to advance the use of NMR methods in near-surface applications.

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SURFACE RELAXIVITY ESTIMATION AND NMR-MICP MATCHING IN DIFFUSIONALLY COUPLED ROCKS

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ABSTRACT

The surface relaxivity, ρ , is one of the most important parameters in NMR petrophysics, since it scales NMR relaxation time measurements of fluid saturated rocks to their pore size distributions. NMR-MICP matching techniques can be used to obtain an estimation of ρ , but in general they are very dependent on an optimal correlation between these two measurements, thus are not appropriate in cases where the NMR measurements are heavily affected by diffusional coupling between micro and macro pores. Simulations of the NMR T_2 relaxation time phenomena were conducted using a coupled dual porosity model calibrated to match real T_2 distributions of three carbonate samples. Once calibrated, the simulations were then repeated with the same parameters, but without allowing diffusional coupling between the micro and macro pores of the model. The results of the decoupled simulations showed better correlation with MICP data for these carbonates, and were used to improve the estimation of ρ using NMR-MICP matching techniques. Furthermore, a diffusional coupling parameter was calculated based on the model parameters, to evaluate the strength of diffusional coupling of these carbonates.

INTRODUCTION

In Nuclear Magnetic Resonance (NMR) petrophysics, transversal relaxation time (T_2) measurements of 100% brine saturated rocks are commonly used to estimate their pore size distribution, which then can provide reasonable estimations for the permeability (k), irreducible water saturation (S_{wi}) and other petrophysical properties. This approach is valid when four main assumptions are applicable: fluid's relaxation within pores occurs in the fast diffusion regime; diffusion and bulk relaxations are negligible; the diffusion coupling effect is negligible; and surface relaxivity is homogeneous throughout the pore walls [1]. By considering these assumptions, the pore size distribution of the pore system is related to T_2 as follows:

$$\frac{1}{T_2} = \rho_2 \left(\frac{S}{V} \right)_{pore} \quad (1)$$

where $\left(\frac{S}{V} \right)_{pore}$ is the surface-to-volume ratio of the pores and ρ_2 is the transverse surface relaxivity. While $(S/V)_{pore}$ is representative of the pore size, the surface relaxivity

parameter is related to the strength of the fluid's magnetic interaction with the pore walls, which causes the relaxation phenomena [2].

Equation 1 shows the importance of ρ_2 for NMR petrophysics, since it scales T_2 measurements with pore size. For simple lithologies, ρ_2 is commonly considered to be homogeneous for each formation and several NMR predictions, such as the classical SDR permeability equation (k_{SDR}) and Swi, can be reasonably estimated without knowing the actual value for ρ_2 . However, in heterogeneous rocks with complex lithologies, like carbonates, variations of ρ_2 may occur in small scales within a formation and the lack of information of this parameter may result in poor accuracy of the NMR predictions.

One of the first methods developed for estimating ρ_2 from rock cores consists in correlating in a common dimension T_2 distributions measurements with pore throat distributions obtained from the Mercury Injection Capillary Pressure technique (MICP). In this case, it is assumed that the pore geometry in the pore system is predominantly cylindrical, thus T_2 and throat radius (R_t) can be correlated through equation 1 by [3]:

$$\left(\frac{S}{V}\right)_{cylinder} = \frac{2\pi R_t l}{\pi R_t^2 l} = 2^* / R_t \rightarrow R_{t_{cylinder}} = 2\rho_s T_2 \quad (2)$$

where ρ_s is an effective surface relaxivity and l is the length of the cylinder. The value of 2^* is the number that converts the $(S/V)_{pore}$ term from equation 1 into the pore radius, called pore shape factor. The effective relaxivity ρ_s is then adjusted as the value that maximizes the cross-correlation (CC) between the T_2 and R_t distributions, using:

$$CC(\rho_2) = \sum_i A_{Hg}(r_i) \cdot A_{NMR}(r_i = 2\rho_s T_2) \quad (3)$$

where A_{Hg} is the amplitude of the R_t distribution, and A_{NMR} is the amplitude of the T_2 distribution scaled by $2\rho_s$. Although the NMR-MICP matching provides a reasonable estimation of ρ , it relies on the optimal correlation between these two techniques. So, it is dependent on the ability each technique has to reflect the actual pore size and pore throat distributions of the rock. Considering this, directly matching T_2 and R_t may not be the best approach in cases where the pore's saturating fluid dynamics is under a strong diffusional coupling between micro and macro pores, a situation that enhances the diffusional coupling effect. In these cases, diffusional coupling mixes the NMR magnetic response of coupled pores with different sizes, causing it to lose its correspondence to the actual pore size. Because of this, one must consider the strength of the diffusional coupling effect before applying the NMR-MICP matching and calculating ρ_s .

NMR SIMULATIONS IN DUAL POROSITY SYSTEMS

In order to understand the diffusional coupling effect in complex porous systems, simulations of the NMR T_2 relaxation time, developed by Ramakrishnan et al [6], were applied in porous media models with porosity mixed between interconnected micro- and macro-pores. These simulations use random walk techniques to mimic the saturating fluid's diffusion, in which each particle (called "walker") represents a group of fluid

molecules moving randomly inside pores. The simulation starts with the walkers being placed randomly in the model's porous space, with an initial energy that represents the magnetic moment. The random walking is then performed in a stepwise fashion, where the distance traveled by each step takes an amount of time related to the self-diffusion coefficient of the fluid. When a walker meets a pore wall, this energy can be decreased by a factor or annihilated by chance, where both approaches depend on the value of ρ [6,7].

The dual porosity model used in this work is the Micro-Grain Consolidation Model (μ GC) [6,7], which is a cubic packing of overlapping spherical small grains with radius R_{small} , that composes bigger spherical grains centered at each corner of the cube, with radius R_{big} . This emulates the rock matrix as a network of bigger grains with intra-granular micro porosity, while the macro porosity consists in the inter-granular space of the bigger grains. Micro- and macro- porosity fractions are controlled by R_{small} and R_{big} , while the sizes of the micro- and macro- grids control the interconnectivity of the pores. An advantage of using this model is that we can introduce an artificial 'barrier' that prevents walkers from entering or exiting the micro or macro pores. This makes it possible to simulate the NMR experiment with or without diffusional coupling between the two domains. This approach is useful to infer how the MICP and NMR relaxation times correlate with each other when the diffusional coupling phenomenon is turned off.

DIFFUSIONAL COUPLING STRENGTH

Diffusional coupling between micro and macro pores has been modeled by Anand et al [4], in which the fluid's molecules relaxes at the surface of flake-like micro-pores while diffusing between micro and macro pores. The model's coupled pore geometry is defined by its half-length and half-width, named $L1$ and $L2$ respectively, and the micro-porosity fraction β . In the work, the authors introduced a coupling parameter α , described by:

$$\alpha = \beta\eta\mu \quad (4)$$

where η is $L2/L1$ and μ is the Brownstein Number [5]. μ can be estimated by calculating $L2\rho/D$, where D is the self-diffusion coefficient of the fluid ($2.5 \times 10^{-9} \text{ } \mu\text{m}^2/\text{s}$ for water at 25°C). The study also related this diffusional coupling model to the μ GC by having $L1 = R_{small}$ and $L2 = R_{big}$, in order to apply the model for carbonates. Furthermore, α is described as being inversely proportional to diffusional coupling strength.

RESULTS

To evaluate the described methodology, three carbonate rock cores were chosen, MICP measured, NMR T_2 measured (as described in [2]) and simulated using the μ GC model described above. The random walk simulations were written in Fortran™ language and the micro and macro grains radius R_{small} and R_{big} were determined by matching the micro- and macro-porosity fractions from the μ GC model with the ones observed in the MICP distributions. These values are shown in Table 1. Other parameters of the simulation, such as micro- and macro-grid sizes, were adjusted so that the simulation results best match with the experimental T_2 distributions. Figures 1, 2 and 3 show the distributions of

measured and simulated T_2 and MICP from the selected carbonate samples, named respectively as H11, H23 and Silurian Dolomite. All samples showed (in the left side of the figures) excellent agreement between experimental (full blue curves) and simulated (called “coupled”, the green dashed curves) distributions. As described above, the random walk algorithm was executed again using the same parameters, but restricting the diffusive process of the walker: now, it is not allowed to leave the pore that they are randomly probing. The results of the decoupled simulations are shown on the figures as the black dashed curves called “uncoupled”. The difference between decoupled and real T_2 distributions suggests that when coupled, the T_2 distribution’s modes of micro- and macro-porosities (small and long T_2 ’s, respectively) are averaged in a degree that indicates diffusional coupling. Through this approach and comparing it with the shape of the MICP distributions (right side of the figures), it is clear that samples H11 and H23 have higher diffusional coupling strength, while the Silurian Dolomite have lower effect.

In order to estimate ρ_g , both decoupled and real T_2 distributions were matched with MICP distributions using the method described in equation 3. The matched distributions are shown together in the right side of the figures, while the values of ρ_g obtained from this matching are presented in Table 1. The results show that the decoupled distributions have much better correlation with MICP than the experimental T_2 distributions, suggesting that matching decoupled T_2 distributions with MICP R_t distributions provides a more robust and accurate ρ_g values. Both the H23 and Silurian Dolomite samples showed excellent correlations between decoupled T_2 and R_t , while the micro- and macro-modes from the decoupled simulation of the sample H11 did not match exactly with the ones from MICP. This fact can be explained if both porosities did not match for a unique choice of ρ_g and pore shape factor, or the pore system have a variable pore body-to-throat ratio (BTR). On the other hand, the excellent correlations in the Silurian Dolomite and H23 samples suggest that ρ_g and the pore shape factor are constants throughout the cores (not being a function of pore size). These are interesting cases where the analysis of decoupled T_2 distributions can provide significant insight about the properties of the porous system.

After matching the distributions, the α parameter was calculated using the R_{small} and R_{big} from the simulations, the micro porosity fraction observed in the MICP distribution and the Brownstein number estimated with ρ_g from the matching of the decoupled distributions (with equation 3). The results showed that the initial supposition regarding the comparison between strengths of diffusional coupling on the samples were correct, as H11 and H13 shows more coupling effect (α equals to 1.38 and 2.37 respectively), while the Silurian Dolomite shows much less effect ($\alpha = 12.99$).

CONCLUSION

The presented study showed that simulated decoupled T_2 distributions produces better matching with R_t distributions obtained from MICP curves, thus providing a more reliable estimation for surface relaxivity for carbonates. In cases where this approach

does not generate an exact correlation between decoupled T_2 and MICP, it was inferred that different pore size families on complex samples may present different pore shapes, BTR or surface relaxivities between them. This subject is important to the study of pore systems. The diffusional coupling parameter was calculated successfully using sample's parameters, showing indeed an inverse relation with the strength of the diffusional coupling effect on T_2 distributions. This shows that the α parameter can be used to estimate the coupling strength and that it helps to validate the T_2 distribution as a reliable estimation of pore size.

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Table 1: Characteristic parameters used to, and extracted from, the experimental analysis and simulations performed.

Sample Name	H11	H23	Silurian Dolomite
Porosity (p.u.)	28.8	28.1	18.3
MICP micro-porosity fraction (%)	40.0	45.0	41.8
R_{small} (μm)	1.29	1.27	1.34
R_{big} (μm)	122.68	124.46	129.75
NMR-MICP matched ρ_e (coupled) ($\mu\text{m/s}$)	1.72	1.76	9.78
NMR-MICP matched ρ_e (decoupled) ($\mu\text{m/s}$)	0.74	1.08	6.17
Estimated Brownstein number	0.036	0.054	0.320
α	1.38	2.37	12.99

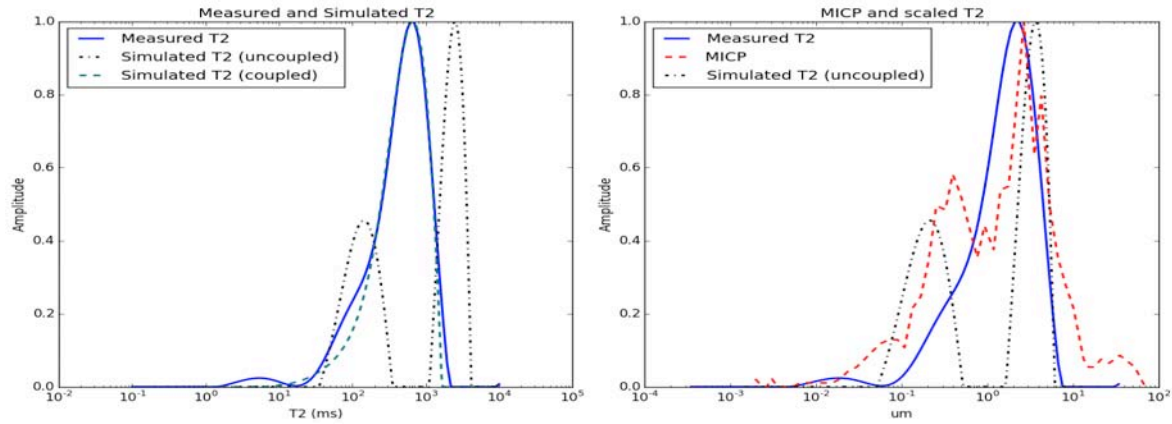


Figure 1: Data from the sample "H11". Left: measured and simulated T_2 distributions and right: MICP distribution with T_2 measured and simulated, scaled by their optimal ρ_E .

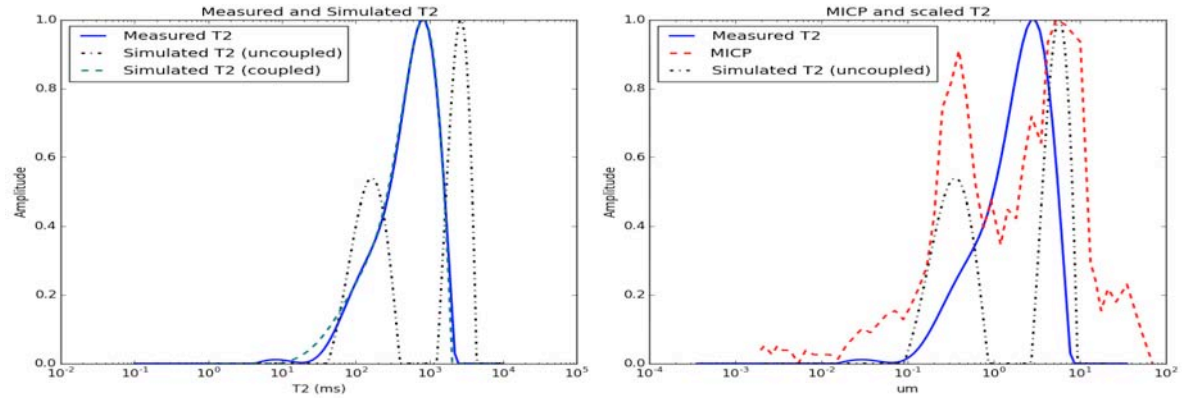


Figure 2: Data from the sample "H23". Left: measured and simulated T_2 distributions and right: MICP distribution with T_2 measured and simulated, scaled by their optimal ρ_E .

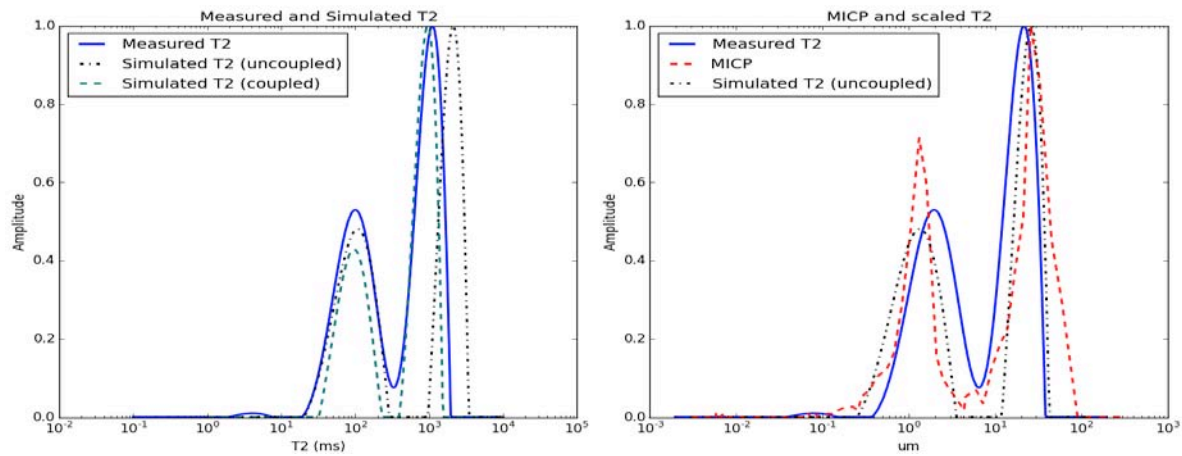
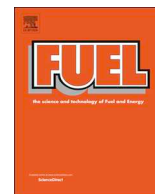


Figure 3: Data from the sample "Silurian Dolomite". Left: measured and simulated T_2 distributions and right: MICP distribution with T_2 measured and simulated, scaled by their optimal ρ_E .



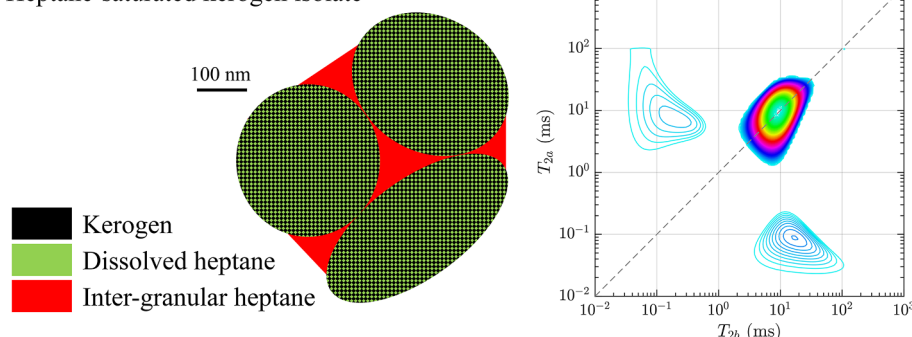
Diffusive coupling in heptane-saturated kerogen isolates evidenced by NMR T_1 - T_2 and T_2 - T_2 maps

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GRAPHICAL ABSTRACT

Heptane-saturated kerogen isolate



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Surface relaxation

ABSTRACT

It is well known that in unconventional shale, the ^1H NMR T_1/T_2 ratio for light producible hydrocarbons in the kerogen ($T_1/T_2 \approx 3 \leftrightarrow 6$) is larger than water in the clays ($T_1/T_2 \approx 2$). While such a contrast in T_1/T_2 is potentially useful for fluid typing and saturation in shale, the underlying mechanism remains elusive. Our previous NMR reports at $f_0 = 2.3$ MHz of heptane-saturated kerogen isolates distinguished two signals (a) inter-granular heptane between the kerogen grains with small $T_1/T_2 \approx 2.4$, and (b) dissolved heptane in the kerogen grain with large $T_1/T_2 \approx 70$. In this report we present new evidence of diffusive coupling effects between the inter-granular heptane and the dissolved heptane using T_1 - T_2 correlation maps as a function of saturation and T_2 - T_2 exchange maps. Our results indicate a phenomenon not previously reported; namely, different diffusive-coupling regimes between T_1 (with intermediate coupling) and T_2 (with weak coupling). We find that T_1/T_2 of the inter-granular heptane increases from $T_1/T_2 \approx 2.1 \rightarrow 6.5$ with decreasing saturation of the inter-granular pore-space, which we attribute to increasing confinement and the difference in diffusive coupling between T_1 and T_2 . Our findings imply that the large range $T_1/T_2 \approx 3 \leftrightarrow 6$ of the inter-granular (i.e. the more producible) hydrocarbons in shale is due to confinement together with diffusive-coupling effects with the dissolved (i.e. the less producible) hydrocarbons in kerogen.

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1. Introduction

Diffusive coupling is a well known phenomenon in porous media. It was first reported in water-saturated carbonates where macro-pore water was shown to be diffusively coupled to micro-pore water in micro-calcite grains [1]. Diffusive coupling was also shown to occur in water-saturated sandstones where macro-pore water couples to the clay flakes lining the macro-pores [2,3], and in water-saturated microporous glass beads [4]. One of the consequences of diffusive coupling is that in the strong diffusive-coupling regime, the NMR relaxation times $T_{1,2}$ for macro and micro-pore water merge into one peak, making them indistinguishable. Consequently, in the strong diffusive-coupling regime the $T_{1,2}$ distributions do not reflect the pore-size distribution. On the other hand, in the weak diffusive-coupling regime, the macro and micro-pore water have distinct $T_{1,2}$ peaks which can be interpreted as a pore-size distribution. Models have been proposed that predict the apparent surface relaxivity of the macro pores in the weak diffusive-coupling regime [1,3,5]. However these models do not explicitly account for pore-size or saturation of the macro-pores, although a semi-quantitative account of macro-pore saturation was reported in Ref. [1].

The difference in $T_{1,2}$ distributions for strong versus weak diffusively-coupled pores was recently reported for an organic-rich chalk [6,7]. In this study, bitumen blocks the diffusive coupling between macro-pore and micro-pore water, resulting in weak to intermediate diffusive-coupling. Meanwhile at shallower depths in the formation, the less mature organic-rich chalk has less bitumen and therefore no blocking mechanism between macro to micro-pore water, resulting in strong diffusive-coupling.

In the weak diffusive-coupling regime, the macro and micro-pore peaks are clearly separated in the T_2 distribution, and one can use T_2 - T_2 exchange maps to measure the exchange time between macro and micro-pores. As one increases the storage time in the T_2 -store- T_2 measurement, one observes the build up of off-diagonal peaks (i.e. exchange peaks) in the T_2 - T_2 map [8–13]. Exchange peaks in T_2 - T_2 maps have been observed in various systems including sandstones [8], cement pastes [9,11], and carbonates [14]. T_2 - T_2 exchange maps have also been used to study the effect of asphaltene deposition on pore-space morphology and wettability for sandstones [15].

More recently, exchange peaks have been observed in heptane-saturated organic-rich shales [16]. In this case, some of the exchange peaks in the T_2 - T_2 maps are postulated to be from magnetization transfer between the heptane (with longer T_2) and kerogen/bitumen (with shorter T_2). In other words, some of the exchange peaks are a result of exchange in nuclear magnetization rather than from diffusion of molecules between two pore-types. The results in Ref. [16] imply that ^1H - ^1H dipole-dipole relaxation dominates over surface paramagnetism in unconventional shales. Evidence for ^1H - ^1H dipole-dipole relaxation in kerogen was also reported in Ref. [17] by comparing T_1 - T_2 of water-saturated kerogen versus heavy-water saturated kerogen with $\text{H}_2\text{O}:\text{D}_2\text{O}$ proportions of 1:9, a technique previously used in water-saturated carbonates [18]. Understanding which NMR relaxation mechanism dominates in shale is of fundamental interest when trying to understand the origin of the contrast in T_1/T_2 ratio, where T_1/T_2 is reported to be larger for light producible hydrocarbons ($T_1/T_2 \approx 3 \leftrightarrow 6$) compared to water in clays ($T_1/T_2 \approx 2$) [16,17,19–52].

We previously investigated the ^1H NMR relaxation mechanism in heptane-saturated kerogen isolates from an immature Kimmeridge shale outcrop, measured at a resonance frequency of $f_0 = 2.3\text{ MHz}$ [17,38,39]. The motivation for isolating the kerogen from the shale is that there is no interference from clay-bound water or hydroxyls [17,31,53,39,54]. We found two distinct features in the heptane-saturated kerogen (a) inter-granular heptane between the kerogen grains with long relaxation times $T_{2>} \approx 15\text{ ms}$ and small $(T_1/T_2)_{>} \approx 2.4$, and, (b) dissolved heptane in the kerogen with short relaxation times $T_{2\mu} \approx 0.24\text{ ms}$ and large $(T_1/T_2)_{\mu} \approx 70$. We postulate that the large $(T_1/T_2)_{\mu} \approx 70$ ratio of the dissolved peak is due to nano-confinement in

the organic pores of kerogen/bitumen. We tested this hypothesis on a polymer-heptane mix, where the high-viscosity polymer ($\approx 333,000\text{ cP}$) acted as a model for immature kerogen. Both NMR measurements [55–57] and molecular dynamics (MD) simulations [58] confirmed that the $T_{1s}/T_{2s} \approx 4$ for the surface relaxation of heptane was enhanced due to nano-confinement in the polymer matrix. These results strongly suggest that ^1H - ^1H dipole-dipole relaxation dominates over paramagnetism in organic nano-pores.

Characterizing the dissolved hydrocarbons in kerogen is also important for estimating the total gas in place in shale-gas reservoirs. Direct evidence of dissolved gas in kerogen comes from pressure-decay measurements of methane in shale, where it has been reported that dissolved gas constitutes $\approx 20\text{ wt\%}$ of the total gas [59], while the remaining gas is free ($\approx 40\text{ wt\%}$) and adsorbed ($\approx 40\text{ wt\%}$). Dissolved methane in shale has also been reported with in situ NMR measurements [40]. The presence of dissolved gas in kerogen could increase the reserve estimates in shale gas, and may be recoverable by pressure depletion, however it is generally not taken into account. The production of dissolved gas in kerogen could also be enhanced by CO_2 which preferentially dissolves in kerogen over methane [60].

In this report we extend our measurements of heptane-saturated kerogen isolates from an immature Kimmeridge shale outcrop. We show that the T_1/T_2 ratio of the inter-granular heptane increases from $T_1/T_2 \approx 2.1 \rightarrow 6.5$ as we de-saturate the inter-granular pore-space by controlled evaporation. We posit that the only explanation for an increase in T_1/T_2 with de-saturation is due to increasing confinement and a difference in diffusive coupling for T_1 (with intermediate coupling) versus T_2 (with weak coupling). To our knowledge, this is the first report of a difference in diffusive-coupling for T_1 versus T_2 . T_2 - T_2 exchange maps concur that T_2 is in the weak diffusive-coupling, from which we quantify the characteristic exchange time between the dissolved (i.e. the less producible) heptane and the inter-granular (i.e. the more producible) heptane. Our findings indicate that confinement together with diffusive-coupling play a crucial role in the large range $(T_1/T_2)_{>} \approx 3 \leftrightarrow 6$ previously reported for light producible hydrocarbons in shale [20,32].

2. Methodology

2.1. Kerogen preparation and characterization

The Kimmeridge shale originated from the coast of Dorset in the United Kingdom, and is the same as in Refs. [17,38,39,61]. LECO and RockEval measurements found a total organic carbon of $\text{TOC} \approx 10.1\text{ wt\%}$ (or $\approx 17.6\text{ vol\%}$ equivalently), a hydrogen index of $\text{HI} \approx 684\text{ mg-HC/g-TOC}$, an oxygen index of $\text{OI} \approx 12\text{ mg-CO}_2/\text{g-TOC}$, and a $T_{\text{max}} \approx 424^\circ\text{C}$, indicating Type II immature kerogen.

The isolation procedure was carried out at Stratum Reservoir as such:

1. Extraction of “free oil” (and some bitumen) with dichloromethane (DCM)
2. Acid digestion of calcite and dolomite with hydrochloric acid (HCl)
3. Multiple rinses with tap water
4. Acid digestion of silicate with hydrofluoric acid (HF)
5. Multiple rinses with tap water
6. Acid digestion with HCl
7. Multiple rinses with tap water
8. Sink-float separation with a ZnBr_2 solution of density 1.8 g/cm^3
9. Multiple rinses with tap water
10. Multiple rinses with deionized (DI) water
11. Acid digestion with HF
12. Multiple rinses with tap water
13. Multiple rinses with DI water
14. Freeze dry (low temperature dehydration)
15. Vacuum oven at 60°C

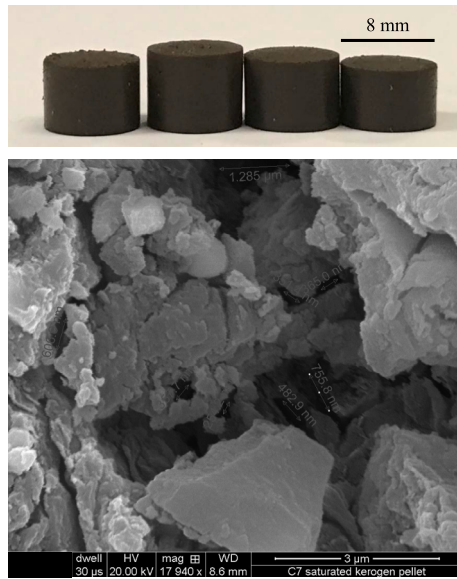


Fig. 1. Picture of kerogen pellets (top) pelletized at 77 kpsi stress; total mass of pellets is ≈ 1 g. SEM (scanning electron microscope) image of the crushed pellets after heptane saturation (bottom), taken from Ref. [17].

Inductively coupled plasma (ICP) of the isolates indicated a pyrite concentration of ≈ 8.2 wt% (or ≈ 3.5 vol% equivalently). We did not attempt to remove the pyrite using the chromous chloride method [62,63] in order to avoid contamination from paramagnetic Cr^{3+} ions on the kerogen surface. According to SEM (scanning electron microscopy) the pyrite was clustered in large $\sim 3 \mu\text{m}$ sized crystals [17], with no evidence of pyrite on the kerogen surfaces.

The kerogen powder was then further dried in a vacuum oven at 140°C at 0.3 bar for three days, then stored in a desiccator. The dry powder was then pelletized in an 8 mm diameter pellet press with an (axial) pelletizing stress of 26 kpsi, 51 kpsi, 77 kpsi, or 154 kpsi, where “kpsi” is short for 1000 psi. An example of the pellets is shown in Fig. 1 (top) at 77 kpsi, where the total dry mass was ≈ 1 g. The bulk density of the dry pellets $\rho_{b,0} = M_{\text{dry}}/BV_{\text{un-swollen}}$ was determined from the mass of the dry pellets M_{dry} , and the bulk volume of the un-swollen pellets before heptane saturation $BV_{\text{un-swollen}}$. The results listed in Table 1 indicate $\rho_{b,0} \approx 1.15 \text{ g/cm}^3$ at 51 kpsi, 77 kpsi, and 154 kpsi stress, and lower $\rho_{b,0} \approx 0.9 \text{ g/cm}^3$ at 26 kpsi stress. This indicates that a critical stress for compaction lies between 26 kpsi and 51 kpsi.

The pellets were then dried again in the vacuum oven, then NMR T_2 was measured on the dry pellets. As detailed in Ref. [39], the dry pellets had negligible NMR signal compared to background noise, indicating negligible moisture on the kerogen grains, and negligible signal from the kerogen or remaining bitumen. The pellets were then vacuum-saturated by applying a high vacuum for 2 h followed by saturation with *n*-heptane at 5000 psia hydrostatic pressure for 4 days.

Also included in this study are previous results after bitumen

extraction of the kerogen powder. The data in this report corresponds to “Extracted B” in Ref. [39], where extraction with toluene was carried out for 24 days before pelletizing and saturating with heptane.

The bulk density of the dry swollen pellets $\rho_b = M_{\text{dry}}/BV_{\text{swollen}}$ was determined from the mass of the dry pellets before heptane saturation M_{dry} , and the bulk volume of the swollen pellets after heptane saturation BV_{swollen} . The swelling factor δBV was then computed as such:

$$\delta BV = \frac{\rho_{b,0} - \rho_b}{\rho_b}. \quad (1)$$

As listed in Table 1, samples at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress indicated large swelling factor of $\delta BV \approx 50\%$, which is a direct consequence of heptane dissolving in the kerogen grains [39,60]. The swelling was less $\delta BV \approx 40\%$ for the pellet formed at 26 kpsi stress, indicating a critical stress for compaction between 26 kpsi and 51 kpsi. The swelling was even less for the bitumen extracted sample $\delta BV \approx 23\%$, which is due to less dissolved heptane in the bitumen-extracted kerogen-grains (see below). We note that most of swelling took place along the axial direction of the pellet, i.e. in the direction of the axial stress during pelletization. We also note that the pellets kept the same bulk volume, i.e. they did not “de-swell”, after evaporation of heptane.

N_2 gas adsorption data at 77 K and BET (Brunauer, Emmett, Teller) analysis was performed on the pellets after completion of the heptane-saturated NMR measurements. The pellets were crushed into powder and dried at 110°C for 16 h before the gas-adsorption measurement of specific surface area S_{BET} . The inter-granular pore diameter $d_>$ was then determined using the following relation, which assumes spherical pores [39]:

$$\frac{\rho_b S_{\text{BET}}}{\phi_>} = \frac{6}{d_>}, \quad (2)$$

where $\phi_>$ is the inter-granular porosity according to NMR T_1 - T_2 (see below). Implicit in Eq. (2) is that S_{BET} only accounts for the surface area of the kerogen grain, in other words N_2 has limited solubility in kerogen at 77 K. Error analysis of the gas-adsorption data indicated that variation in S_{BET} between samples (Table 1) was within experimental uncertainties. As such, we use the average value $S_{\text{BET}} \approx 14 \text{ m}^2/\text{g}$ when computing $d_>$ in Eq. (2) for all the samples. As shown in Table 1, the pore diameter is $d_> \approx 200 \text{ nm}$, except of the 26 kpsi sample where $d_> \approx 300 \text{ nm}$, indicating a critical stress for compaction between 26 kpsi and 51 kpsi.

The grain diameter d_g is given by the following relation, which assumes spherical grains:

$$d_g = \frac{1 - \phi_>}{\phi_>} d_>. \quad (3)$$

We find that the average grain diameter for all the samples $d_g \approx 340 \text{ nm}$ is consistent with SEM images of the crushed pellets after heptane saturation shown in Fig. 1(bottom).

2.2. NMR measurement

The NMR data were acquired on an Oxford Instruments GeoSpec2 rock-core analyzer at a resonance frequency of $f_0 = 2.3 \text{ MHz}$ (similar to wireline NMR logging tools) using a customized 18 mm probe. The pellets were placed in a sealed glass tube, and left for one hour in the spectrometer to equilibrate to 30°C before data acquisition. The 2D T_1 - T_2 data were acquired with 32 log-spaced inversion recovery steps ranging from 0.1 to 1000 ms for T_1 encoding, followed by a CPMG (Carr, Purcell, Meiboom, Gill) train with an echo spacing of $t_E = 0.1 \text{ ms}$ for T_2 encoding. The 2D T_2 - T_2 data were acquired with 32 log-spaced decay times ranging from 0.1 to 500 ms for encoding T_{2a} , followed by storage time ranging from $t_\Delta = 0.1 \text{ ms}$ to 30 ms, followed by a CPMG train with $t_E = 0.1 \text{ ms}$ for encoding T_{2b} .

Table 1

Results of pore-size analysis including bulk density of dry un-swollen pellet ($\rho_{b,0}$), bulk density of dry swollen pellet (ρ_b), swelling of pellet bulk volume (δBV), inter-granular porosity from NMR ($\phi_>$), BET surface area (S_{BET}), inter-granular pore diameter ($d_>$), and grain diameter (d_g).

Sample	$\rho_{b,0}$ (g/cm ³)	ρ_b (g/cm ³)	δBV (%)	$\phi_>$ (pu)	S_{BET} (m ² /g)	$d_>$ (nm)	d_g (nm)
26 kpsi	0.90	0.64	40	53	17	360	320
51 kpsi	1.12	0.73	53	34	14	200	390
77 kpsi	1.14	0.78	47	37	13	210	350
154 kpsi	1.16	0.75	55	32	13	190	390
Extr. 77 kpsi	1.21	0.99	23	40	12	180	260

The 2D maps were processed using an inverse Laplace transform [64,65] with 120 log-spaced bins ranging from 10^{-2} ms to 10^3 ms for both T_1 and T_2 . Porosity is reported in “pu”, i.e. porosity units, or percent of bulk volume, equivalently, where the NMR hydrogen index of $HI = 1.0028$ for *n*-heptane is taken into account. The units of the relaxation distributions are in pu/div, i.e., pu per division, where for instance, for T_2 , “div” = $\log_{10}(T_{2,i+1}) - \log_{10}(T_{2,i})$ is independent of the bin index i . Saturation is reported in “su”, i.e. saturation units, or, percent of inter-granular pore-volume, equivalently. Note that while porosity and saturation are reported in “pu” and “su” units respectively, all the equations use porosity and saturation in fractions. The amplitude of the exchange peaks in the T_2 - T_2 map were determined by integrating the off-diagonal peaks using a polygon to encircle the exchange peaks, thereby limiting interference from the on-diagonal peaks.

There was no NMR signal on the scale of the plots for the dry kerogen pellets at $t_E = 0.1$ ms [39]. Increasing the echo-spacing from $t_E = 0.1$ ms to 0.2 ms did not alter the T_2 distribution of the heptane-saturated and partially-saturated pellets above $T_2 \geq 0.2$ ms, implying no internal magnetic-field gradient effects on T_2 [17].

At ambient conditions, *n*-heptane has a bulk relaxation time of $T_{1B} = T_{1B} = 1910$ ms (not de-oxygenated), a diffusion coefficient of $D_0 = 3.43 \mu\text{m}^2/\text{ms}$, a viscosity of $\eta = 0.390$ cP, and a density of $\rho_{C7} = 0.680$ g/cm³. All the measured relaxation times are shorter than ≤ 80 ms, implying that bulk relaxation is negligible and therefore $T_{1,2}$ are dominated by surface-relaxation. As such, the customary subscript S for $T_{1,2}$ indicating surface-relaxation is implied and omitted for clarity.

3. Results

3.1. T_1 - T_2 of fully saturated pellets

The T_1 - T_2 correlation maps of the heptane-saturated kerogen pellets are shown in Fig. 2 for different pelletizing stress, plotted as T_1 versus T_2 in (a) and T_1/T_2 versus T_2 in (b). The cutoff separating the inter-granular heptane from the dissolved heptane is chosen to be $T_2 = 1.7$ ms (dashed line). The inter-granular porosity $\phi_>$ above $T_2 > 1.7$ ms, the intra-granular porosity $\phi_<$ below $T_2 < 1.7$ ms, and the total NMR porosity ϕ_t (listed in the legend) are related by:

$$\phi_t = \phi_< + \phi_> \quad (4)$$

$$\phi_\mu = \frac{\phi_<}{1 - \phi_>}, \quad (5)$$

and ϕ_μ is the “micro-porosity”, i.e. the dissolved-heptane porosity of the kerogen grain. Fig. 3 (left) shows an illustration of three kerogen grains (black) of size $d_g \approx 340$ nm, dissolved heptane in the grains (green), and the inter-granular heptane (red).

Results of the dissolved heptane are listed in Table 2. We find that the dissolved heptane porosity is $\phi_\mu \approx 13$ pu at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress. $\phi_\mu \approx 10$ pu is slightly lower at 26 kpsi, which may be due to experimental uncertainties. The bitumen extracted kerogen has significantly less dissolved heptane $\phi_\mu \approx 4$ pu, which indicates that heptane dissolves to a greater extent in bitumen than kerogen [39]. $T_{1\mu} \approx 16$ ms, $T_{2\mu} \approx 0.23$ ms, and $(T_1/T_2)_\mu \approx 70$ is found for all samples, indicating that the organic nano-pores in the kerogen/bitumen are the same size, as expected. The slight difference for 26 kpsi is likely due to uncertainties.

Results of the inter-granular heptane are listed in Table 3. We find that the inter-granular porosity is $\phi_> \approx 36$ pu at 51 kpsi, 77 kpsi, and 154 kpsi pelletizing stress, which is consistent with a random packing of mono-disperse spheres. $\phi_> \approx 53$ pu is higher at 26 kpsi, indicating a critical stress for compaction between 26 kpsi and 51 kpsi. $T_{1>} \approx 35$ ms, $T_{2>} \approx 15$ ms and $d_{>} \approx 200$ nm are the same at 51 kpsi, 77 kpsi, and 154 kpsi. $T_{1>} \approx 82$ ms and $T_{2>} \approx 39$ ms are larger for 26 kpsi, which is a result of larger pore-size $d_{>} \approx 300$ nm. $T_{1>} \approx 80$ ms and $T_{2>} \approx 35$ ms are larger for the bitumen extracted

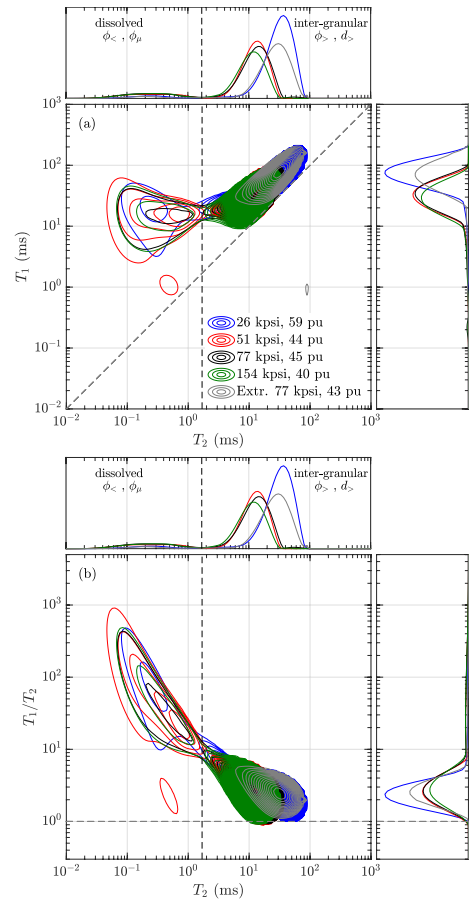


Fig. 2. (a) T_1 - T_2 correlation map for heptane-saturated kerogen pellets made using different pelletizing stresses. “Extr.” refers to kerogen with additional bitumen extraction using toluene. Subplots show 1D projections. Legend shows total NMR porosity ϕ_t in pu (i.e. porosity units). Cutoff at $T_2 = 1.7$ ms separates dissolved heptane from inter-granular heptane. Dry kerogen has negligible signal on the scale of this plot. (b) Same data plotted as a T_1/T_2 - T_2 correlation map.

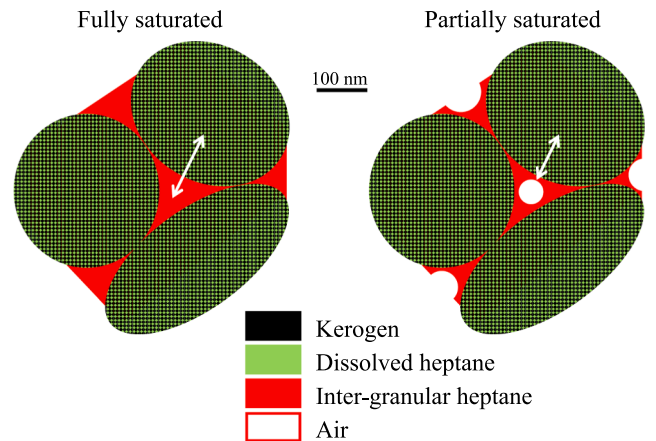


Fig. 3. Illustration of three kerogen grains (black), dissolved heptane in the kerogen (green), and inter-granular heptane (red) for fully saturated inter-granular pores (left), and partially-saturated inter-granular pores (right) with air (white) assuming heptane is the wetting fluid. Arrows illustrate the exchange between inter-granular and dissolved heptane.

sample, which is a result of less dissolved heptane (i.e. lower ϕ_μ) together with diffusive-coupling effects [39].

One can postulate from Fig. 2(a) that T_1 may be in the intermediate

Table 2

Results for dissolved heptane including intra-granular porosity $\phi_{<}$ taken below $T_2 < 1.7$ ms, and micro-porosity (i.e. amount of dissolved heptane) in the kerogen grain ϕ_{μ} , plus $T_{2\mu}$ and $(T_1/T_2)_{\mu}$ taken at the mode (i.e. peak) of the 2D map in Fig. 2, all as a function of pelletizing stress and for bitumen extracted “Extr.” kerogen.

Sample	$\phi_{<}$ (pu)	ϕ_{μ} (pu)	$T_{1\mu}$ (ms)	$T_{2\mu}$ (ms)	$(T_1/T_2)_{\mu}$
26 kpsi	7	10	21	0.21	104
51 kpsi	10	13	17	0.23	73
77 kpsi	8	12	15	0.23	67
154 kpsi	9	12	15	0.23	64
Extr. 77 kpsi	2	4	17	0.24	73

Table 3

Results for inter-granular heptane, including inter-granular porosity $\phi_{>}$ taken above $T_2 > 1.7$ ms, plus $T_{2>}$ and $(T_1/T_2)_{>}$ taken at the mode (i.e. peak) of the 2D map in Fig. 2, and the inter-granular pore-size $d_{>}$ (Eq. (3)), all as a function of pelletizing stress and for bitumen extracted “Extr.” kerogen.

Sample	$\phi_{>}$ (pu)	$T_{1>}$ (ms)	$T_{2>}$ (ms)	$(T_1/T_2)_{>}$	$d_{>}$ (nm)
26 kpsi	53	82	39	2.1	300
51 kpsi	34	35	14	2.4	200
77 kpsi	37	36	15	2.4	220
154 kpsi	32	31	13	2.5	200
Extr. 77 kpsi	40	80	35	2.4	200

diffusive-coupling regime due to the partial overlap between the inter-granular peak and the dissolved peak. One can also postulate from Fig. 2(a) that T_2 may be in the weak diffusive-coupling regimes due to the large separation of the inter-granular peak from the dissolved peak. While these observations from Fig. 2 are only postulates, i.e. they may be just coincidences, the increase in $(T_1/T_2)_{>}$ with de-saturation (see below) can only be explained by diffusive-coupling.

3.2. T_1 - T_2 of partially saturated pellets

The heptane-saturated kerogen pellets were de-saturated by controlled evaporation to air. This was achieved by opening the glass tube for a short period and allowing the heptane to partially evaporate. The effect of de-saturation on the 1D projections of the T_1 - T_2 maps are shown in Fig. 4 for the kerogen at 51 kpsi pelletizing stress. The saturation of the inter-granular pore-space is determined as such:

$$S_{>} = \frac{\phi_t(S_{>}) - \phi_{<}(1)}{\phi_t(1) - \phi_{<}(1)}, \quad (6)$$

where $\phi_t(1)$ and $\phi_{<}(1)$ refer to the total and intra-granular NMR porosities, respectively, at $S_{>} = 100$ su. The above definition for $S_{>}$ avoids having to use a cutoff at $T_2 = 1.7$ ms, which becomes inaccurate below $S_{>} \lesssim 30$ su as $T_{2>}$ approaches $T_2 = 1.7$ ms.

Fig. 4(a) and (b) clearly show that the distribution shifts to shorter T_1 and T_2 with decreasing $S_{>}$, respectively. Moreover, the shift in $T_{2>}$ (bullets) is more pronounced than $T_{1>}$ (bullets), which causes $(T_1/T_2)_{>}$ (bullets) to increase with decreasing $S_{>}$, as shown in Fig. 4(c). The increase in $(T_1/T_2)_{>}$ with decreasing $S_{>}$ poses a conundrum, which to the authors' knowledge has never been reported before. The conventional interpretation of surface relaxation in the fast-diffusion regime [66] is given by the following relation:

$$\frac{1}{T_{1>}} = \rho_{1>} \frac{6}{S_{>} d_{>}} \quad (7)$$

$$\frac{1}{T_{2>}} = \rho_{2>} \frac{6}{S_{>} d_{>}}, \quad (8)$$

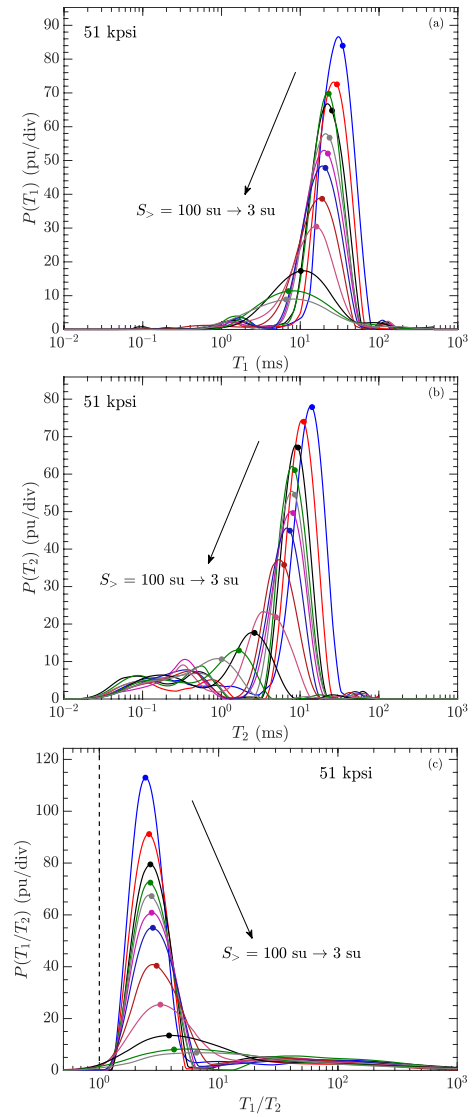


Fig. 4. (a) T_1 distributions (b) T_2 distributions and (c) T_1/T_2 distributions for heptane-saturated kerogen isolates pelletized at 51 kpsi stress, as a function of saturation of the inter-granular pore-space $S_{>}$ defined in Eq. (6). Bullets on the distributions correspond to modes (i.e. peaks) of the 2D T_1 - T_2 maps, defined as $T_{1>}$, $T_{2>}$ and $(T_1/T_2)_{>}$, respectively.

where the volume-to-surface ratio as a function of saturation is expressed as $S_{>} d_{>}/6$ since $d_{>}$ is computed assuming spherical pores (Eq. (2)). The surface-relaxivities $\rho_{1>}$ and $\rho_{2>}$ should not depend on saturation $S_{>}$, i.e. they are only functions of the fluid-surface interactions [66]. The resulting $(T_1/T_2)_{>}$ ratio should then be:

$$\left(\frac{T_1}{T_2}\right)_{>} = \frac{\rho_{2>}}{\rho_{1>}} \quad (9)$$

which should not depend on $S_{>}$. The only way for $(T_1/T_2)_{>}$ to depend on $S_{>}$ is for $\rho_{1>}$ and $\rho_{2>}$ to be (different) functions of $S_{>}$. We posit that the decrease in $S_{>}$ causes $T_{1>}$ to go from intermediate $\rightarrow$ strong diffusive-coupling, and $T_{2>}$ to go from weak $\rightarrow$ intermediate diffusive-coupling. Such changes in the extent of the diffusive-coupling cause $\rho_{1>}$ and $\rho_{2>}$ to depend on saturation $S_{>}$ (see below), and moreover for $\rho_{1>}$ and $\rho_{2>}$ to be different functions of $S_{>}$. We note that a semi-quantitative account of the effects of $S_{>}$ on $T_{2>}$ in Ref. [1] are in line with those reported here.

Fig. 5(a) shows the dependence of $(T_1/T_2)_{>}$ on $S_{>}$. All the samples show a monotonic increase in $(T_1/T_2)_{>}$ with decreasing $S_{>}$. The 51 kpsi

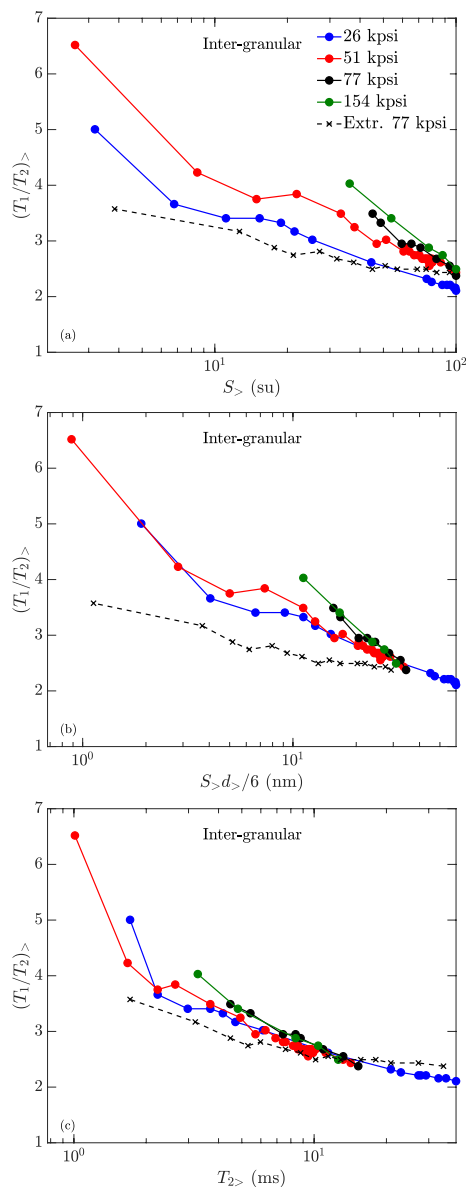


Fig. 5. $(T_1/T_2)_>$ of the inter-granular heptane in the kerogen pellets, as a function of inter-granular (a) saturation $S_>$ in su (i.e. saturation units), (b) volume-to-surface ratio as a function of saturation $S_> d_>/6$, and (c) $T_{2>}$. Data is shown for kerogen pellets made using different pelletizing stresses, and bitumen extracted “Extr.” kerogen.

sample, which has the most de-saturation data available, shows an increase from $(T_1/T_2)_> \approx 2.4 \rightarrow 6.5$ as saturation decreases from $S_> \approx 100 \rightarrow 3$ su. 77 kpsi and 154 kpsi agree with 51 kpsi over the limited range shown. 26 kpsi shows a similar trend, but with a lower $(T_1/T_2)_>$ at a given $S_>$. The bitumen extracted data has the least increase in $(T_1/T_2)_>$ with decreasing $S_>$, which corroborates with the weakest diffusive-coupling due to less dissolved heptane ($\phi_\mu \approx 4$ pu) [39].

As illustrated for the partially saturated case in Fig. 3 (right), in the presence of air the volume-to-surface ratio of the inter-granular space is $S_> d_>/6$, provided heptane remains the wetting fluid [15,67,68]. A large volume-to-surface ratio corresponds to the least confinement, while a small volume-to-surface ratio corresponds to the most confinement. Fig. 5(b) shows the dependence of $(T_1/T_2)_>$ on the volume-to-surface ratio $S_> d_>/6$ of the inter-granular space. Plotting against $S_> d_>/6$ causes the 26 kpsi data to collapse onto the 51 kpsi, 77 kpsi and 154 kpsi data, which suggests that the volume-to-surface ratio is

the underlying variable in $(T_1/T_2)_>$. In other words, decreasing the volume-to-surface ratio (i.e. increasing confinement) in the inter-granular pores enhances the diffusive-coupling effects for both $T_{1>}$ and $T_{2>}$. A similar trend is expected when decreasing the inter-granular pore-size $d_>$ with fixed $S_> = 100$ su, as previously been reported for $T_{2>}$ [3].

Finally, Fig. 5(c) shows the dependence of $(T_1/T_2)_>$ on $T_{2>}$, where all the 26 kpsi, 51 kpsi, 77 kpsi and 154 kpsi data collapse onto one curve. From Figs. 5(b) and (c) we can conclude that inter-granular pores with the least confinement have $T_{2>} \approx 40$ ms and $(T_1/T_2)_> \approx 2.1$, while inter-granular pores with the most confinement have $T_{2>} \approx 1$ ms and $(T_1/T_2)_> \approx 6.5$. This range is consistent with the range $(T_1/T_2)_> \approx 3 \leftrightarrow 6$ postulated for light producible hydrocarbons in shale [20,32]. We have therefore shown that the range $(T_1/T_2)_> \approx 3 \leftrightarrow 6$ for light hydrocarbons in shale is caused by the degree of confinement of the inter-granular (i.e. the more producible) pores, together with diffusive-coupling with dissolved (i.e. the less producible) hydrocarbons in the kerogen. Specifically, we find that nano-confinement of the order $S_> d_>/6 \approx 1$ nm causes the producible hydrocarbons to have large $(T_1/T_2)_> \approx 6.5$.

We note that while the above measurements and interpretation are for heptane/air saturated inter-granular pores, the heptane response should in principle be similar for heptane/water saturated inter-granular pores [1], provided heptane remains the wetting fluid [15,67,68]. In other words, these results are in principle applicable to hydrocarbon/water saturated shales.

3.3. $\rho_{1>}$ and $\rho_{2>}$ of partially saturated pellets

Fig. 6 shows the surface relaxivities $\rho_{1>}$ and $\rho_{2>}$ determined from Eqs. (7) and (8), respectively. Above $S_> \gtrsim 30$ su, $\rho_{1>}$ is roughly constant while $\rho_{2>}$ increases with decreasing $S_>$, which causes $(T_1/T_2)_>$

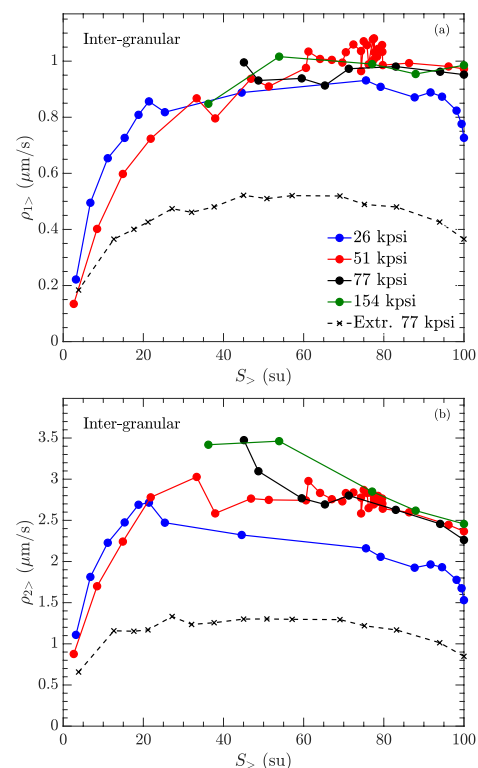


Fig. 6. Surface relaxivities (a) $\rho_{1>}$ and (b) $\rho_{2>}$, defined in Eqs. (7) and (8) respectively, of the inter-granular heptane in the kerogen pellets, as a function of inter-granular saturation $S_>$ in su (i.e. saturation units). Data is shown for kerogen pellets made using different pelletizing stresses, and bitumen extracted “Extr.” kerogen.

(Eq. (9)) to increase with decreasing S_v . Below $S_v \lesssim 30$ su, both $\rho_{1>}$ and $\rho_{2>}$ decrease with decreasing S_v . The decrease below $S_v \lesssim 30$ su is more pronounced for $\rho_{1>}$ than for $\rho_{2>}$, which causes $(T_1/T_2)_{>}$ to increase with decreasing S_v . In other words, if the extent of the decrease for $\rho_{1>}$ and $\rho_{2>}$ were the same, $(T_1/T_2)_{>}$ would be constant. Note that the cross-over at $S_v \simeq 30$ su corresponds to a volume-to-surface ratio of $S_v d_{>}/6 \simeq 10$ nm, see Fig. 5(b).

An expression for the “apparent” surface-relaxivity ρ_{2app} (which is equivalent to $\rho_{2>}$ in this report) has previously been proposed in the weak diffusive-coupling regime [1,3,5]. However, the apparent relaxivity theory for $\rho_{2>}$ does not take pore-size or saturation of the intergranular pore space into account, nor does it work in the intermediate or strong diffusive-coupling regimes. We note however that the apparent relaxivity theory does account for the lower $\rho_{1>}$ and $\rho_{2>}$ for the bitumen extracted sample at $S_v = 100$ su in Fig. 6, which is a result of less dissolved heptane $\phi_\mu \simeq 4$ pu (Table 2) in kerogen [39]. The slightly lower $\rho_{1>}$ and $\rho_{2>}$ for the 26 kpsi sample at $S_v = 100$ su could also be due to less dissolved heptane $\phi_\mu \simeq 10$ pu (Table 2), or could be from experimental uncertainties.

We also note that according to the apparent relaxivity theory, for a given ϕ_μ , a lower surface-relaxivity leads to stronger diffusive-coupling [1,3,5]. This is consistent with our interpretation of Fig. 6 where the decrease in $\rho_{1>}$ and $\rho_{2>}$ below $S_v \lesssim 30$ su is a result of stronger diffusive-coupling due to increasing confinement in the inter-granular pore-space.

We note in passing that the fast-diffusion limit [66] applies throughout, i.e. $\rho_{1>} d_{>}/D_0 \ll 1$ and $\rho_{2>} d_{>}/D_0 \ll 1$, which validates the use of Eqs. (7) and (8), respectively.

3.4. T_2 - T_2 of fully saturated pellet

T_2 -store- T_2 is a powerful technique for determining the exchange time τ_{ex} between diffusively coupled macro and micro-pores [8–13,16], or in our case inter-granular and dissolved heptane, respectively. The first requirement for observing the exchange peaks and determining τ_{ex} is weak diffusive-coupling since the inter-granular and dissolved heptane must have distinct (i.e. well separated) T_2 peaks. The second requirement is that the exchange time τ_{ex} must be comparable to or less than the average (i.e. log-mean) T_{1LM} , i.e. $\tau_{ex} \lesssim T_{1LM}$. This condition arises because there is T_1 decay during the storage time t_Δ in the T_2 -store- T_2 measurement, and the signal cannot decay to the noise level before exchange is observable.

In Fig. 7 we see that both of these conditions are fulfilled. At the shortest storage time of $t_\Delta = 0.1$ ms in Fig. 7(a), we see two distinct peaks on the diagonal, with a hint of an exchange peak. At a longer storage time of $t_\Delta = 20$ ms in Fig. 7(b), we clearly see the exchange peaks symmetrically distributed off-diagonally.

In Fig. 8(a) we plot the total NMR porosity ϕ_t as a function of t_Δ , which (in part) decays as a result of T_1 decay during storage [13]. Meanwhile, the integral of the NMR exchange peaks ϕ_{ex} increases initially up to $t_\Delta \simeq \tau_{ex,ef} \simeq 9$ ms, then decays due to T_1 decay during storage. The dissolved peak in Fig. 7(b) disappears from the on-diagonal location at $T_{2\mu} \simeq 0.23$ ms, which according to Ref. [12] is a direct consequence of T_1 decay during storage.

In Fig. 8(b) we plot the ratio ϕ_{ex}/ϕ_t as a function of t_Δ , and we find an exponential-like recovery. A simplified expression to the t_Δ dependence in ϕ_{ex}/ϕ_t is given by [13]:

$$\frac{\phi_{ex}}{\phi_t} = A \left[1 - \exp\left(-\frac{t_\Delta}{\tau_{ex,ef}}\right) \right] \quad (10)$$

$$\frac{1}{\tau_{ex,ef}} = \frac{1}{T_{1LM}} + \frac{1}{\tau_{ex}}. \quad (11)$$

The two free parameters in the fit are the effective exchange time $\tau_{ex,ef} \simeq 9$ ms and asymptotic fraction of the exchange-peak porosity

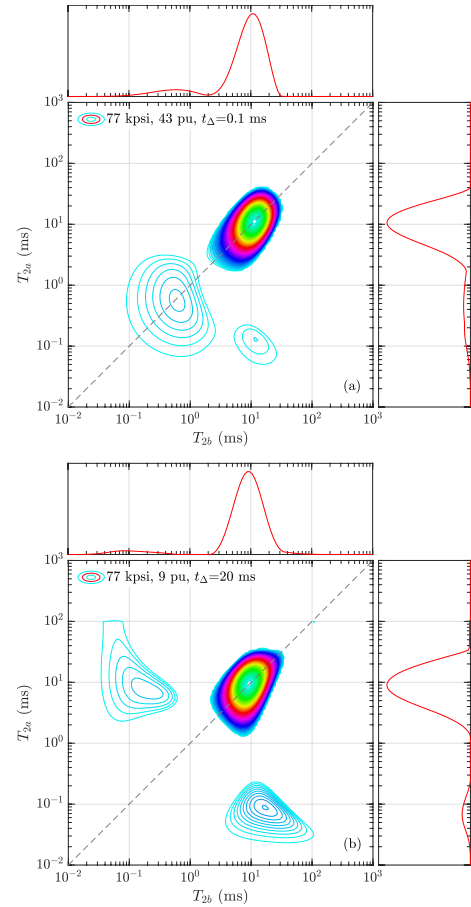


Fig. 7. T_2 - T_2 exchange maps for heptane-saturated kerogen isolates pelletized at 77 kpsi stress, using a storage time of (a) $t_\Delta = 0.1$ ms, and (b) $t_\Delta = 20$ ms, all fully saturated ($S_v = 100$ su). Subplots show 1D projections. Legend shows total NMR porosity ϕ_t in pu (i.e. porosity units). Dry kerogen has negligible signal on the scale of this plot.

$A \simeq 13\%$, which are determined using non-linear least squares. According to Eq. (11) [8], the intrinsic exchange time $\tau_{ex} = 15$ ms can then be derived from $\tau_{ex,ef} \simeq 9$ ms and $T_{1LM} \simeq 27$ ms.

An estimate of the strength in diffusive-coupling can then be estimated from the following expressions [14]:

$$C_1 = \frac{T_{1\mu}}{\tau_{ex}} \simeq \frac{16 \text{ ms}}{15 \text{ ms}} \simeq 1 \quad (\text{intermediate}) \quad (12)$$

$$C_2 = \frac{T_{2\mu}}{\tau_{ex}} \simeq \frac{0.23 \text{ ms}}{15 \text{ ms}} \simeq 0.02 \quad (\text{weak}) \quad (13)$$

where C_1 is the strength in diffusive coupling for T_1 , and C_2 is the strength in diffusive coupling for T_2 . According to Ref. [14], $C_{1,2} < 0.1$ indicates weak diffusive-coupling, and $C_{1,2} > 10$ indicates strong diffusive-coupling, with intermediate diffusive-coupling for $0.1 < C_{1,2} < 10$. Eqs. (12) and (13) therefore indicate intermediate diffusive-coupling for T_1 , and weak diffusive-coupling for T_2 . This is in agreement with the qualitative observations from Fig. 2(a) where inter-granular and dissolved heptane have partially overlapping peaks in T_1 (i.e. intermediate coupling), and distinct peaks in T_2 (i.e. weak coupling).

Another interpretation for the exchange peaks in T_2 - T_2 maps is magnetization transfer between kerogen/bitumen (with short T_2) and heptane (with long T_2) [16]. We note however that in Ref. [16], the short T_2 peak from kerogen/bitumen is present before heptane-saturation, i.e. it is present in the “as-received” shale. In our kerogen isolates, there is no NMR signal in the T_1 - T_2 or T_2 - T_2 maps before heptane

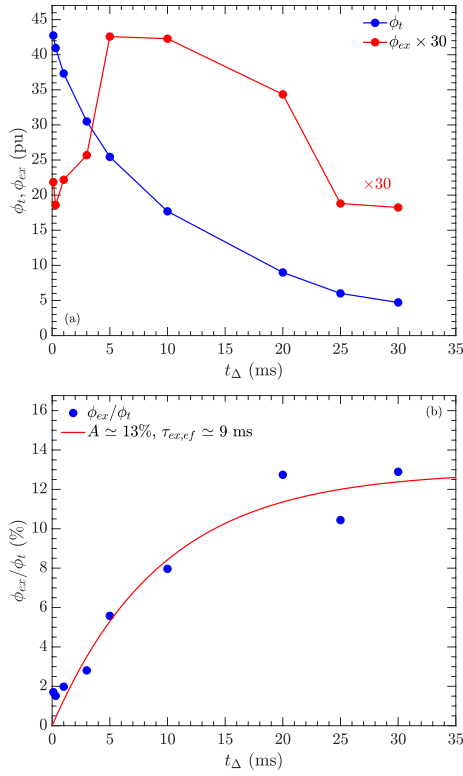


Fig. 8. (a) Total NMR porosity (ϕ_t), and porosity of off-diagonal NMR exchange peaks (ϕ_{ex}) multiplied by $\times 30$ for clarity, as a function of storage time t_{Δ} . (b) ϕ_{ex}/ϕ_t ratio (bullets) as a function of storage time t_{Δ} , and fit using Eq. (10) (curve).

saturation, implying that the short T_2 peak in Figs. 2 and 7 is heptane dissolved in kerogen. We also note that in our heptane-saturated kerogen isolates, there is no ambiguity about which fluid (i.e. water or hydrocarbon) is saturating which pore-type (i.e. organic or inorganic).

From mass balance [17], we can also conclude that the short T_2 peak in Figs. 2 and 7 is not bitumen which becomes visible (i.e. its T_2 becomes detectable) because its viscosity is lowered by dissolved heptane. As such, our interpretation of the exchange peaks in Fig. 7 is diffusive coupling, and not magnetization transfer.

As a consistency check, we calculate the penetration depth $L_{D,ex}$, or restricted diffusion [31,69] length, of heptane in kerogen during exchange time $\tau_{ex} = 15 \text{ ms}$:

$$L_{D,ex} = \sqrt{D_0 \tau_{ex} \phi_{\mu}^{m-1}} \approx 65 \text{ nm} \approx d_g/5 \quad (14)$$

where $m \approx 5.6$ is the Archie cementation exponent of dissolved heptane in kerogen [39]. We note that in Ref. [39] a cementation exponent of $m \approx 6.3$ was derived using the log-means of T_1 and T_2 , however using the mode (i.e. peak) of the 2D T_1 - T_2 instead results in $m \approx 5.6$.

For a spherical grain of diameter $d_g \approx 340 \text{ nm}$, Eq. (14) suggests that heptane penetrates $L_{D,ex} \approx 65 \text{ nm}$ into the grain surface during time τ_{ex} , which corresponds to $(d_g^3 - (d_g - 2L_{D,ex})^3)/d_g^3 \approx 75\%$ of the grain volume. In other words, $\approx 75\%$ of the kerogen grain, i.e. the majority of the kerogen, is penetrated during $\tau_{ex} \approx 15 \text{ ms}$, which is reasonable for the definition of an exchange time. The arrow in Fig. 3 (left) illustrates the exchange between inter-granular and dissolved heptane, and the penetration depth $L_{D,ex} \approx 65 \text{ nm}$ (i.e. the length of the arrow) into the kerogen grain during time τ_{ex} .

A similar analysis to Eq. (14) indicates that the penetration depth during time $T_{1\mu} \approx 16 \text{ ms}$ is given by:

$$L_{D,1} = \sqrt{D_0 T_{1\mu} \phi_{\mu}^{m-1}} \approx 68 \text{ nm} \quad (15)$$

$$L_{D,2} = \sqrt{D_0 T_{2\mu} \phi_{\mu}^{m-1}} \approx 8 \text{ nm} \quad (16)$$

In other words, $\approx 77\%$ of the kerogen grain is penetrated during $T_{1\mu}$, which is reasonable for intermediate diffusive-coupling for T_1 . On the other hand, the penetration depth during time $T_{2\mu} \approx 0.23 \text{ ms}$ is small $L_{D,2} \approx 8 \text{ nm}$, where only $\approx 13\%$ of the kerogen grain is penetrated, which is consistent with weak diffusive-coupling for T_2 .

Fig. 3 (right) illustrates the effect of partial saturation of the inter-granular pore-space, provided heptane is the wetting fluid. The exchange time τ_{ex} between inter-granular and dissolved state will shorten with decreasing $S_{>}$ since the time needed to diffuse from the center of the inter-granular pore to the grain surface will shorten (see shortening of arrows in the inter-granular pore). According to Eqs. (12) and (13), a shorter τ_{ex} leads to stronger diffusive-coupling for both T_1 and T_2 . This trend is consistent with the decrease in $\rho_{1>}$ and $\rho_{2>}$ with decreasing $S_{>}$ in Fig. 6, which is also interpreted as due to stronger diffusive-coupling.

3.5. Effects of reservoir conditions

All of the above data were acquired at laboratory conditions of 30°C and 1 atmosphere. It is worth postulating what the above results would be under reservoir conditions, as measured by an NMR logging tool downhole. Reservoir conditions will typically involve higher temperature, higher pressures, and multiple components in the reservoir oil. In the case of shale oil with a high gas-to-oil ratio, the reservoir oil will have components lighter than heptane such as methane and natural gas liquids.

It is therefore reasonable to expect that reservoir conditions will decrease viscosity η and increase the bulk diffusion coefficient D_0 of the producible oil, given that $D_0 \propto (\eta/T)^{-1}$ [70]. Given the relation $L_{D,ex} \approx d_g/5$ (Eq. (14)), this will tend to decrease the exchange time $\tau_{ex} \propto 1/D_0$ between the dissolved and inter-granular fluid, resulting in larger $C_1 \propto D_0$ and $C_2 \propto D_0$ coefficients (Eqs. (12) and (13)), and therefore stronger diffusive coupling for both T_1 and T_2 . By analogy with viscous fluids [57], we also postulate that $T_{1\mu}$ of the dissolved hydrocarbon is roughly independent of η/T and D_0 , while $T_{2\mu} \propto (\eta/T)^{-1/2} \propto D_0^{1/2}$, resulting in $C_1 \propto D_0$ and $C_2 \propto D_0^{3/2}$. In other words, reservoir conditions will result in stronger diffusive-coupling for both T_1 and T_2 , thereby increasing $T_{1>}$ and $T_{2>}$ for inter-granular pores, perhaps more so for $T_{2>}$.

Another effect of reservoir conditions is that the reservoir oil will have components lighter than heptane, which will decrease the dissolved porosity ϕ_{μ} since lighter components have lower solubility in kerogen [60]. One also expects lower ϕ_{μ} with increasing kerogen maturity [60]. The effect of lower ϕ_{μ} will be similar to the bitumen extracted results, although probably not as severe.

4. Conclusions

We present two evidence for diffusive-coupling effects between inter-granular heptane and heptane dissolved in the isolated kerogen grains of immature Type II kerogen from the Kimmeridge. The first evidence is from T_1 - T_2 correlation maps as a function of saturation, and the second is from T_2 - T_2 exchange maps.

From T_1 - T_2 correlation maps we find that $(T_1/T_2)_{>}$ for inter-granular heptane increases from $(T_1/T_2)_{>} \approx 2.1 \rightarrow 6.5$ with decreasing saturation $S_{>} = 100 \rightarrow 3 \text{ su}$ (saturation units) of the inter-granular pore-space. To the authors' knowledge, this phenomenon has never been reported before, and can only be explained if T_1 and T_2 are in different diffusive-coupling regimes; namely, if T_1 is in the intermediate diffusive-coupling regime while T_2 is in the weak diffusive-coupling regime.

Using the volume-to-surface ratio as a function of saturation of the inter-granular pores ($S_{>} d_{>}/6$), we show that inter-granular pores with $S_{>} d_{>}/6 \approx 60 \text{ nm}$ (i.e. the least confinement) have small $(T_1/T_2)_{>} \approx 2.1$, while inter-granular pores with $S_{>} d_{>}/6 \approx 1 \text{ nm}$ (i.e.

the most confinement) have large $(T_1/T_2)_> \approx 6.5$. Similar behavior is expected when decreasing the inter-granular pore-size $d_>$ with fixed $S_> = 100$ su. These results are consistent with the range $(T_1/T_2)_> \approx 3 \leftarrow 6$ previously observed for light producible hydrocarbons in shale, which we conclude is a function of the degree of confinement for the inter-granular (i.e. the more producible) hydrocarbons, together with diffusive-coupling effects with the dissolved (i.e. the less producible) hydrocarbons in kerogen.

From T_2 - T_2 exchange maps we find off-diagonal exchange peaks as the storage time is increased, which is a evidence for weak diffusive-coupling for T_2 . Our analysis shows that the characteristic exchange time for heptane to diffuse between inter-granular pores and the dissolved state is $\tau_{ex} \approx 15$ ms. Using the dimensionless constants $C_1 = T_{1\mu}/\tau_{ex}$ and $C_2 = T_{2\mu}/\tau_{ex}$, where $T_{1\mu}$ and $T_{2\mu}$ are the relaxation times of dissolved heptane, we confirm that T_1 is in the intermediate diffusive-coupling regime ($C_1 \approx 1$), while T_2 is in the weak diffusive-coupling regime ($C_2 \approx 0.02$). We also postulate how reservoir conditions would affect the results.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Experimental Identification of Diffusive Coupling Using 2D NMR

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Spin relaxation based nuclear magnetic resonance (NMR) methods have been used extensively to determine pore size distributions in a variety of materials. This approach is based on the assumption that each pore is in the fast diffusion limit but that diffusion between pores can be neglected. However, in complex materials these assumptions may be violated and the relaxation time distribution is not easily interpreted. We present a 2D NMR technique and an associated data analysis that allow us to work directly with the time dependent experimental data without Laplace inversion to identify the signature of diffusive coupling between different pores. Measurements on microporous glass beads and numerical simulations are used to illustrate the technique.

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NMR has been used to study a wide array of biological, geological, and synthetic porous materials [1,2]. In these systems the decay of the spin magnetization is controlled by a combination of diffusion and enhanced relaxation at the pore-solid interface. If the surface relaxation is relatively weak, then each pore will exhibit single exponential decay with a lifetime determined by its size; this is the fast diffusion regime, and the net decay signal is a sum of exponentials that reflects the distribution of pore sizes. However, these assumptions can break down, either because the surface relaxation is large enough that relaxation within a given pore involves several modes or because inter-pore diffusion is important. For example, in materials with a mixture of small and large pores, the 1D relaxation time spectrum can be either very narrow or very broad, depending on the connectivity between pores of different sizes [3]. The situation is further complicated by the presence of spatial heterogeneity of pore sizes over distances beyond the diffusion length (typically $\sim 100 \mu\text{m}$) [4]. This Letter presents a 2D NMR relaxation method to identify the diffusion regimes.

Two-dimensional NMR experiments involve a correlated measurement of longitudinal (T_1) and transverse (T_2) relaxation and a 2D correlation map (e.g., T_1 - T_2) is obtained via a 2D Laplace inversion [5–9]. In these experiments diffusive coupling and pore size distributions can exhibit qualitatively different features. For example, when all of the pores are in the fast diffusion regime, the signal from each pore will exhibit a single T_1 and T_2 value and thus appears as a single point on the 2D correlation plot. A distribution of pore sizes will lead to signals along a ridge in the 2D plot with the specific ratio $r \equiv T_1/T_2$. On the other hand, if the fast diffusion regime does not apply, the 2D T_1 - T_2 cross plot can be considerably more complicated with significant contributions away from the $T_2 = rT_1$ ridge. Indeed, it has been shown theoretically that in certain regions the signal can have small negative

amplitude peaks, a fact that severely limits the stability of the 2D Laplace inversion [7,10,11]. Therefore, a new approach to the T_1 - T_2 measurement is needed for complex systems. We introduce a way of working directly with the time-domain data that can sense the small negative signals that are associated with more complex diffusion phenomena. This technique is illustrated with a study of packed microporous glass beads where diffusion occurs between the intergranular pores and the micropores inside the grains. Our measurements and numerical simulations of the diffusion dynamics are shown to identify the presence of the negative correlation peaks. Our experiments were exclusively performed at low magnetic field (~ 500 G) to avoid the complication of higher magnetic fields [12].

The evolution of the nonequilibrium magnetization, $m(\mathbf{r}, t)$, is described by the diffusion equation [13],

$$D_0 \nabla^2 m(\mathbf{r}, t) = \frac{\partial m(\mathbf{r}, t)}{\partial t}, \quad (1)$$

subject to the boundary condition on the interface between the pore fluid and the solid grains,

$$[D_0 \hat{\mathbf{n}} \cdot \nabla m(\mathbf{r}, t) + \rho m(\mathbf{r}, t)]_S = 0. \quad (2)$$

Here D_0 is the intrinsic diffusion coefficient for the fluid saturating the pores, and the parameter ρ measures the strength of the enhanced relaxation at the interface, S . For a pore of size a , a key dimensionless parameter is $k \equiv \rho a/D_0$ [14]. In the fast diffusion regime, where the fluid molecules diffuse extensively before being relaxed at the interface, we have $k \ll 1$ and the magnetization decays as a single exponential with lifetime $T \sim a/\rho$. For $k \geq 1$ (i.e., larger pores or stronger surface relaxation) spin relaxation becomes a multiexponential process.

An example in Fig. 1 shows the T_2 spectra of two samples of randomly packed solid borosilicate glass beads of diameter 55 and 150 μm , respectively. From the measured

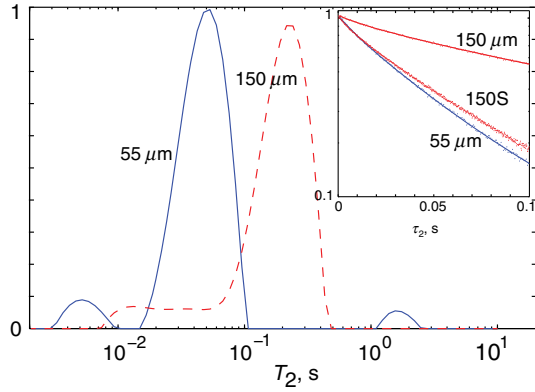


FIG. 1 (color online). T_2 spectrum for two packings of solid borosilicate beads with diameters of 55 and 150 μm . The inset shows the initial echo decay data for the two samples. The trace marked 150S is a scaled decay of the 150 μm sample to match the initial decay rate of the 55 μm sample demonstrating the higher curvature of the decay of the large-grain sample.

relaxation we estimate $k = 1.7$ and 4.4 for the two samples. Thus the 150 μm system is certainly not expected to exhibit fast diffusion behavior, while the 55 μm sample is in an intermediate regime. This is confirmed by the measured T_2 distributions; the 150 μm sample shows a more pronounced tail at short T_2 than does the 55 μm sample. Such behavior can also be seen in the time-domain decay (Fig. 1 inset) where the larger-grain data show more curvature. We emphasize that the short- T_2 tail does not directly represent the small pores; instead, it reflects the complicated diffusion dynamics in the pore space [14]. Even though this forward problem is well understood, the inverse problem, on the other hand, is much more difficult. For example, given a broad T_2 distribution from a natural material, it is impossible to know from this 1D experiment whether slow diffusion is the source of the distribution's width or if the width simply reflects a wide distribution of pore sizes.

This Letter focuses on the T_1 - T_2 correlation experiment which uses the inversion-recovery-Carr-Purcell-Meiboom-Gill pulse sequence [8],

$$\text{RD} - \pi - \tau_1 - \frac{\pi}{2} - t_e/2 - [\pi - t_e/2 - \text{Acq} - t_e/2]_N, \quad (3)$$

where N is the echo number, RD the recycle delay, and Acq represents data acquisition. The signal is obtained in a 2D matrix $\mathbf{m}(\tau_1, \tau_2 \equiv Nt_e)$. We define $\mathbf{M}(\tau_1, \tau_2) \equiv \mathbf{m}(\tau_1 \rightarrow \infty, \tau_2) - \mathbf{m}(\tau_1, \tau_2)$, which is the deviation of signal from its equilibrium. $\mathbf{M}$ is related to the T_1 - T_2 correlation function $\mathbf{S}(T_1, T_2)$ by

$$\mathbf{M}(\tau_1, \tau_2) = \int dT_1 dT_2 \mathbf{S}(T_1, T_2) \exp(-\tau_1/T_1 - \tau_2/T_2) \quad (4)$$

The formalism used by Brownstein and Tarr has been extended to characterize 2D experiments [7,10]. The magnetization solution is written as

$$m(\mathbf{r}, t) = \sum_{n=0}^{\infty} a_n \phi_n(\mathbf{r}) e^{-t/T_n}, \quad (5)$$

and the detected signal is the volume integral of m . Here ϕ_n and T_n are the eigenfunctions and eigenvalues of Eq. (1) and the boundary condition [Eq. (2)]. It is important to note that the surface relaxation parameters governing longitudinal and transverse relaxation, ρ_1 and ρ_2 ($\rho_1 < \rho_2$), are different [15]. Thus two distinct eigensystems are required to describe the T_1 - T_2 experiment. We denote these as $\{\phi_{1,n}, T_{1,n}\}$ and $\{\phi_{2,n}, T_{2,n}\}$. The difference between the two eigensystems is critical to the phenomena discussed in this Letter. The T_1 - T_2 correlation spectrum can be written as [7]

$$\begin{aligned} \mathbf{S}(T_1, T_2) = & \sum_{p,q} \langle f | \phi_{2,q} \rangle \langle \phi_{2,q} | \phi_{1,p} \rangle \langle \phi_{1,p} | i \rangle \\ & \times \delta(T_1 - T_{1,p}) \delta(T_2 - T_{2,q}), \end{aligned} \quad (6)$$

where $\langle \phi | \psi \rangle \equiv \int d^3\mathbf{r} \phi^*(\mathbf{r}) \psi(\mathbf{r})$ and $|i\rangle$ and $|f\rangle$ represent the initial and final distributions which are taken to be constant. While $\langle f | \phi \rangle$ and $\langle \phi | i \rangle$ are associated with the initial and final states, the cross terms $\langle \phi_{2,q} | \phi_{1,p} \rangle$ are related to the overlap of the eigenfunctions in the transverse and longitudinal manifolds and are not detected in 1D experiments [14]. For each value of T_1 and of T_2 , there may be a nonzero value of $\mathbf{S}(T_1, T_2)$, even if $T_1 < T_2$. This simply reflects the ability of a longitudinal mode with relaxation time T_1 to couple into a transverse mode with relaxation time T_2 . For example, terms in Eq. (6) with $p \neq q$ are referred to as off diagonal because they contribute to $\mathbf{S}(T_1, T_2)$ away from the $T_1 = T_2$ diagonal line, which is dominated by terms with $p = q = 0, 1, 2, \dots$. Figure 2 shows an example from Ref. [7], a single, one-dimensional pore with $k_1 \equiv \rho_1 a/D = 0.5$ and $k_2 \equiv \rho_2 a/D = 5$. In this case the eigenmodes were evaluated analytically and $\mathbf{S}(T_1, T_2)$ was obtained from Eq. (6), without Laplace inversion. In addition to contributions near the diagonal line, Fig. 2(a) clearly shows off-diagonal terms corresponding to $p = 0$ and $q = 1$, etc. Note that the amplitude of the off-diagonal terms is small and that the peaks above the diagonal are negative; thus, the conventional algorithm [8] cannot apply.

Small off-diagonal signals are difficult to detect because of the ill-conditioned nature of Laplace inversion techniques [8]. However, the negative character of such peaks has a signature that can be seen directly in the time domain signal without using Laplace inversion, as shown by the dashed line in Fig. 2(b). Note that, for a larger τ_2 , $\mathbf{M}$ initially increases as τ_1 increases and then decays. The slope of $\mathbf{M}$ at small τ_1 is $-\int dT_1 \mathbf{S}(T_1, T_2)/T_1$. Since $T_1 > 0$, a positive initial slope (as observed in Fig. 2) indicates that the amplitude $\mathbf{S}(T_1, T_2)$ contains negative peaks which can only arise in systems that are not in the fast diffusion regime.

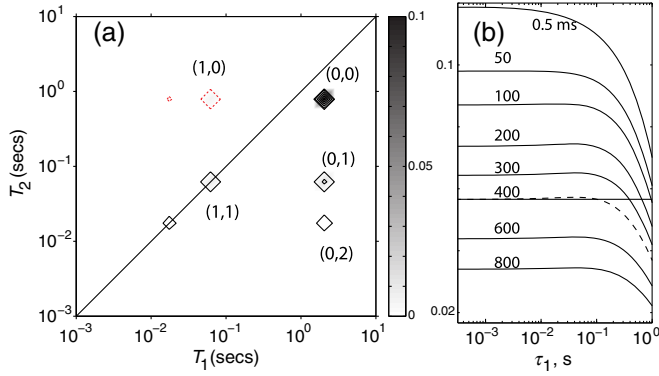


FIG. 2 (color online). T_1 - T_2 data from Fig. 1 of Ref. [7]. (a) T_1 - T_2 spectrum showing several peaks $[(p, q), \text{marked}]$ and negative signals above the diagonal (dashed contour lines). (b) The time domain signal M is plotted as a function of τ_1 at several τ_2 values (marked in the panel). Note the small initial rise of the signal at longer echoes highlighted by the dashed line.

T_1 - T_2 measurements were carried out on a water saturated pack (1 in. diameter and 1 in. long) of about $150 \mu\text{m}$ microporous beads (CPG3000A, Millipore), and on a bulk water sample for comparison (Fig. 3). The CPG beads are themselves porous, with an average micropore diameter of $0.3 \mu\text{m}$. For this sample the total porosity is 80% and is divided equally between the micro- and macropores. The NMR experiments were performed using a Rock Core Analyzer (Magritek Limited, Wellington, NZ) at ^1H NMR frequency of 2 MHz. Because of the large surface area, spin relaxation for the water inside the grains is substantially enhanced. The water molecules in the macropores (between the grains) are relaxed by diffusion into the micropores. Because diffusion between the micropores and the macropores is an essential part of the overall relaxation process, this system cannot be described with a picture based on fast diffusion.

In Figs. 3(b) and 3(c) the decay data, $M(\tau_1, \tau_2)/M(0, \tau_2)$, is graphed as a function of τ_1 for the different echo times, and each trace is shifted vertically in order to highlight the details at short τ_1 . For the bulk water sample [panel (c)], we see a continuously decreasing signal for all echo times. By contrast, in Fig. 3(b), the signal is observed to increase for $\tau_2 > 240$ ms. As discussed above, this behavior is a clear indication of negative peaks in the T_1 - T_2 spectrum. Furthermore, since the positive slope in M occurs when τ_1 is small, the negative amplitude in $S(T_1, T_2)$ must be at low T_1 values. Similarly, since the positive slope is seen only at longer echo times, we expect the negative amplitudes to be associated with large T_2 values (≥ 240 ms). Thus, looking only at the time-domain data, without any Laplace inversion, we can argue that the negative contributions have to be in the upper left corner of the T_1 - T_2 spectrum, consistent with the analytic result of Fig. 2.

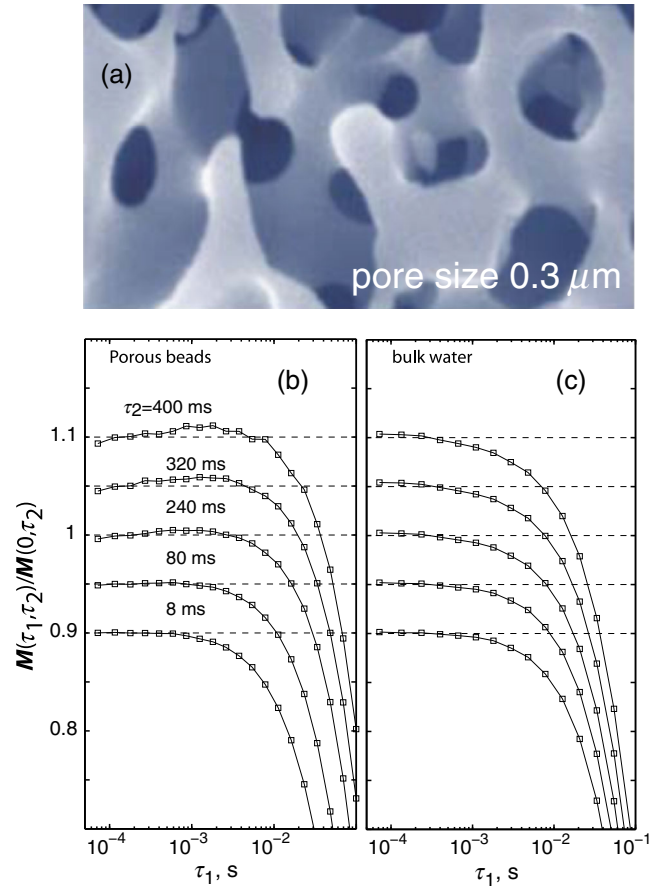


FIG. 3 (color online). (a) A micrograph shows the internal structure of the porous beads. The light gray is the solid phase. Average pore diameter is $0.3 \mu\text{m}$. (b), (c) T_1 - T_2 measurements on porous beads (b) and water (c). $M(\tau_1, \tau_2)$ is plotted as a function of τ_1 for the different echo times shown above each data set. The data are scaled to the average of the first eight data points and then shifted away from unity for display. An initial rise is observed for echo times larger than 240 ms. This rise corresponds to negative components in $S(T_1, T_2)$ associated with diffusive coupling between the micro- and macropores. No initial rise is observed for the bulk water signal.

Relaxation typically describes signal loss as time increases; thus, it is surprising to observe signal increase as τ_1 lengthens. To understand the physical origin of the observed phenomenon, consider the highly simplified model shown in Fig. 4, with a microporous region (shaded) to the left of the macropore. The qualitative behavior of the T_1 modes [4(a)] is central to this phenomenon. The slowest mode, $\phi_{1,0}$, which would be spatially uniform in the limit $\rho_1 \rightarrow 0$, has the same sign in both regions. Because the source of relaxation is the micropores, the amplitude of $\phi_{1,0}$ is higher in the macropore and lower in the micropore. The second mode, $\phi_{1,1}$, is asymmetric and positive in the micropore and negative in the macropore due to its orthogonality with the first mode. Figure 4(a) shows the contribution to the initial magnetization (m_0) from $\phi_{1,0}$ (the long dashed line) and $\phi_{1,1}$ (the short dashed line), and they

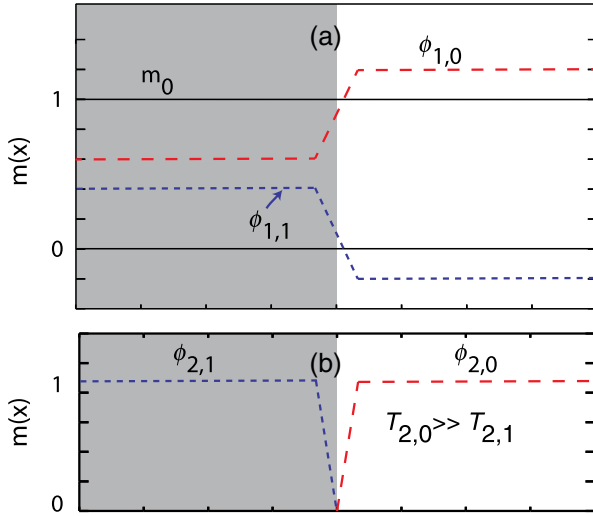


FIG. 4 (color online). Qualitative illustration of the first two eigenmodes of T_1 (a) and T_2 (b) in macro- and micropores. The T_2 modes are mostly isolated to macro- and micropores due to the larger ρ_2 .

add up to the constant m_0 , $m_0 = a_0\phi_{1,0} + a_1\phi_{1,1}$. Note that $|a_0\phi_{1,0}| > m_0$ in the macropore region [Fig. 4(a)].

We further assume ρ_2 is much larger than ρ_1 , and thus the T_2 modes behave as if the two pores are mostly decoupled [Fig. 4(b)]. As a result, at long τ_2 the micropore signal decays and the detected signal only comes from the macropore, thus

$$m \propto e^{-(\tau_1/T_{1,0})}|a_0\phi_{1,0}| - e^{-(\tau_1/T_{1,1})}|a_1\phi_{1,1}|, \quad (7)$$

evaluated within the macropore. Note that $T_{1,0} > T_{1,1}$. As τ_1 increases, the weight of $|\phi_{1,1}|$ decreases and thus $\mathbf{M}$ increases. For our sample, the contrast between ρ_1 and ρ_2 is finite; thus, the resulting increase is small.

To further validate our argument, we have performed numerical simulations based on a three-dimensional microporous model whose geometrical parameters can be matched to those of the bead pack discussed in connection with Fig. 3. The simulations employ variable step-size random walks in a packing of microporous spherical grains [16,17]. The large grains are arranged on a face centered cubic lattice; by adjusting their radius we can fix the amount of macroporosity at 0.40. We have examined two ways of representing the microporous structure within these grains based on periodic arrangements of either spherical micrograins or overlapping spherical micropores. In both cases the porosity within the large grains is adjusted to 0.67 so that the porosity contribution from the macro- and micropores are both 0.40. The grain sizes are adjusted so that the effective pore volume to surface area ratio, $V_P/S = 0.31 \mu\text{m}$. The diffusion coefficient for the bulk fluid is $2.5 \mu\text{m}^2/\text{ms}$. First, the comparison of the measured diffusion coefficient $[D(t)]$ with the simulation [16],

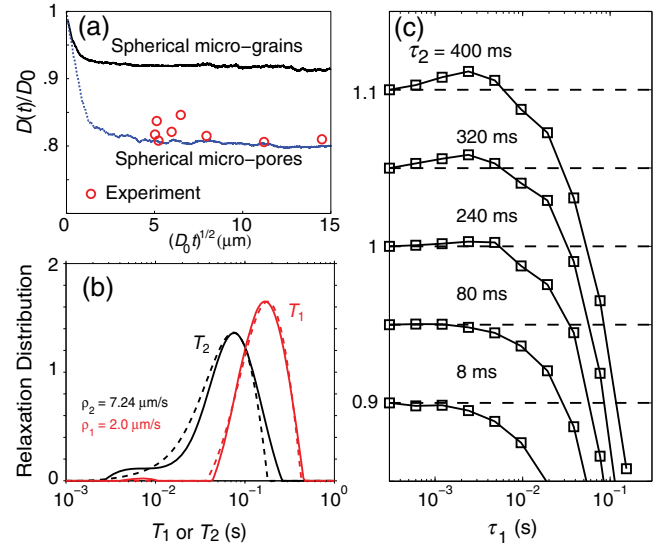


FIG. 5 (color online). Simulation results for microporous bead models. (a) Comparison of the measured time dependent diffusion coefficient, $D(t)$, with calculations based on two models for the microstructure. (b) Comparison of the simulated T_1 and T_2 distributions (dashed lines) with the measurements (solid lines). (c) Simulation of the T_1 - T_2 experiment. Data are plotted as in Fig. 3(b), showing the initial rise for $\tau_2 > 240$ ms.

Fig. 5(a), shows that the model with spherical micropores provides a much better fit to the experimental diffusion data. Second, the simulation of the T_1 and T_2 distributions is compared with our experiment [Fig. 5(b)] in order to determine ρ_1 and ρ_2 . Next, the micropore model was used to simulate the T_1 - T_2 experiment and the results are presented in Fig. 5(c). The calculations are presented in the same fashion as the experimental data [Fig. 3(b)] and a small positive slope is observed for $\tau_2 > 240$ ms, as in the experiment.

Consistency between our measurements, simulation, and analysis demonstrates that the T_1 - T_2 experiment can identify deviations from the fast diffusion regime. Evidence for this effect can be seen directly in the time-domain data without the complication of the Laplace inversion. In the presence of a broad spectrum of pore sizes, this method should continue to work well because $\mathbf{S}(T_1, T_2)$ integrates contributions from different pore sizes and the negative contributions will remain. The position of the negative structure is likely to depend on the pore size and will produce a strip of off-diagonal negative amplitude (instead of individual peaks). The location and magnitude of this structure will hold the information about pore coupling in such systems.

In conclusion, we have demonstrated that time dependent 2D NMR data can be used directly to identify complex diffusion regimes and diffusive coupling between pores in porous media. We plan to extend our analysis to other 2D experiments (e.g., T_2 -storage- T_2 experiments) [7,17] and natural samples with a wide distribution of pore sizes.

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Communication

Detection of intermolecular homonuclear dipolar coupling in organic rich shale by transverse relaxation exchange

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Magnetization exchange

ABSTRACT

The mechanism behind surface relaxivity within organic porosity in shales has been an unanswered question. Here, we present results that confirm the existence of intermolecular homonuclear dipolar coupling between solid and liquid phases in sedimentary organic matter. Transverse magnetization exchange measurements were performed on an organic-rich shale saturated with liquid hydrocarbon. Liquid and solid constituents were identified through both sample resaturation and through their T_1/T_2 ratios. Extensive cross peaks are observed in the T_2 - T_2 exchange spectra between the solid and liquid constituents, indicating an exchange of magnetization between the two phases. This result cannot arise from physical molecular diffusion, and the dissolution energies are too high for chemical exchange, such that the magnetization exchange must arise from intermolecular homonuclear dipolar coupling. These results both confirm a possible source of surface relaxivity in organic matter and emphasize caution in the use of standard porous media interpretations of relaxation results in shales because of coupling between different magnetization environments.

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1. Introduction

In the past decade, numerous investigations of unconventional petroleum resources [1–13] by nuclear magnetic resonance (NMR) have been conducted. The low permeability of shales makes traditional core analysis characterization difficult and unreliable [14], such that a non-invasive method such as NMR is appealing. Despite its extensive use, many unanswered questions still exist regarding the interpretation of the NMR results in shale and mudrocks.

One of these questions is the mechanism of surface relaxivity in organic matter [15]. Surface relaxivity in conventional petroleum reservoirs [16–23] has been well established as arising from the presence of paramagnetic impurities on the pore surface. When a fluid molecule encounters a paramagnetic ion, the interaction between the electron and nuclear spins causes the rapid relaxation of the nuclear spin. Shales may contain a significant quantity of paramagnetic impurities; however, in organic matter itself, the amount of paramagnetic impurities is extremely low [24,25]. Given their low concentrations, it does not seem plausible that

the dominant mechanism of surface relaxivity in organic porosity is due to fluid interaction with paramagnetic impurities.

A significant difference between unconventional shale resources and conventional petroleum resources is the presence and potentially high concentrations of organic hydrogen in solids. For conventional resources, the pore matrix consists predominantly of either sandstone or carbonate, neither of which contain significant quantities of hydrogen. Therefore, any hydrogen in the sample is assumed to be in fluid-filled porosity. However, organic matter is rich in hydrogen and, in many cases, it is the organic matter that contains most of the porosity in the sample [26]. Several researchers have hypothesized that the surface relaxivity arises from the homonuclear dipolar coupling [2,9,15] between the hydrogen in the organic matter and the saturating fluid. Korringa, SeEVERS, and Torrey presented the theory of longitudinal surface relaxation for the interaction between fluid molecules and paramagnetic ions [16]. This was later adapted for transverse relaxation by Kleinberg et al. [17]. This has been adapted to surface relaxation due to homonuclear dipolar coupling [27]. If two spins are undergoing dipolar coupling, this will cause an enhancement in their relaxation rate. For a fluid molecule adsorbed on the pore surface, the two spins will be in the rigid lattice regime and this enhancement will be inversely proportional to the strength of their coupling [28]:

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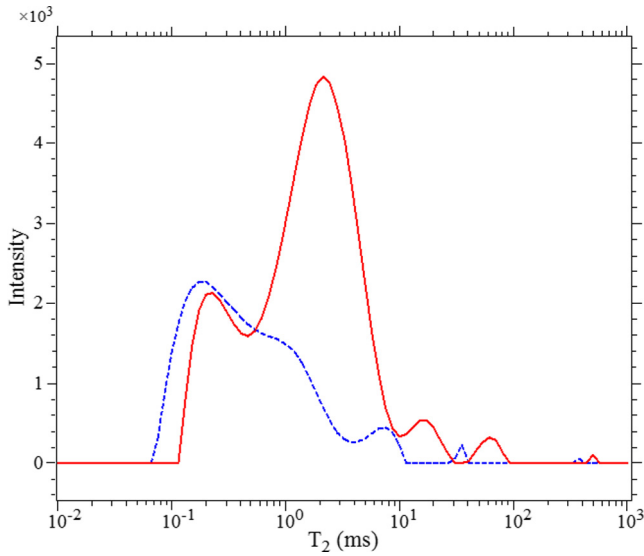


Fig. 1. T_2 distribution for the (a) as-received state (blue dashed) and (b) resaturated state (solid red) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

$$\frac{1}{T_{2D}} = \|D_{ij}\| = \left\| -\frac{\mu_0 \gamma^2 \hbar}{4\pi r_{ij}^3} \left(\frac{3 \cos^2 \theta_{ij} - 1}{2} \right) \right\| \quad (1)$$

where T_{2D} is the relaxation time of two hydrogen atoms undergoing dipolar coupling, D_{ij} is the dipolar coupling constant, the brackets denote the absolute value, μ_0 is the permeability of a vacuum, γ is the gyromagnetic ratio, $\hbar$ is Planck's constant, r_{ij}^3 is the distance between the coupled spins, and θ_{ij} is the angle they make with the applied magnetic field. While Korringa, Seever, and Torrey theory contains a complete description of how surface correlation time affected the measured relaxation time, Kleinberg et al. [17] later found that for transverse relaxation due to paramagnetic ions, the correlation term could be neglected because the relaxation time between the nuclear spin of the fluid molecule and the electron spin of the paramagnetic ion was significantly more rapid than the surface residence time. However, the relaxation time from the dipolar coupling cannot be assumed to be much greater than the surface

residency of the fluid molecule, such that the correlation time must be taken into consideration. Therefore, the transverse relaxation rate for a single pore in the fast diffusion limit is described as:

$$\frac{1}{T_2} = \frac{1}{T_{2B}} + \left(\frac{Sh}{V} \right) \left(\frac{n_H}{T_{2D} + \tau_c} \right) \quad (2)$$

where T_2 is the measured relaxation time, T_{2B} is the relaxation time of the bulk fluid, S is the surface area, V is the pore volume, h is the layer thickness, n_H is the relative fraction of hydrogen sites on the surface of the pore surface, and τ_c is the correlation time of how long a fluid molecule remains at the surface before exchanging with the bulk. Eq. (2) can be simplified by rewriting it to include a surface relaxivity term:

$$\rho_D = \frac{h \times n_H}{T_{2D} + \tau_c} \quad (3)$$

where ρ_D is surface relaxivity due to dipolar coupling. Then, assuming the system contains a low viscosity fluid, the bulk term can be neglected and the system can be described in a form similar to that of transverse relaxation due to paramagnetic ions:

$$\frac{1}{T_2} = \rho_D \frac{S}{V} \quad (4)$$

Although there were many indications that homonuclear dipolar coupling was the source of surface relaxivity in organic matter, there was no direct evidence. For example, if the surface relaxivity is caused by homonuclear dipolar coupling, because of the presence of τ_c in ρ_D , the measured transverse relaxation time will become longer with increased temperature due to shorter correlation times. Longer transverse relaxation times with increased temperature in organic rich shales were observed [29], but other possible causes for the longer relaxation times could not be ruled out, such as that the change in relaxation time was caused by a decrease in oil viscosity with temperature. Here, the molecular rotational correlation time decreases with temperature, leading to weaker dipolar interactions within the oil molecules themselves and a longer relaxation time would be observed regardless of whether the source of surface relaxivity in the porous sample were due to interactions with paramagnetic impurities or homonuclear dipolar coupling.

Recent work by Singer et al. [30] has shown the existence of intramolecular homonuclear dipolar coupling in fluid-filled organic matter. The method used was an adaptation of the tech-

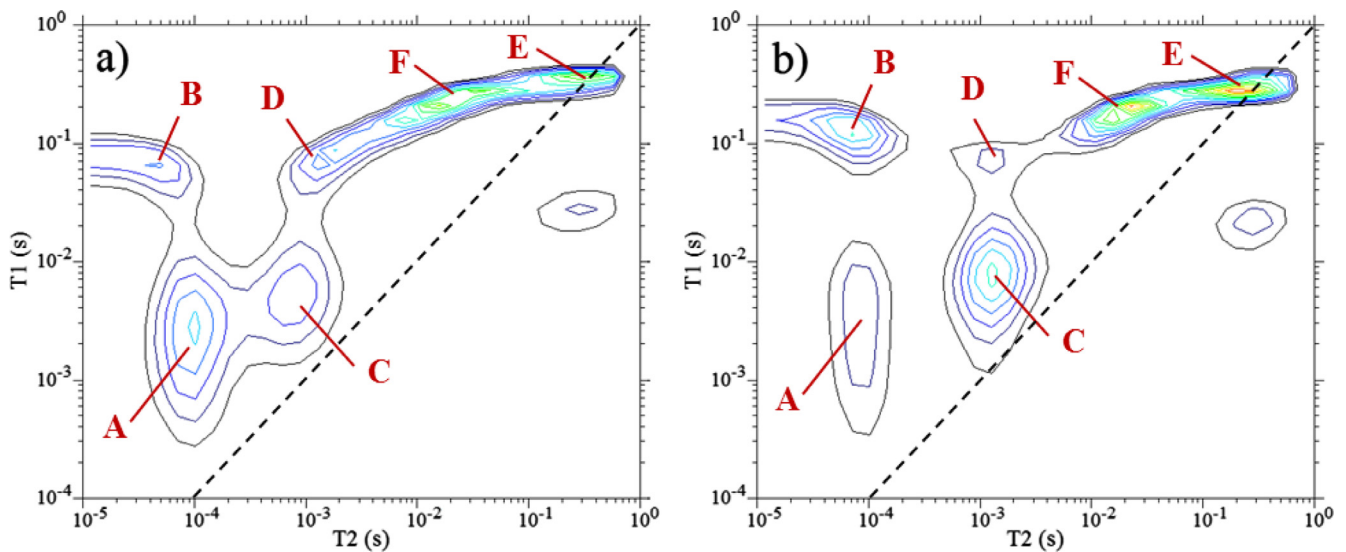


Fig. 2. T_1 - T_2 correlation for the (a) as received state sample and (b) resaturated state sample. Dashed line indicates parity.

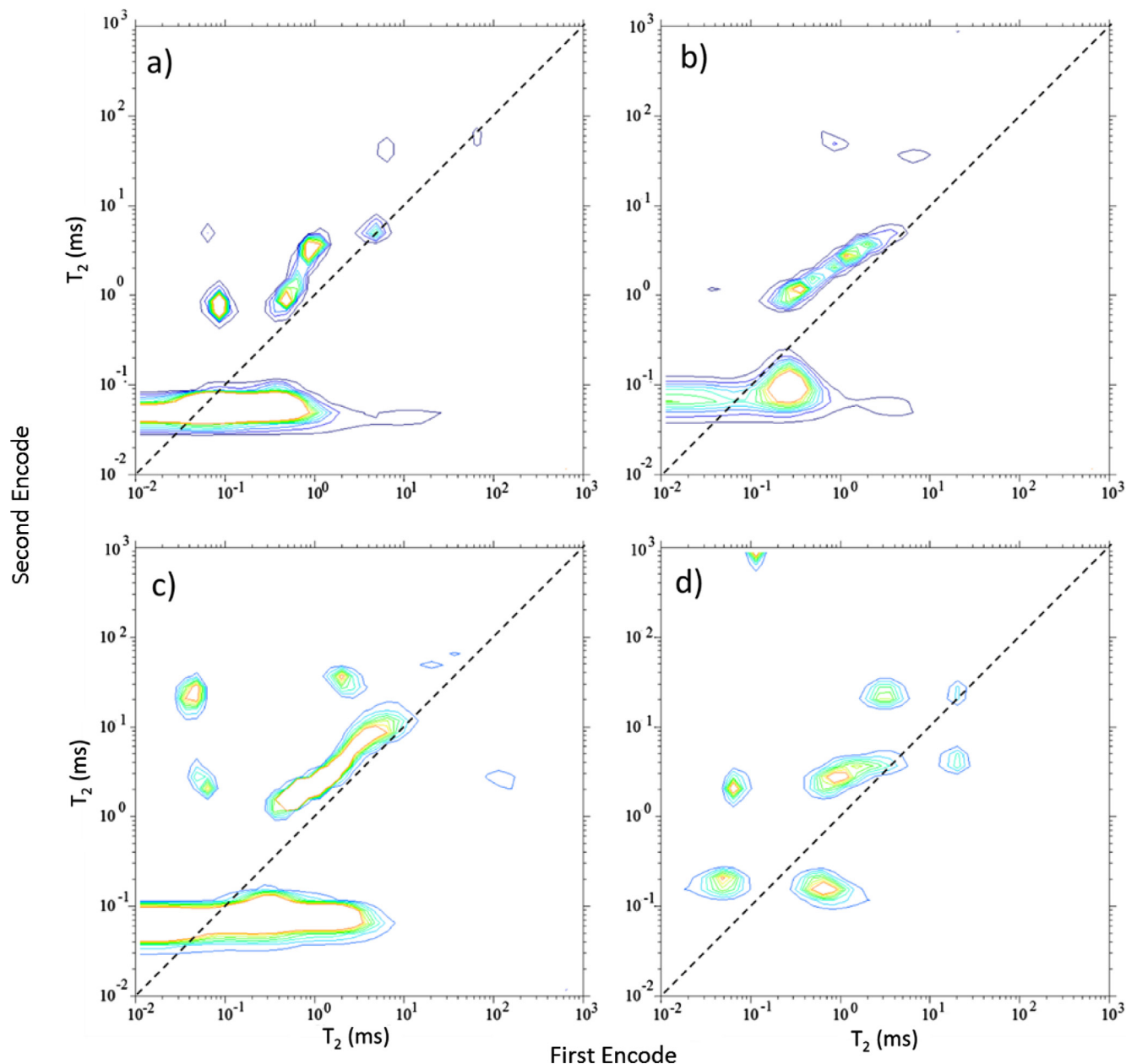


Fig. 3. T_2 - T_2 exchange plots for the as-received state sample at (a) 100 μ s (b) 1 ms (c) 5 ms (d) 10 ms mixing times.

nique by Straley [31] to confirm the existence of homonuclear dipolar coupling in carbonates. To avoid an influence from inorganic pores, the organic matter was isolated from the shale matrix. The kerogen was then saturated with both normal distilled water and a mix of normal and deuterated water. If the relaxation was due to paramagnetic impurities, the relaxation time of the two systems should be identical. However, if the relaxation was due to homonuclear dipolar coupling, there should be an increase in the relaxation time of the partially deuterated system, which was observed. This method confirms the existence of intramolecular homonuclear dipolar coupling; however, it does not confirm the presence of intermolecular homonuclear dipolar coupling.

This manuscript presents evidence to confirm the existence of intermolecular homonuclear dipolar coupling between fluid molecules and organic matter. Transverse relaxation exchange, also known as T_2 - T_2 exchange [32,33], is a pulse sequence that measures the movement of magnetization between different environ-

ments. The pulse sequence begins with a partial Carr Purcell Meiboom Gill (CPMG) [34,35] sequence to encode for T_2 , and then the magnetization is stored along the z-axis for a period of time.

During this mixing time, the system does not experience transverse relaxation, only longitudinal relaxation, and the fluid molecules are allowed to diffuse between environments. After the mixing time, the magnetization is returned to the transverse plane, and a full CPMG measurement is acquired. The experiment is repeated, increasing the number of 180° pulses in the initial partial CPMG before the storage time. The experiment is then analyzed by a two-dimensional inverse Laplace transform [36,37]. If magnetization remains in its original environment, then its signal occurs on the diagonal line of the 2D T_2 - T_2 map, indicating that the T_2 environment for the magnetization was the same at the end of the experiment as in the beginning. If there is no exchange between T_2 environments or if the mixing time is much shorter than the rate or rates of exchange in the system, all signal in the

T_2 - T_2 map will be located along the diagonal [38]. The formation of cross peaks indicates that magnetization has moved from one T_2 environment to another T_2 environment during the mixing time.

Although the method is generally used to investigate the diffusion or flow of spin-bearing fluid molecules between different pore environments, it can be used to observe the movement of magnetization between any environments with different transverse relaxation times. To this end, we sought to find evidence of intermolecular homonuclear dipolar coupling by performing a T_2 exchange measurement on an organic-rich shale saturated with liquid hydrocarbon. Besides molecular diffusion, another mechanism that can lead to the exchange of transverse relaxation is spin diffusion. When two spins are undergoing dipolar coupling, there is a possibility that they may swap energy states, a process called spin diffusion. The T_2 - T_2 exchange pulse sequence can as readily measure the change of the relaxation environment due to spin dif-

fusion as it can measure an exchange resulting from molecular diffusion. The presence of cross peaks between solids and liquids in the spectra provides a confirmation that the constituents are exchanging magnetization. While there is a third possible mechanism for an exchange of magnetization between T_2 environments, chemical exchange [33], the dissolution energies make chemical exchange between solid organic matter and liquid hydrocarbon extremely unlikely [39]. Therefore, we are confident that cross peaks between signal associated with organic solids and liquid hydrocarbon indicate the two constituents are experiencing intermolecular homonuclear dipolar coupling.

2. Experimental

Measurements were performed on a core plug from an organic-rich, liquid hydrocarbon-bearing reservoir. The sample was taken

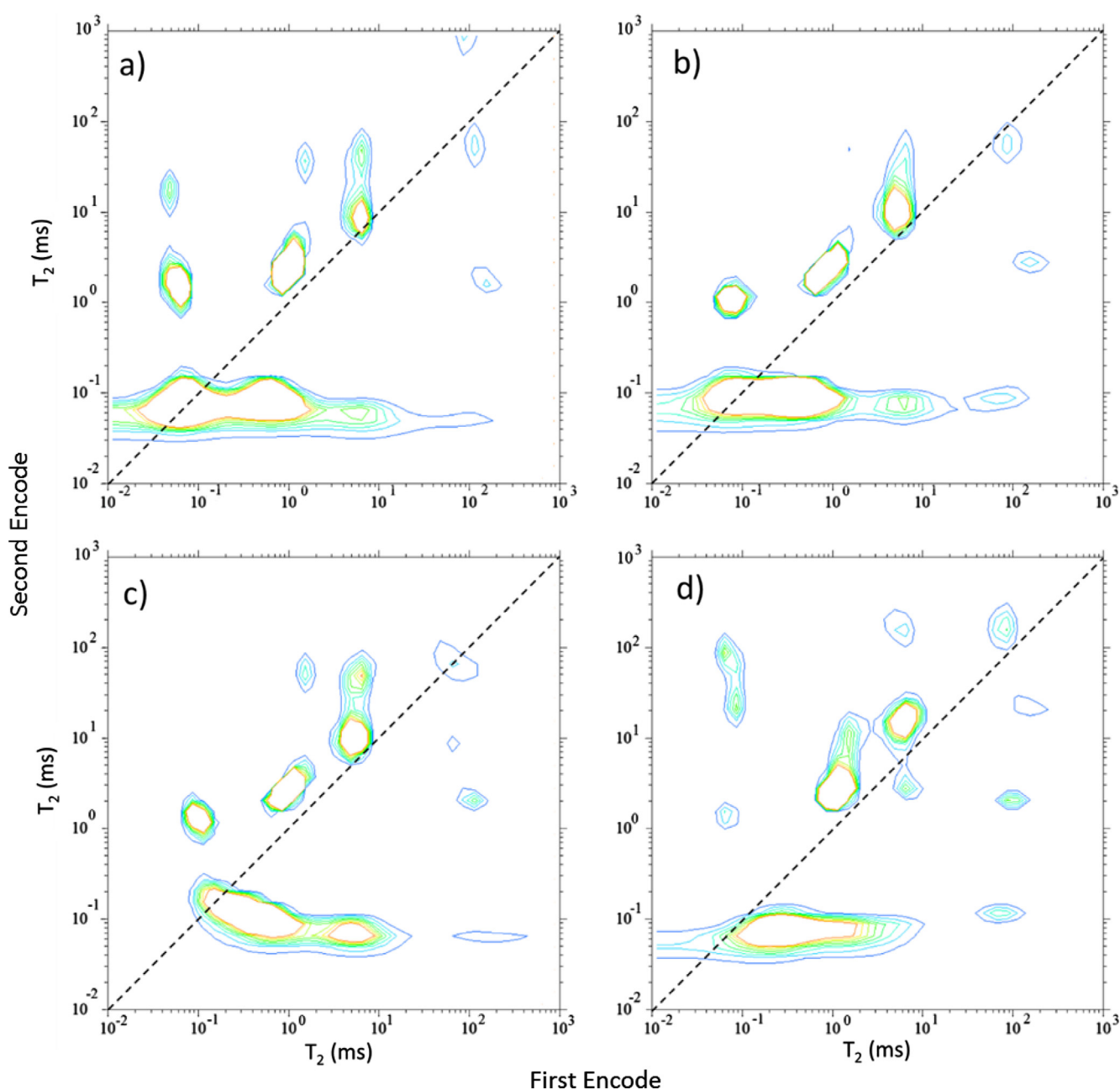


Fig. 4. T_2 - T_2 exchange plots for the resaturated state sample at (a) 100 μ s (b) 1 ms (c) 5 ms (d) 10 ms mixing times.

from a zone that had, on average, 9.8% organic matter by weight, and 2D scanning electron microscope (SEM) imaging confirmed the majority of porosity ($\sim 90\%$) to be located in the organic matter. Focused Ion Beam scanning electron microscopy (FIB-SEM) found average pore sizes ranged from 25 nm to 1 μm and approximately 95% of the pore space was connected with a horizontal permeability of 12.3 μD and a vertical permeability of 0.4 μD . Programmed pyrolysis measurements [40] of a sample end trim confirmed that the sample contained large quantities of both forms of organic matter: kerogen, solid organic matter, and bitumen, semi-solid organic matter. The region from which the sample was taken contained almost no clay ($<5\%$), as confirmed by X-ray diffraction (XRD) and Fourier transform infrared spectroscopy (FTIR). The core plug was measured in both the as-received state and the resaturated state. In the as-received state, no purposeful changes have been made to the state of the core, for example – to clean it, but it is expected that there may be unavoidable changes to the core as compared to its condition in the reservoir, such as evaporation of liquids or empty pore space as a result of gas loss during the core retrieval. Resaturation was performed by placing the sample under a vacuum and then saturating using heptane at a pressure of 2000 psig for a week.

NMR measurements were performed on a GeoSpec (Oxford Instruments, Oxford UK) using GIT System software (Green Imaging Technologies, New Brunswick, Canada). T_2 was measured with the standard CPMG method and 20,000 echoes per measurement were acquired. For the T_2 - T_2 exchange experiments, mixing times of 100 μs , 1 ms, 5 ms, 10 ms, 20 ms, 50 ms, and 100 ms were used. The initial partial CPMG in the T_2 - T_2 exchange pulse sequence was encoded from 0.1 to 250 ms with 32 steps spaced equally on a logarithmic scale. At each step, 2500 data points were acquired. The inversion-recovery method was used for T_1 - T_2 measurements. A total of 32 recovery times were used, ranging from 0.04 to 2277 ms on an equally spaced logarithmic scale. For each recovery time, 20,000 data points were acquired. A 2 s wait time and 0.1 ms echo time were used for all NMR measurements.

Experimental data were analyzed with Prospa© software (Magritek Ltd., Wellington New Zealand) using a Butler-Reed-Dawson inversion for the one-dimensional measurements [41] and a maximal entropy inversion for the two-dimensional inverse Laplace transform [37]. Although previous work has shown that analyses of organic shale relaxation data are best performed with an inversion method that can solve for the Gaussian relaxation of solids and the exponential relaxation of liquids [42], to date a two-dimensional method has not been developed. Despite the appearance of peaks at artificially low relaxation times for solid constituents using a standard two-dimensional inverse Laplace transform, the results are stable and repeatable. Furthermore, all other inversions in the literature are performed with a standard two-dimensional inverse Laplace transform, and therefore this approach best enables a comparison of our results to those from previous researchers.

3. Results and discussion

In previous work to observe the magnetization exchange in organic-rich samples [15,43], it could not be conclusively stated that the exchange was occurring between liquids and solids, as opposed to solids and semi-solids, such as between kerogen and bitumen. To confirm the identity of the solid and liquid constituents in the sample, we compared both the T_2 distributions (Fig. 1) and the T_1 - T_2 correlation plots (Fig. 2) for the sample in the as-received state and the resaturated state.

A comparison of the T_2 distributions for the as-received and resaturated states shows a large increase in the signal located at

~ 2 ms after resaturation, which suggests that this region is predominantly associated with hydrocarbons in organic pores. This result agrees with previous work, as other researchers have observed that this region of the T_2 distribution is associated with hydrocarbon in organic porosity at 2 MHz [44,45]. Two weak peaks exist at relaxation times above 10 ms, a region of the T_2 distribution attributed to hydrocarbon in inorganic porosity. Both peaks shift to slightly to longer relaxation times after resaturation. We interpret this shift to be due to air occupying a portion of the pore space in the as-received sample. The short peak at $\sim 100 \mu\text{s}$, a region of the T_2 spectrum typically associated with organic matter, decreases slightly in intensity and there is a slight shift to a longer relaxation time. This may be due to an averaging of the relaxation time if magnetization exchange occurs between liquids in solids, or it may be simply an artifact of the inversion.

We see in the T_1 - T_2 correlation measurements that there exist two different constituents at the short T_2 relaxation time. Peak A has a T_1/T_2 ratio of approximately 20–30, which is on the same order as previously attributed to bitumen [45]. Surprisingly, a peak also exists at a T_1/T_2 ratio of approximately 1000 (Peak B), which is a ratio associated with solid kerogen at 2 MHz [46]. It was not expected that the kerogen would be detected using a 100 μs echo spacing. However, if there is significant magnetization exchange occurring, it may be that the kerogen T_2 value is being averaged with that of the liquid hydrocarbons, enabling it to be detected. The intensity of the kerogen peak increases slightly in the resaturated sample, which supports this theory. The liquid hydrocarbon peak at $T_2 \sim 2$ ms (Peak C) has a T_1/T_2 ratio of approximately 4, which is also in line with ratios found in the literature. A smaller peak appears above the main oil peak with a longer T_1 time (Peak D). This result may arise from the averaging of magnetization with solid sample constituents during the measurement. As with the kerogen, this peak increases in intensity in the resaturated state.

Two peaks were observed in the region of the T_2 spectra typically attributed to hydrocarbon present in inorganic matter. The peak with the longer T_2 value (Peak E) has a T_1/T_2 ratio close to unity, which is in line with the literature value for hydrocarbon in inorganic porosity. The T_1/T_2 ratio of the oil peak at 20 ms (Peak F) was approximately 10; this result was higher than expected because literature values generally put the ratio of hydrocarbons in inorganic porosity in the range of 1–2 [44]. We believe that, despite the longer relaxation time, this peak arises from hydrocarbon associated with organic porosity. There appears to be a weak peak at nonphysical time ($T_2 > T_1$) in both the as-received and resaturated spectra. We are uncertain as to its origin. It may arise from magnetization exchange between constituents with different T_1/T_2 ratios in the sample with during the course of measurement [47] or perhaps from non-random noise (e.g. noise from electronic circuitry) picked up by the probe.

Based on the T_2 and T_1 - T_2 correlation measurements, we are confident in the identification of the different T_2 peaks as associated with organic solids, hydrocarbons in organic pores, and hydrocarbons in inorganic pores. We then apply this identification to the interpretation of the transverse relaxation exchange plots. Fig. 3 shows the T_2 exchange plots for the as-received state, and Fig. 4 shows the resaturated state. The intensity ranges were kept the same for both samples, such that some of the weaker peaks for the as-received state plots may not be visible. Even at the shortest exchange time, extensive exchange peaks are observed between solid ($T_2 \sim 0.1$ ms) and liquid ($T_2 \sim 2$ ms) constituents in both the as-received and the resaturated sample. This result is in line with previous research, which showed that magnetization exchange occurred as rapidly as the measurement could be made [15,43]. This exchange of magnetization between the solid and liquid constituents confirms the presence of homonuclear dipolar coupling between them, as the exchange could not be the result of molecu-

lar diffusion of spin-bearing fluid molecules. Although spin diffusion rates are slow, the exchange depends on the surface area between the liquids and solids [48]. In shales, the specific surface area is frequently very large [49], although no specific surface area measurement is available for this particular sample. We observe that exchange rapidly occurs between the solids ($T_2 \sim 0.1$ ms) and the peak located at approximately $T_2 \sim 20$ ms, which supports the theory that, despite the long T_2 time, the peak is associated with hydrocarbons in organic porosity.

At the longer exchange times, the as-received and resaturated states develop cross peaks that indicate an exchange of magnetization between the inorganic ($T_2 \sim 100$ ms) and organic porosity ($T_2 \sim 2$ ms and $T_2 \sim 20$ ms) and the two organic porosity environments ($T_2 \sim 2$ ms and $T_2 \sim 20$ ms). This exchange is the result of molecular diffusion of the spin-bearing molecules between pores and suggests strong diffusional coupling in the systems. There also begins to develop cross peaks at longer exchange times between the inorganic porosity ($T_2 \sim 100$ ms) and the organic solids ($T_2 \sim 0.1$ ms). As there is not a direct mechanism for magnetization to exchange occur between organic solids to inorganic porosity, this suggests that multiple exchanges of magnetization are occurring on the time scale of the NMR measurement. These results underscore that the interpretation of NMR relaxation data from organic-rich shales must be treated differently than the standard interpretation for conventional sandstone and carbonate reservoirs. If diffusional coupling is present, the transverse relaxation distribution will not be reflective of the relative pore sizes in the sample. Although the inverted data for the 20 ms, 50 ms, and 100 ms exchange times showed exchange peaks, the signal-to-noise ratio (SNR) was too low (<100 SNR) for the inversion to be considered reliable, and significant T_1 weighting was evident.

Significant asymmetry was observed in some of the exchange spectra, which has been associated with multisite exchange [50]. Given the rapid exchange that occurs between the constituents, it is highly likely that magnetization is exchanging environments multiple times during the course of the measurement. This also suggests that exchange is occurring not only during the mixing time, but during the CPMG encode periods as well. This has been observed to lead to shifts in the diagonal peaks away from the diagonal and asymmetry in the cross peaks [51]. The asymmetry may also be a result of the large T_1/T_2 ratio of the solid sample constituents. Because of the complexity of the exchange in the spectra, no quantitation of exchange rates was attempted here. Additional work involving simulation would be necessary to fully understand the effect of exchange between constituents with low and high T_1/T_2 ratios on the resulting T_2 - T_2 exchange plots.

4. Conclusions

This manuscript presents the first direct evidence for intermolecular homonuclear dipolar coupling in organic porosity. Sample constituents were identified by hydrocarbon resaturation and T_1 - T_2 correlation. Transverse relaxation exchange experiments show clear movement of magnetization between the solid and liquid sample constituents. This, along with recent research on the presence of intramolecular dipolar coupling, helps to understand the source of surface relaxivity in fluid-filled organic matter. In addition to the presence of intermolecular homonuclear dipolar coupling, the transverse relaxation results show diffusional coupling of fluid molecules between organic and inorganic porosity in the sample.

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Quantitative analysis of diffusional pore coupling from T_2 -store- T_2 NMR experiments

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ABSTRACT

Low field NMR T_2 distribution is an efficient tool for characterizing pore size distribution in porous media. In NMR relaxation experiments, we measure the magnetization decay characterized by the so-called T_2 relaxation time, resulting essentially from liquid–solid interactions of the spins carried by molecules and exploring the pore space by diffusion. For multimodal systems, a new technique called T_2 -store- T_2 allows the analysis of diffusional pore to pore exchange that is extremely useful for the characterization of connectivity. This technique uses 2D inverse Laplace techniques and produces T_2 – T_2 maps. Qualitatively, a system is coupled when off-diagonal peaks are observed.

Based on an analytical solution describing diffusional coupling between two pore populations, we propose defining a coupling factor that quantify the degree of coupling between the two populations, providing an easy understanding of the complex analytical solution. This analysis allows understanding 1D T_2 experiments as well, and indicates some limitations of the T_2 characterization when interpreted as a pore size distribution.

We provide examples of bimodal pore structures in which we applied our methodology: a clay gel system, a shaly sandstone and two double porosity carbonates. These systems are also described by conventional techniques (mercury injection, SEM visualization) and illustrate weak, intermediate and strong coupling. Despite the presence of distribution of pore sizes, the two pore system exchange model gives satisfactory results for the quantitative analysis of the coupling and T_2 -store- T_2 experiments. In carbonates, more complex exchanges can occur between micro-, meso- and macropores.

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1. Introduction

Low field NMR relaxation is a unique non-invasive and non-destructive technique to characterize pore size distribution of porous media. Spin bearing molecules explore the pore space by diffusion and provide a unique information about the volume to surface ratio of a given compartment. This principle was first evidenced by Brownstein and Tarr [1] in biological cells and relies on a contrast between surface and bulk relaxivity. It is used extensively for the interpretation of NMR relaxation data in natural porous media especially in the field of petrophysics. Without considering the additional difficulty of the fast and slow diffusion regimes of relaxation [2], i.e. we consider typical pore sizes and surface relaxivity values, the pore size interpretation is thought to be usually valid because the diffusion length during the lifetime of the magnetization is of the same order of the distance between grains. Hence, the exchange of molecules between pores should be limited. However, there are situations where this is not the case.

Diffusional pore coupling is an issue of practical importance, especially in the field of logging interpretation. NMR relaxation data should not depend on temperature and this fact was put forward to justify the surface dominated regime [3]. However, in carbonates, a significant temperature dependence can occur, typically when comparing laboratory and log data performed, respectively, at 20 and 100 °C. This is a direct effect of pore coupling and a solution to predict the shift of the T_2 distributions was proposed [4]. At the extreme situation, a multimodal system may appear unimodal and this has important interpretation consequences since the NMR log is the only in situ measurement sensitive to the pore structure. Later, Anand et al. [5,6] proposed a 1D geometric model and a coupling parameter to describe the pore to pore coupling phenomenon and applied it to various situations. For bimodal pore systems, the shift of macro- and micropore NMR peaks can be predicted as a function of the coupling parameter comparing the characteristic relaxation rate to the diffusion rate.

From the NMR point of view, the methods for observing the modification of the environment of spin carrying molecules has been evidenced very early in the development of NMR [7]. At high field, the analysis of the chemical exchange is now a common practice, for example in the exchange spectroscopy technique EXSY [8]

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that uses very similar principle as used here. The results of these experiments is a 2D map of chemical shift in which off-diagonal peaks signify a change of chemical environment after a chosen exchange time. At low field, the observation of the modification of the spin carrying molecules in terms of relaxation is recent. Simultaneously, two groups [9,10] proposed new NMR sequences to detect the relaxation exchange of molecules in the case of a cement paste and a sandstone. The experiment consists of two T_2 relaxation periods with a variable exchange time sandwiched between them. When analyzing the resulting 2D T_2 - T_2 maps, the pore coupling is evidenced by off-diagonal peaks. The motions of molecules can be observed only if a surface relaxivity contrast exists between T_1 and T_2 [9,11] which is usually the case in porous media. More generally, the existence of off-diagonal peaks can also be analyzed from the resolution of eigenvalues of the underlying Bloch-Torrey diffusion equations and some general behavior can be deduced [11].

In this work, we first analyze the analytical solution proposed by Montheillet et al. [9] and propose a coupling parameter based uniquely on NMR measurable factors to quantify the degree of coupling between two pore sizes. Then, the method is applied to different systems with a large range of sizes, from nanometers to tens of microns. We study the pore to pore exchange in a natural bimodal model system composed of flocculated clay particles, and samples typical of oil bearing rocks: a high permeability sandstone containing clays and two complex carbonates.

2. Theory: coupling parameter

We first consider the case of complete coupling using a simple fast exchange model. Consider two porous zones in a porous media (Fig. 1), typically representative of micro- (subscript a) and macropores (subscript b). For simplicity, we will assume a unique pore size in each zone. In these conditions, the NMR relaxation time in the surface dominated regime in the absence of coupling will be proportional to the volume to surface ratio of the macropores $T_{2b} \propto V_b/S_b$ and micropores $T_{2a} \propto V_a/S_a$. The measured distribution will be bimodal. In this case, diffusive molecular motions are such that the magnetization has decreased to zero before the molecules can travel to a different porosity zone.

In the case of complete coupling, molecules explore both environments during the typical life time of the magnetization. Assuming that relaxation is still in the fast diffusion regime, the observed relaxation time T_2^+ (the notation is linked to the theory described later) is then proportional to the total volume to surface ratio according to:

$$T_2^+ \propto \frac{V_T}{S_T} = \frac{V_a + V_b}{S_a + S_b} \quad (1)$$

Each volume V_a and V_b corresponds to a magnetization M_a^0 and M_b^0 . Assuming a unique surface relaxivity for both a and b zones, the observed relaxation time T_2^+ is unique and smaller than the macropore peak T_{2b} according to:

$$\frac{T_2^+}{T_{2b}} = \frac{M_a^0 + M_b^0}{T_{2b} M_a^0 + M_b^0} \quad (2)$$

For example, when the pore sizes are different by a factor of 10 ($T_{2b}/T_{2a} = 10$), the observed single relaxation time $T_2^+ = 0.2T_{2b}$ when $M_a^0 = M_b^0$. The relaxation time T_2^+ is closer to the micro-pore peak than the macro-pore because there is much more surface in the microporosity.

If the coupling is intermediate, we now use the theory developed by Monteillet et al. [9] describing the full details of the diffusional coupling between two zones a and b , and we show that a single parameter can be build to characterize the degree of coupling. For spin-spin relaxation, the system of coupled differential equations is given by:

$$\begin{aligned} \frac{dM_a}{dt} &= -k_a M_a + k_b M_b - R_{2a}^0 M_a \\ \frac{dM_b}{dt} &= -k_b M_b + k_a M_a - R_{2b}^0 M_b \end{aligned} \quad (3)$$

where $k_{a,b}$ is the exchange rate, $R_{2a,b}^0$ is the nuclear spin-spin relaxation rate ($= 1/T_{2a,b}^0$) of each zone in the absence of coupling. At $t = 0$, the magnetization in each reservoir is $M_{a,b}^0$; the exchange rates $k_{a,b}$ are linked by the balance equation:

$$k_a M_a^0 = k_b M_b^0 \quad (4)$$

The system of Eq. (3) was solved analytically [9] in the specific context of the T_2 -store- T_2 sequence detailed in Appendix A. Essentially, the relaxation is first observed using a CPMG sequence of variable length. Then, the magnetization is stored along the z -axis (T_1 relaxation) to allow spin bearing molecules to move by diffusion in a different environment (a different pore system). Also, because the surface relaxivity is slightly different, the relaxation decay during the storage or exchange period is different. Finally, the magnetization is returned to the x - y plane and the decay recorded. The result of the experiment is a T_2 - T_2 map containing four peaks in the case of coupling (Fig. 2). The observed relaxation rates $s_{1,2}^{+/-}$ ($= 1/T_{1,2}^{+/-}$) are given by:

$$\begin{aligned} s_{1,2}^{+/-} &= -\frac{1}{2} (R_{1,2}^a + k_a + R_{1,2}^b + k_b) \\ &\quad \pm \frac{1}{2} \sqrt{(R_{1,2}^a + k_a + R_{1,2}^b + k_b)^2 - 4 \{ (R_{1,2}^a + k_a)(R_{1,2}^b + k_b) - k_a k_b \}} \end{aligned} \quad (5)$$

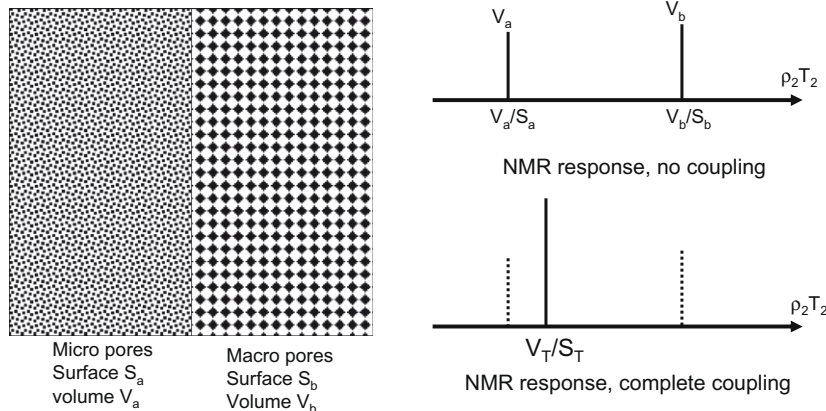


Fig. 1. Illustration of the pore coupling effect on 1D T_2 relaxation time distributions. The pore structure information may be lost; micro- and macropores can no longer be identified in the case of complete coupling.

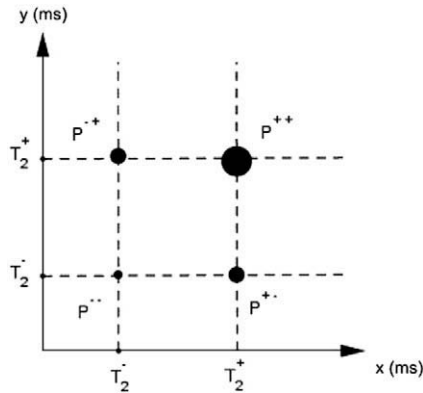


Fig. 2. Illustration of the positions and amplitudes of diagonal and off-diagonal peaks as a result of diffusional coupling between two reservoirs, as measured in a T_2 -store- T_2 experiment.

The observed amplitudes of the diagonal peaks P^{++} and P^{--} , and off-diagonal peaks P^{+-} and P^{-+} , are function of the exchange time t_e that can be varied experimentally:

$$\begin{aligned} P^{++} &= C_1 \exp(s_1^+ t_e) + C_2 \exp(s_1^- t_e) \\ P^{--} &= C_3 \exp(s_1^+ t_e) + C_4 \exp(s_1^- t_e) \\ P^{+-} &= P^{-+} = C_5 \exp(s_1^+ t_e) + C_6 \exp(s_1^- t_e) \end{aligned} \quad (6)$$

where the coefficients from C_1 to C_6 are complicated functions of $s_{1,2}^{\pm}$, $M_{a,b}^0$, $R_{1,2}^{a,b}$ and $k_{a,b}$ (see Appendix B). Note that the ratio T_1/T_2 is included in these coefficients. These peaks can be observed experimentally using 2D inverse Laplace techniques described later. The above model can be used to show the shift of the micro- and macropore peaks in a 1D NMR CPMG relaxation experiment, and to explore the behavior of 2D experiments for different degrees of coupling.

Due to the complicated solution of Eq. (3), it is rather difficult to analyze the effect of the numerous parameters included in the system. For this purpose, we suggest to use the following coupling parameter:

$$\alpha = \frac{(\text{average exchange rate})}{(\text{micropore relaxation rate})} = 2(k_a + k_b)T_{2a} \quad (7)$$

There are other possibilities for defining a coupling parameter. Here, we base our parameter on the micropore relaxation rate (or the smallest pores) because microporosity is a major source of relaxation due the much larger amount of surface compared to macroporosity. We will also see that the use of this parameter greatly simplifies the analysis of the system. Note that α is directly proportional to the diffusivity of the saturating liquid because we expect a direct dependence of exchange rate with diffusivity. Using Eqs. (4)–(6), we calculated four quantities of practical interest by varying the exchange rate k_a by several orders of magnitude; random choices of the pore size ratio T_{2b}/T_{2a} and magnetization ratio M_a^0/M_b^0 were selected in the intervals [5,50] and [0.05,1], respectively. This means that we consider situations where the micro- and macropores are well separated and microporosity does not exceeds macroporosity. We used $T_1/T_2 = 1.5$, a typical value for sandstones and carbonates at a Larmor frequency of 2 MHz.

The four quantities of interest are plotted in Fig. 3:

- The ratio P^{--}/M_a^0 calculated at $t_e = 0$ represents the observed microporosity amplitude relative to the true one in a 1D experiment. It mostly follows a S-shaped curve that has been found using a very different approach [5]. To go from a weak coupling ($\alpha < 0.1$) to a strong coupling ($\alpha > 10$), the parameter α must vary by two orders of magnitude.

- The ratio $2P^{+-}/P_{\text{tot}}$ calculated at $t_e = T_{2b}$ represents the amplitude of the relative off-diagonal peak intensity in a 2D experiment. With the choice $T_1/T_2 = 1.5$, it has a maximum at $\alpha \cong 1$; this maximum will occur at smaller α when increasing the ratio T_1/T_2 (at the extreme value of 4, the maximum occurs at 0.6). For strongly coupled systems ($\alpha > 10$), it means that the 2D T_2 -store- T_2 experiment is unable to evidence the coupling. One must lower the value of α using a fluid of lower diffusivity.
- Concerning relaxation times, the ratio T_2^+/T_{2b} expresses the deviation of the macropore relaxation time relative to its true value without coupling. It depends both on α and the magnetization and relaxation time ratio M_a^0/M_b^0 and T_{2b}/T_{2a} , respectively. We also verified that, for large α :

$$\frac{T_2^+(\alpha \rightarrow \infty)}{T_{2b}} = \frac{M_a^0 + M_b^0}{\frac{T_{2b}}{T_{2a}} M_a^0 + M_b^0} \quad (8)$$

which corresponds to the value given in Eq. (2) using the simple volume to surface model explained above.

- Finally, the ratio T_2^-/T_{2a} expresses the shift of the micropore relaxation time relative to its value without coupling. Surprisingly, the micropore peak is shifted to smaller T_2 values (Fig. 3) as the coupling coefficient increases, whereas one would have intuitively expected a gradual merging of the micro- and macropore peaks. This behavior has been clearly observed experimentally in model systems and predicted [5].

3. Experimental methods

We describe here some important aspects when performing 2D inverse Laplace transform (ILT) and T_2 -store- T_2 experiments.

From the theory described above, we can extract the amplitudes of the off-diagonal peaks to test the feasibility of the method in various configuration, and gather important information on how to design the experiments. Using the same random fluctuations of the ratio T_{2b}/T_{2a} and M_a^0/M_b^0 as previously, we plotted the relative off-diagonal peak amplitudes $2P^{+-}/P_{\text{tot}}$ ($P_{\text{tot}} = (P^{++} + P^{--} + 2P^{+-})$) as a function of the storage time t_e (that can be varied experimentally) for a fixed value of the coupling parameter $\alpha = 1$ (Fig. 4). It is seen that the total off-diagonal peak relative amplitude does not exceed 10% for a typical value of $T_1/T_2 = 1.5$, and this has important implications for data acquisition and processing. The graph also shows that it is sufficient to observe the off-diagonal peaks for storage time t_e at maximum up to the observed dominant peak T_{2+} . If no off-diagonal peak is observed at $t_e = T_{2+}$, it means that the bimodal system is either not coupled, or totally coupled. To describe the coupling, four independent parameters need to be calculated: k_a , T_{2b} , T_{2a} and M_a^0 . The other parameters k_b and M_b^0 are deduced from Eq. (4) and by normalizing the data set by $M_a^0 + M_b^0$. For this back calculation, we suggest to use in addition the diagonal peak amplitudes that also depends on the storage time t_e (Fig. 4). For the present conditions, the normalized variation of the macro- and micropore peaks ($P_n^{++} = P^{++}/P^{++}(t_e = 0)$ and $P_n^{--} = P^{--}/P^{--}(t_e = 0)$) are mostly described by a single exponential decay.

3.1. Off-diagonal peak detection

It is well known that ILT must be used carefully [12] to avoid spurious peak and over interpretation of the results. For 2D calculations, it is obviously more crucial. Based on efficient computing algorithms [13], we build our own software and made extensive tests to evaluate possible short comings. Essentially, one must find a matrix of relaxation times A fitting the data, such as to minimize the cost function SE:

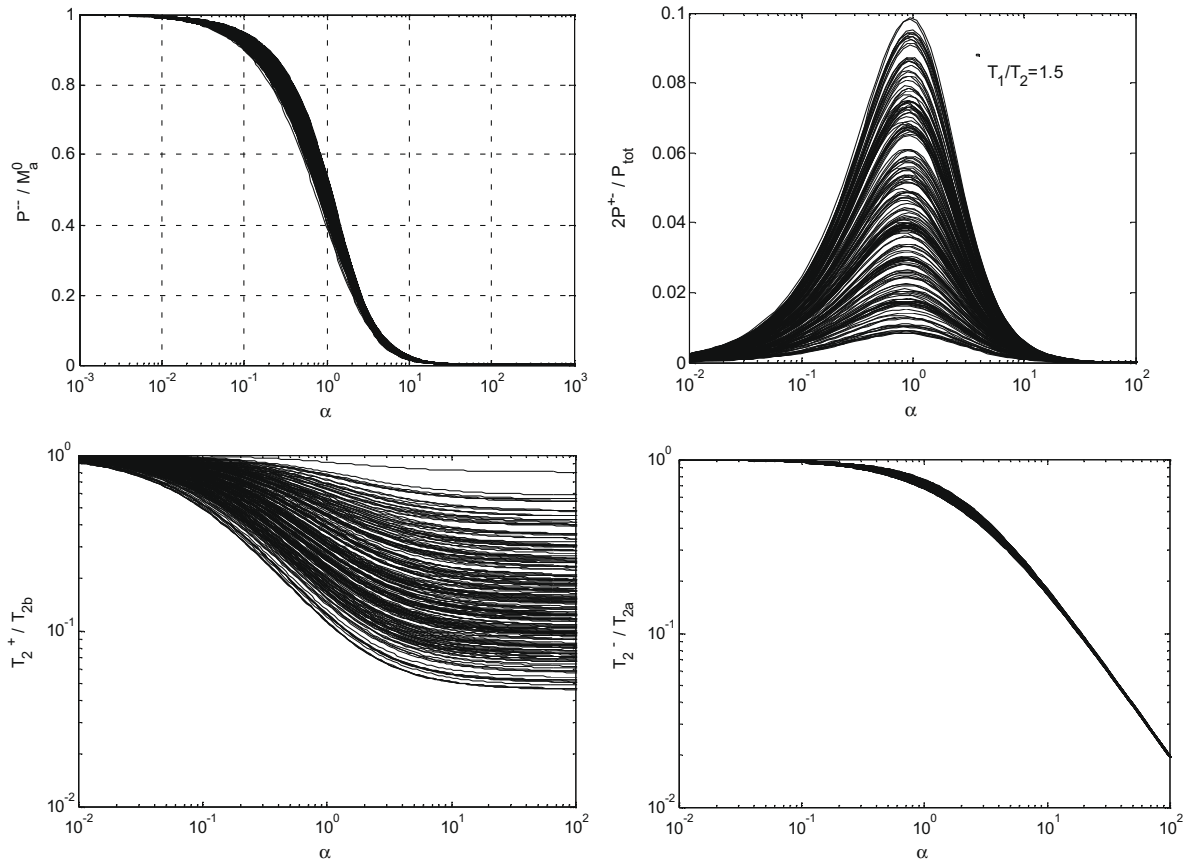


Fig. 3. Study of four quantities of interest as a function of the coupling parameter α when the ratio T_{2b}/T_{2a} varies from 5 up to 50, and M_a^0/M_b^0 from 0.05 up to 1 ($T_1/T_2 = 1.5$). Random combinations of these two ratios were chosen to calculate the curves. P^{+-}/M_a^0 : amplitude ratio of the observed microporosity peak to the true amplitude at $t_e = 0$; $2P^{+-}/P_{tot}$: amplitude ratio of the off-diagonal peak to the macroporosity peak at $t_e = T_{2b}$; T_2^{+}/T_{2b} : deviation of the macropore relaxation time relative to its true value; T_2^{-}/T_{2a} : deviation of the micropore relaxation time relative to its true value.

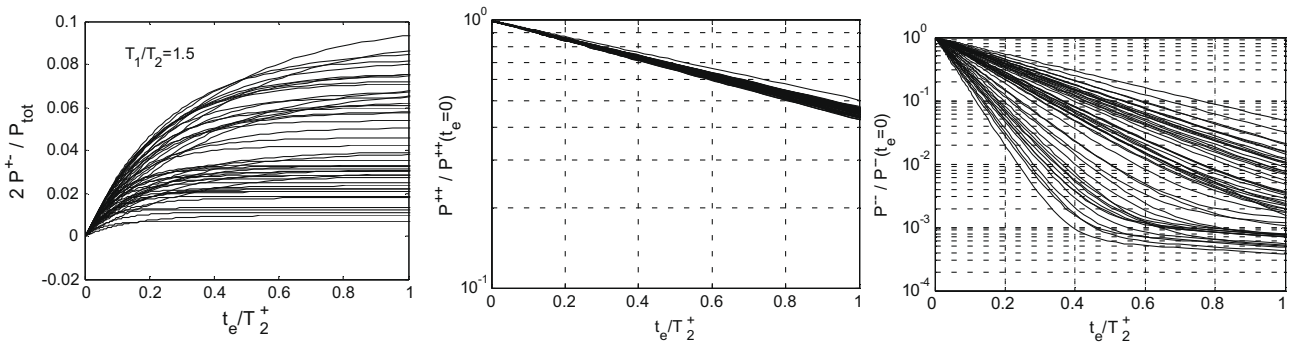


Fig. 4. On- and off-diagonal relative amplitude as a function of the storage time t_e for a coupling parameter $\alpha = 1$. The ratio T_{2b}/T_{2a} varies randomly from 5 up to 50, and M_a^0/M_b^0 from 0.05 up to 1.

$$SE(\lambda) = \|X \cdot A - S\|^2 + \lambda \|A\|^2 \quad (9)$$

where S is the measured signal and X is the exponential matrix function such that a component of the product $X \cdot A$ is $A_{ij} \exp(-t_1/T_{2i}) \exp(-t_2/T_{2j})$. The above equation is solved using matrix calculations only, involving single value decomposition (SVD) and data compression not detailed here. We focus here on the choice of the regularization parameter λ which is important for the detection of small amplitude peaks. For clarity, we illustrate the problem using a simple 1D experiment because the same procedure is used for 1D and 2D calculations. To obtain the optimum regularization value, the standard procedure is to plot the $SE(\lambda)$ curve and chose

the largest value of λ giving the lowest SE value (Fig. 5). This procedure is more important than one would think in terms of statistical significance of the small peaks. For this purpose, using a distribution containing a small peak characterizing micropores, we calculated a synthetic signal. Then we added a white noise at a ratio $S/N = 100$ and calculated a new solution according to Eq. (9). This procedure was repeated for 1000 different noise realizations. The results are plotted in Fig. 6 as a distribution band representing the highest and lowest values obtained at a given relaxation time. When the regularization parameter is not optimum and too small, the low limit of the band reaches zero amplitude for the microporosity peak, meaning that the amplitude and position of the peak are very

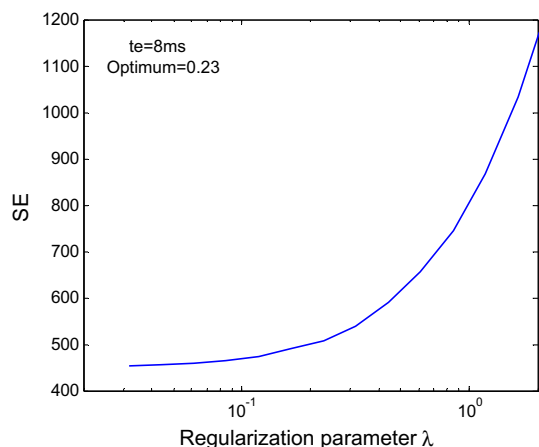


Fig. 5. Illustration of the choice of the optimum regularization parameter when computing 2D relaxation time maps. This curve corresponds to the smectite gel example described in Fig. 9 ($t_e = 8$ ms).

uncertain and depend on the noise realization. When the regularization parameter is optimum in the sense described above, the lowest limit of the error band is non-zero and one can clearly identify the micropore peak. Note that the choice of the optimum value on the $SE(\lambda)$ curve may differ slightly from one user to another because different criteria can be used to define the curve increase. The same tuning applies to the 2D analysis that uses the same algorithm and regularization principle. It is particularly important when trying to detect small amplitude peaks.

The second important aspect is the effect of the signal to noise ratio. This is illustrated on a 2D calculation in which synthetic 2D signals were generated from a known solution (Fig. 7). The square represents the position of the four peaks of amplitude $P^{--} = 0.07$, $P^{+-} = P^{-+} = 2.5$ and $P^{++} = 100$. The calculations performed for three values of signal to noise ratio indicate that small off-diagonal peaks can indeed be detected, but their location may deviate from the theoretical square position, except for the highest signal to noise ratio of 500. However, such a high value may require time consuming experiments. We usually acquire experiments at a signal to noise ratio of 300.

4. Experimental results

We studied four systems showing a bimodal pore structure in which diffusional exchange are suspected to modify the pore structure information inferred from 1D relaxation time distribution. Experiments were performed on a MARAN-ULTRA 23.7 MHz from Oxford Instruments equipped with a 18 mm proton probe. The samples are small cylinders of diameter 15 mm and length 20 mm. For the gels, we used a glass container of similar size.

4.1. Smectite gels

The first system consists of smectite clay particles that have aggregated due to inter-particle interactions occurring at high salinity (0.1 mol/L CaCl_2) and high clay concentration. A full study of diffusion properties [14] shows an exponential decrease of the effective diffusion coefficient as a function of clay fraction. When flocculated, the T_2 distribution is bimodal (Fig. 8); the lowest mode

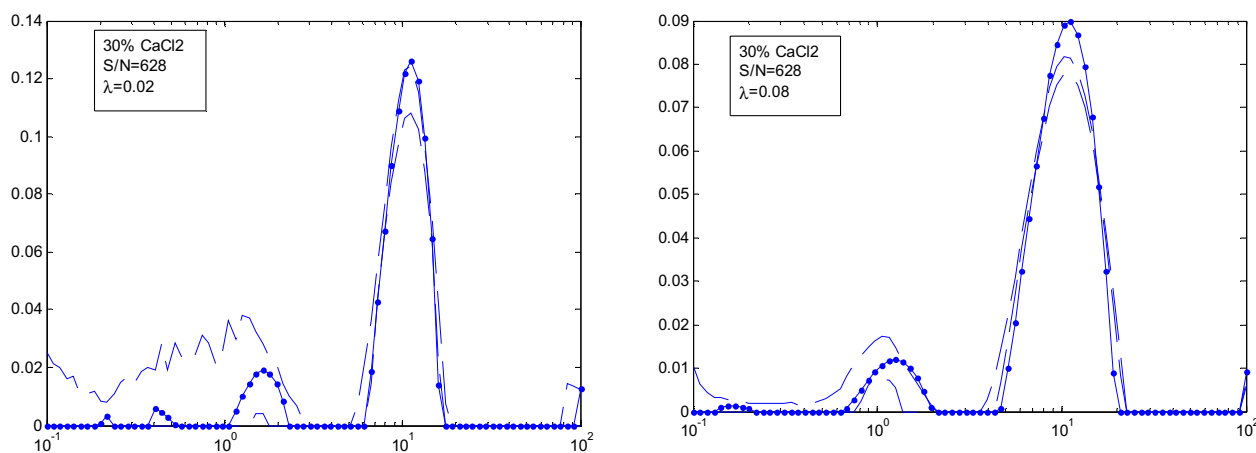


Fig. 6. Error band for two different values of the regularization parameter λ .

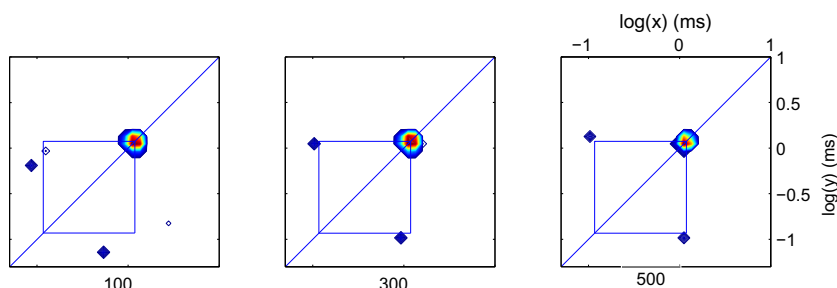


Fig. 7. Effect of the signal to noise ratio (100, 300, 500) on the detection of off-diagonal peaks in a T_2 -store- T_2 experiment. The input signal consists of four peaks of amplitude $P^{--} = 0.07$, $P^{+-} = P^{-+} = 2.5$ and $P^{++} = 100$.

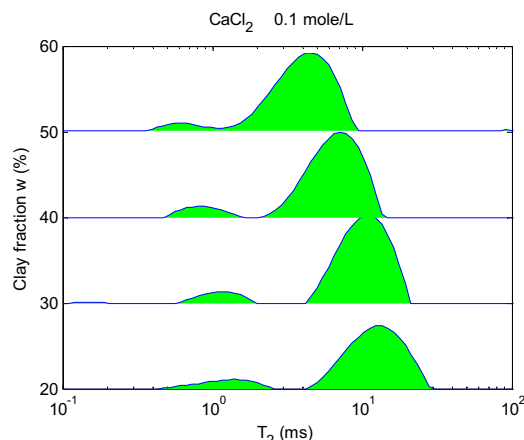


Fig. 8. Bimodal distributions obtained on flocculated smectite gels as a function of clay fraction. From Ref. [14].

corresponds to the intra-flock spacing (about 1 nm, see Fig. 8 in Ref. [14]) and the highest to the inter-flock spacing (about 10 nm), the latest varying with the amount of clay. Given the very small pore sizes in these systems, diffusional coupling may strongly affect the T_2 distribution and result in wrong conclusions about the inter- and intra-flocks fraction.

The T_2 -store- T_2 experiments were performed on the 30% clay fraction sample for increasing values of the storage time t_e , from 0.5 ms up to 10 ms. The 2D maps (Fig. 9) indicate no diagonal peaks at short exchange time ($t_e = 0.5$ ms) but they gradually appear when increasing the exchange time. Note that the amplitude color¹ coding is not the same between the different graphs, such that the amplitude decreases cannot be observed. The four peaks form a square pattern as expected but the square is not perfect for the reasons developed in the previous section. We also measured T_1 and T_2 distributions and determined that $T_1/T_2 = 1.8$.

The analysis of the peak intensities (Fig. 10) allows an estimation of the exchange rates k_a and k_b , the initial magnetization M_a^0 and M_b^0 , and the relaxation times T_{2b} and T_{2a} . We adjusted the four parameters k_a , M_a^0 , T_{2b} and T_{2a} to the curves. The intensity of the main peak is normalized by the amplitude at the smallest imposed t_e value ($P_n^{++} = P^{++}/P^{++}(t_e = 0.5 \text{ ms})$). We found $k_a = 0.06$, $M_a^0/(M_a^0 + M_b^0) = 0.155$, $T_{2b} = 11$ ms, $T_{2a} = 2$ ms. Hence, we have $k_b = 0.011$ and the degree of coupling is $\alpha = 0.28$. According to Fig. 3, this system is weakly coupled and intraflocks porosity fraction is underestimated by about 10% only.

4.2. Shaly sandstone (Vosges sandstone)

This case is typical of natural sandstones containing a variable amount of clays, either filling the pores or lining at their surface. The estimation of the pore volume fraction occupied by water in clays is of practical industrial importance because it is directly related to the potential of an oil reservoir in terms of hydrocarbon in place. Compared to the previous example, the pore sizes are changed by several orders of magnitude. Typically, clay pore sizes in such natural systems is around 1 μm or less and intergranular spacing is of order 10 μm . From a mercury injection experiment on a companion sample, the dominant throat diameter is 7 μm , and the clays are represented as a tail at smaller size extending down to 0.1 μm . However, the mercury injection experiment is somewhat ambiguous because it must be performed in a dry state.

From this point of view, the NMR T_2 characterization appears ideal because performed in a fully saturated state. However, the separation between water trapped in clays (clay bound water) and intergranular water is usually not well defined, as shown by the 1D projection of the 2D map at short exchange time in Fig. 11. It is usually considered that a cut-off around 10 ms gives reasonable estimation of clay bound water [15].

The T_2 - T_2 maps calculated at a long exchange time ($t_e = 90$ ms, Fig. 11) indicate clearly an exchange between clays and intergranular porosity (i.e. between pores at $T_2^- = 2$ ms and $T_2^+ = 50$ ms indicated by the square pattern). In a similar way as for the smectite gel, we analyzed the intensities as a function of the exchange time t_e (Fig. 12). In this case, we subdivided the T_2 maps into four regions to define off and on-diagonal intensities (Fig. 11). In this case, the intensities P^{++} , P^{--} , P^{+-} and P^{-+} are the sum of the amplitudes in each sub-region. We found $k_a = 0.02 \text{ ms}^{-1}$, $k_b = 0.016 \text{ ms}^{-1}$, $M_a^0/(M_a^0 + M_b^0) = 0.45$, $T_{2b} = 200$ ms, $T_{2a} = 10$ ms. The T_1/T_2 ratio was set at 1.8. Hence, the degree of coupling $\alpha = 0.73$ is weak to intermediate. The back calculated values of T_2^- and T_2^+ are, respectively, 8 and 55 ms. Hence, the intergranular pore size (or more exactly the V/S ratio) is underestimated by a factor of 4 on the 1D distribution.

4.3. Double porosity carbonates

Carbonates often contain multiple pore sizes due to various diagenetic processes. Contrary to sandstones, natural carbonates are made of various debris (animals, plants) that are subjected to dissolution/precipitation before, during or after burial. Hence, porosity can be present between, as well as, inside "grains". The samples used here are part of a larger study comparing NMR T_2 distributions and pore throat size distributions deduced from mercury injection experiments (MICP), with the help of visual examination of the pore structure from thin sections and scanning electron microscopy (SEM) [16,17]. In this work, it was established that microporosity defined in the petrographical sense, is the volume accessible through throat diameters smaller than 2 μm and correspond to relaxation times smaller than 200 ms in the T_2 distribution. An example is given in Fig. 13. Macroporosity is located between red algae, that are themselves porous. As a result, the MICP distribution indicates a bimodal distribution. The NMR T_2 distribution indicates similarly a bimodal distribution shifted, as expected, to the left of the MICP distribution, after calculation of the surface relaxivity from BET measurements. The macropore peak is centered around $T_2 = 1000$ ms, and we performed a 2D experiment using an exchange time of $t_e = 1000$ ms that indicated no off-diagonal peak and therefore no pore to pore diffusional exchange. The presence of sparitic cement as well as a cement rim around red algae is the most likely explanation for the lack of coupling. Also, the size of the object containing microporosity is large (300 μm) compared to the diffusion length of molecules during the exchange time (about 100 μm for 1000 ms).

A careful observation of the porous structure may give good indication of the existence of pore coupling. In the case of the Lavoux limestone (Fig. 14, left), microporosity is located in well identified spherical structures (oolithes) without cementation at their surface. In this case, the bimodal MICP signature (Fig. 14, right) is in contradiction with the 1D NMR when saturated with water (Fig. 15, left). Indeed, the MICP indicates that about 50% of the pore volume is accessible through throat diameters below 2 μm . The NMR distribution indicates a wide distribution with a small peak around 2 ms representing only 10% of the pore volume. The 2D map indicates an exchange between population located at 30 and 2 ms. Clearly, there exist very large pores (of the order of 100 μm) that should have a large relaxation time (1000 ms) similar to the previous case. Therefore, this system is strongly coupled. To

¹ For interpretation of the references to color in Fig. 9, the reader is referred to the web version of this paper.

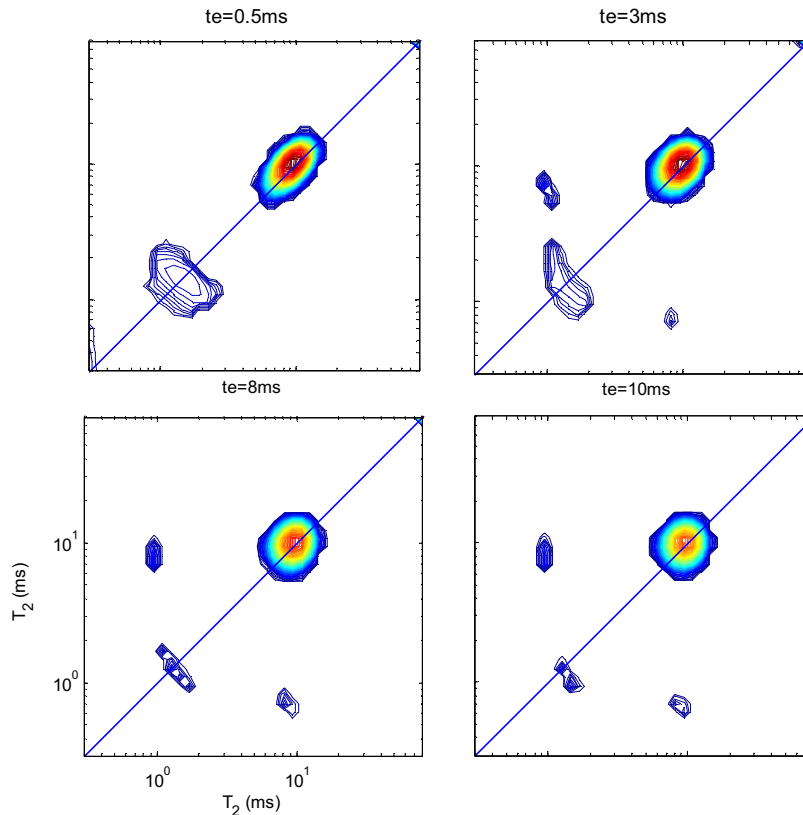


Fig. 9. T_2 -store- T_2 relaxation maps for a smectite gel at a clay fraction of 30%, as a function of the exchange time t_e . The position of the micro- and macropore peaks are independent of t_e ($T_2^+ = 9.7$ ms, $T_2^- = 1.4$ ms).

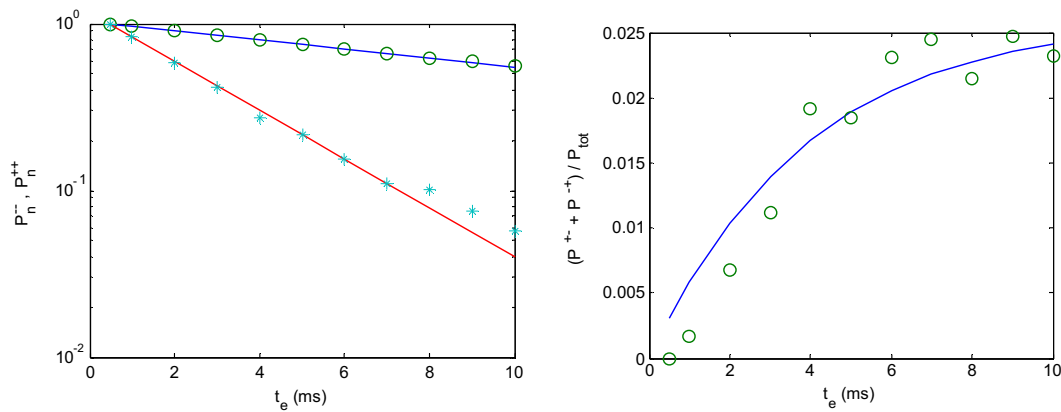


Fig. 10. Peak intensities as a function of exchange time t_e for the smectite gel at a clay fraction of 30%. Left: $P_n^{++} = P^{++}/P^{++}(t_e = 0.5 \text{ ms})$ ($\circ$), $P_n^{--} = P^{--}/P^{--}(t_e = 0.5 \text{ ms})$ (*). Right: normalized relative amplitude of the off-diagonal peak intensities. The full curves indicate a fit with $k_a = 0.060 \text{ ms}^{-1}$, $k_b = 0.011 \text{ ms}^{-1}$, $M_a^0/(M_a^0 + M_b^0) = 0.155$, $T_{2b} = 11$ ms, $T_{2a} = 2$ ms. The degree of coupling is $\alpha = 0.28$.

decrease the degree of coupling, the saturating fluid was replaced by a mixture of water and glycerol to a weight ratio of 36:64. Hence, the diffusivity is reduced by one order of magnitude (from $2.6 \times 10^{-9} \text{ m}^2/\text{s}$ down to $0.18 \times 10^{-9} \text{ m}^2/\text{s}$ at 30°C , as measured by PFG NMR). The liquid bulk relaxation time is also reduced (from 2700 down to 380 ms) but is sufficiently large to allow the observation of large pores. In these conditions, we observed a tri modal 1D distribution more representative of the texture observed by SEM. When performing the exchange experiment using a similar exchange time (30 ms), we see two couples of diagonal peaks indicating an exchange between populations located at 30 and 2 ms, and 300 and 2 ms, but no exchange between populations at 300

and 30 ms. We interpret these observations in terms of the structure revealed by SEM visualization (Fig. 14): oolites contain microporosity but also intermediate size pores (that could be labeled mesoporosity) in their centers. Hence the exchange between meso- and macropores has a large time constant preventing the observation of off-diagonal peaks. It is also consistent with the observation of two distinct peaks in the MICP distribution: mesoporosity is only accessible through microporosity located around the oolites.

For both saturating liquids, we calculated the peak intensities in a similar way as for the Vosges sandstone and attempted to calculate the coupling coefficients and exchange rates. These attempts

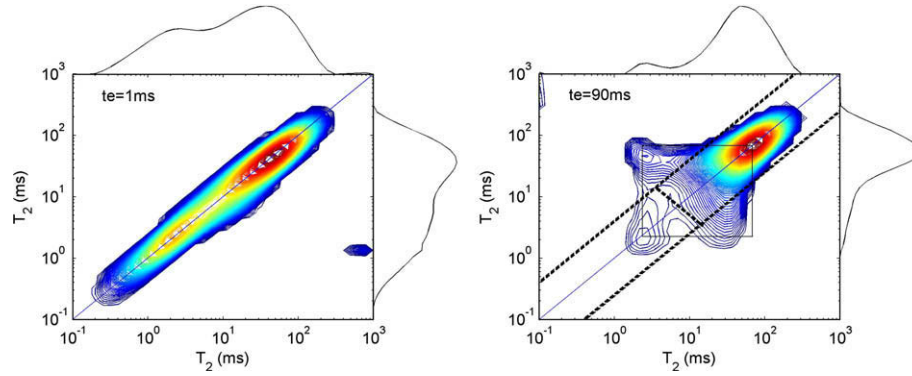


Fig. 11. T_2 -store- T_2 relaxation maps for a Vosges sandstone sample for two values of the exchange time t_e . Porosity 22% and permeability: 210 mD. The clay (micropore) and intergranular (macropore) peaks are centered around $T_2^- = 2$ ms and $T_2^+ = 50$ ms, respectively. The square pattern indicate the coupling, and the dashed lines the sub-regions for calculating peak intensities.

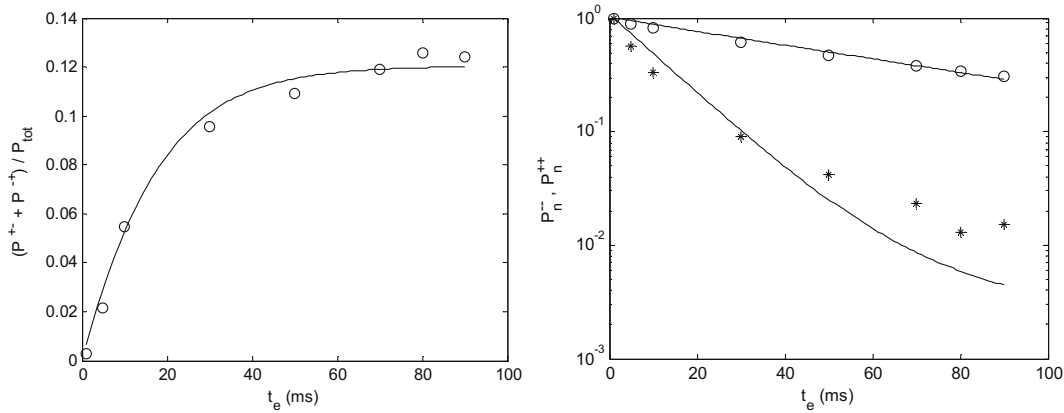


Fig. 12. Peak intensities as a function of exchange time t_e for the Vosges sandstone sample. Left: $P_n^{++} = P^{++}/P^{++}(t_e = 0.5 \text{ ms})$ ($\circ$), $P_n^{--} = P^{--}/P^{--}(t_e = 0.5 \text{ ms})$ ($*$). Right: normalized relative amplitude of the off-diagonal peak intensities. The full curves indicate a fit with $k_a = 0.02$, $M_a^0/(M_a^0 + M_b^0) = 0.45$, $T_{2b} = 200$ ms, $T_{2a} = 10$ ms. The degree of coupling is 0.73.

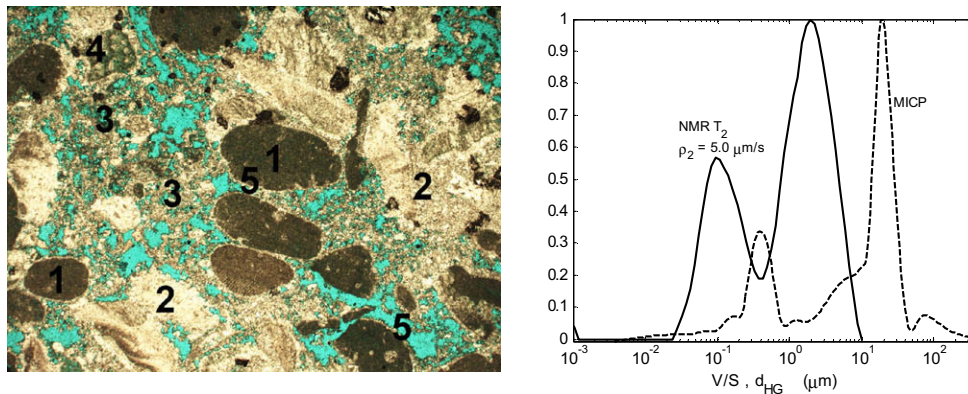


Fig. 13. Left: texture of a sample of Estailades limestone as shown by a thin section; 1: red algae; 2: sparitic cement; 3: partially dissolved cement; 4: brachiopod fragment; 5: early cement rim. Porosity 29% and permeability: 175 mD. The width of the image represents 3125 μm . The green color indicates macropores (filled with resin). Right: NMR T_2 and mercury injection distributions. The surface relaxivity ρ_2 has been calculated using specific surface measurements on the same sample (BET technique). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this paper.)

were not successful and yielded inconsistent values of T_2^- and T_2^+ when compared to the observed peak center position. This not surprising because the model only takes into account two pore populations. However, we expected to be able to estimate the coupling coefficient with the lower diffusivity experiment by analyzing the

peak intensity separately. We encountered this problem in other carbonate samples having large pore size and similar complex structure (not shown). This difficulty may be linked to the existence of a distribution of exchange time. More likely, all the macropore water may not be in exchange with micropore water

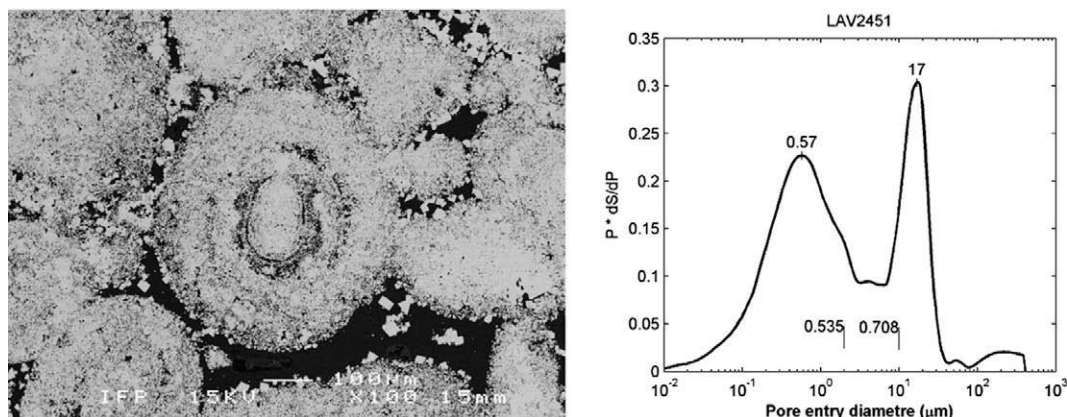


Fig. 14. Characterization of a Lavoux limestone sample. Left: SEM visualization, right: mercury injection capillary pressure curve (MICP). Porosity: 24%, permeability: 19 mD. The two pore size structure is evidenced by the bimodal MICP pore entry size distribution.

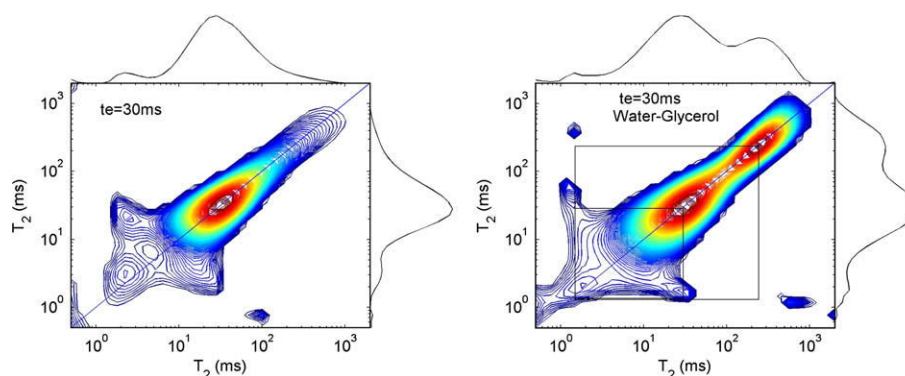


Fig. 15. T_2 -store- T_2 relaxation maps for the Lavoux limestone saturated with water (left, $D = 2.610^{-9} \text{ m}^2/\text{s}$ at 30°C) and a mixture of water and glycerol (right, $D = 0.1810^{-9} \text{ m}^2/\text{s}$, $T_{2\text{bulk}} = 380 \text{ ms}$ at 30°C). The squares highlight the exchange between micro- and mesopores, and micro- and macropores. When saturated with water, the T_2 distribution is wide and quasi-unimodal, in contradiction with the MICP distribution and the SEM visualization in Fig. 14. For a lower molecular diffusion of the saturating fluid, the pore coupling decreases and a multimodal distribution is observed.

because of the large macropore size and an insufficient diffusion length, a situation similar to the diffusion limited regime when considering surface relaxation mechanism [2]. A rough estimation can be made by comparing the experiments with water and water/glycerol: the exchange rate being proportional to diffusivity and decreased by a factor of ten, we have necessarily a coupling coefficient in the range 1–10 for water experiment, and 0.1–1 for the water/glycerol mixture experiment.

5. Conclusion

Diffusional coupling is a very common mechanism, potentially perturbing the interpretation of 1D relaxation time distributions, but providing new information of the connectivity of porous systems. For bimodal systems in which we cover a range of pore sizes from 1 nm up to 100 μm , we defined a coupling parameter α comparing the average exchange rate and the relaxation rate of the smallest pore population (micropore) having the largest specific surface. Based on an analytical solution, we analyzed the behavior of 1D T_2 and 2D T_2 -store- T_2 relaxation experiments as a function of α . In 1D experiments and for increasing values of α , the relaxation time of micropores is shifted to smaller values and their amplitude decreases. The coupling parameter, which is proportional to the diffusivity of the saturating fluid, must vary by two orders of magnitude to switch from a weakly to a strongly coupled state. For 2D

T_2 -store- T_2 experiments, diagonal peak intensities are maximum at intermediate coupling degrees and represent at maximum 10% of the total intensity for usual T_1/T_2 ratio of 1.5. Therefore, these experiments require high signal to noise ratio, of the order of 200 or larger, and a careful use of 2D inverse Laplace transform, as illustrated by various tests of the inversion process. The coupling was evidenced and quantified on a flocculated smectite gel and a shaly sandstone. Although relatively simple, the two pore population model allows the analysis of distributed bimodal systems. On complex carbonate structures, multiple exchanges were observed and explained with the help of SEM visualizations but the two pore population model could not be used for a quantitative analysis of molecular exchange. We suggest that this difficulty is either due a distribution of exchange times or the partial exchange of macropore water with microporosity in the case of large pores. An estimate of the order of magnitude of the degree is performed by changing the diffusivity of the saturating liquid by a factor of ten.

Acknowledgments

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Appendix A. T_2 -store- T_2 sequence

The T_2 -store- T_2 sequence is essentially a succession of two CPMG sequences separated by a storage time t_e :

$$P_{90} - (\tau - P_{180} - \tau - \text{echo} -)_m - P_{90} - t_e - P_{90} \\ - (\tau - P_{180} - \tau - \text{echo}_{\text{acq}} -)_n$$

All echoes intensities in the second part of the sequence are acquired, yielding a two-dimensional data set of size (m, n) such that the recorded magnetization is a two-dimensional function:

$$M(t_1, t_2) = A_{xy} \exp(-t_1/T_{2x}) \exp(-t_e/T_1) \exp(-t_3/T_{2y}), \\ t_1 = 2m\tau, \quad t_3 = 2n\tau \quad (10)$$

It is usually necessary to have small τ values to properly catch short relaxation times but this would result in a huge number of pulses when large and small relaxation times are present simultaneously such as in our carbonate systems. To limit the total number of pulses to about 10,000 (to avoid a potential heating of the sample and unstability of the power amplifier), the inter-pulse time τ is increased gradually from an initial value of 30 μs up to 100 μs (or more depending on the largest relaxation time present in the sample). Then, it stays at a constant value. During the second CPMG sequence, typically 5000 echo maximum are recorded (the value of n is tuned such as to obtain a signal decreasing to zero, as usually performed in a single CPMG sequence). Then this signal is compressed by averaging all echo maximum in 40 predefined bins having a constant log spacing in time (with origin the third P_{90}), yielding a compressed time series t_e . For the first CPMG sequence, only the m values corresponding to the 40 values of t_e are used. Hence, the two-dimensional data set acquired reduces finally to a 40×40 matrix. The advantage of this protocol is the large increase of signal to noise ratio for intermediate and large relaxation times due to compression.

The solutions described in Eqs. (5) and (6) are only valid if a $+/-z$ phase cycling is performed when averaging several signals. If this is not performed, many off-diagonal peaks may be observed, even in a single pore [11]. Adding this phase cycling to the usual CYCLOP phase cycling, a minimum of eight scans is necessary.

Appendix B

Coefficients included in Eq. (6) describing the dependence of the on- and off-diagonal peak amplitudes as a function of storage time t_e .

$$C_1 = (A_2^- - F_2) \left[A_1^- (A_2^- M_a^0 - G_2 M_b^0) - G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] \\ + (B_2^- - G_2) \left[B_1^- (B_2^- M_b^0 - F_2 M_a^0) - F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_2 = (A_2^- - F_2) \left[A_1^+ (A_2^- M_a^0 - G_2 M_b^0) + G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^- - G_2) \\ \times \left[B_1^+ (B_2^- M_b^0 - F_2 M_a^0) + F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_3 = (A_2^+ - F_2) \left[A_1^- (A_2^+ M_a^0 - G_2 M_b^0) + G_1 (B_2^+ M_b^0 + F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^- (B_2^+ M_b^0 + F_2 M_a^0) - F_1 (A_2^+ M_a^0 + G_2 M_b^0) \right]$$

$$C_4 = (A_2^+ + F_2) \left[A_1^+ (A_2^+ M_a^0 + G_2 M_b^0) + G_1 (B_2^+ M_b^0 + F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^+ (B_2^+ M_b^0 + F_2 M_a^0) + F_1 (A_2^+ M_a^0 + G_2 M_b^0) \right]$$

$$C_5 = (A_2^+ + F_2) \left[A_1^- (A_2^+ M_a^0 - G_2 M_b^0) - G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^- (B_2^- M_b^0 - F_2 M_a^0) - F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

$$C_6 = (A_2^+ + F_2) \left[A_1^+ (A_2^+ M_a^0 - G_2 M_b^0) + G_1 (B_2^- M_b^0 - F_2 M_a^0) \right] (B_2^+ + G_2) \\ \left[B_1^+ (B_2^- M_b^0 - F_2 M_a^0) + F_1 (A_2^- M_a^0 - G_2 M_b^0) \right]$$

where

$$A_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_{1,2}^a + k_a}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \quad B_{1,2}^{+/-} = \frac{s_{1,2}^{+/-} + R_{1,2}^b + k_b}{s_{1,2}^{+/-} - s_{1,2}^{-/+}}, \\ F_{1,2} = \frac{k_a}{s_{1,2}^- - s_{1,2}^+}, \quad G_{1,2} = \frac{k_b}{s_{1,2}^- - s_{1,2}^+}.$$

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